

Power Electronics

6th Sem, ECE, NIT Silchar

Syllabus

Unit 1	Introduction: need for power conversion with efficient, high frequency, light weight converters; Power electronic converters classifications and scope;
Unit 2	<i>Power semiconductor switches</i> : Power/fast diodes, BJT, MOSFET, SCR, Triacs, GTOs, IGBT and MCT, Ratings, static and dynamic characteristics, principle of operation, drive and switching aid circuits and cooling; isolation; protection.
Unit 3	<i>AC-DC converters (Rectifiers)</i> : Diode rectifier, phase control rectifiers; effect of source inductance, single/three phase rectifiers, semi/full rectifiers, power factor, harmonics.
Unit 4	<i>DC-AC converters (inverters)</i> : Concept of switched mode inverters, PWM switching, voltage and frequency control of single/three phase inverters, harmonics reduction, other switching schemes-square wave pulse switching, programmed harmonic elimination switching, current regulated modulation switching- tolerance band control, fixed frequency control, voltage source inverter (VSI),current source inverter(CSI).
Unit 5	<i>DC-DC Converter (Chopper)</i> : Principle, buck, boost and buck-boost converters, analysis with continuous and discontinuous loads, flyback converters
Unit 6	<i>AC voltage controllers</i> : Principle of ON-OFF control and phase control, single/three phase controllers, PWM AC voltage controller. ;Single phase AC static switches; transient-free switching of inductive loads; cycloconverter
Unit 7	Basic concepts of adjustable speed dc and ac drives and industrial application, Switched mode power supply

Syllabus for MidSEM (approx.)

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Introduction to Power Electronics

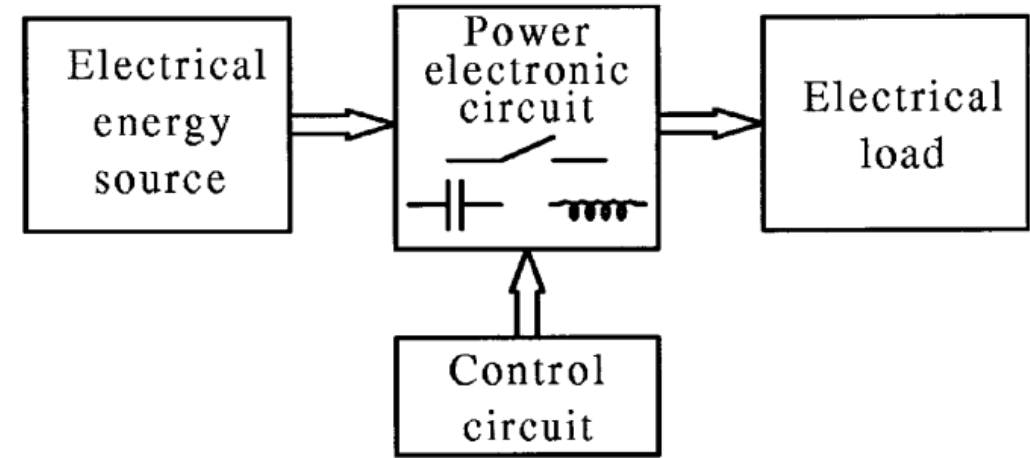
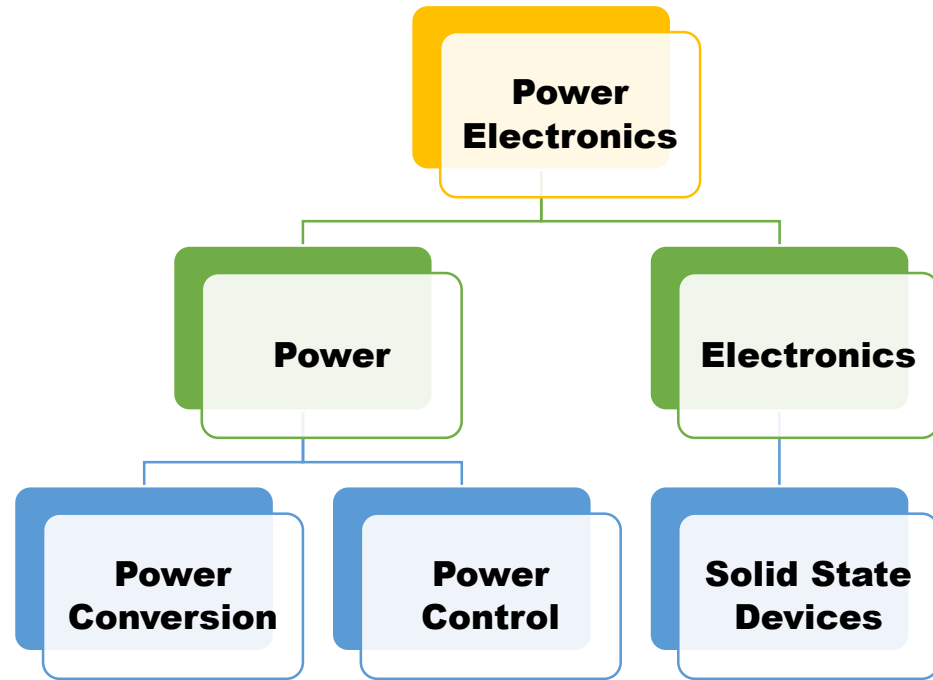


FIGURE 1.3 A basic power electronic system. (From Reference [2], copyright © 1998, Oxford University Press, Inc.; used by permission.)

- Power electronics involves the study of electronic circuits intended to control the flow of electrical energy. These circuits handle power flow at levels much higher than the individual device ratings.
- Power Electronics is a field which combines Power (electric power), Electronics and Control systems.

Need for Power Conversion

❖ Mismatch Between Source and Load

- Power generation may be **AC or DC**
- Voltage levels may be **high or low**
- Frequency may be **fixed or variable**
- Loads may require **constant voltage, constant current, or variable power**

Examples:

- A thermal power plant generates **11 kV AC at 50 Hz**, but your laptop needs **19 V DC**.
- A solar panel produces **variable DC**, but the grid requires **fixed-frequency AC**.
- A battery provides **DC**, but an induction motor needs **AC with variable frequency**.

This mismatch creates the *fundamental need for power conversion*.

- Changing **voltage level** (step-up or step-down)
- Changing **current level**
- Changing **AC to DC, DC to AC, AC to AC, or DC to DC**
- Changing **frequency** and **waveform**

Importantly, in power electronics, we aim to do this with:

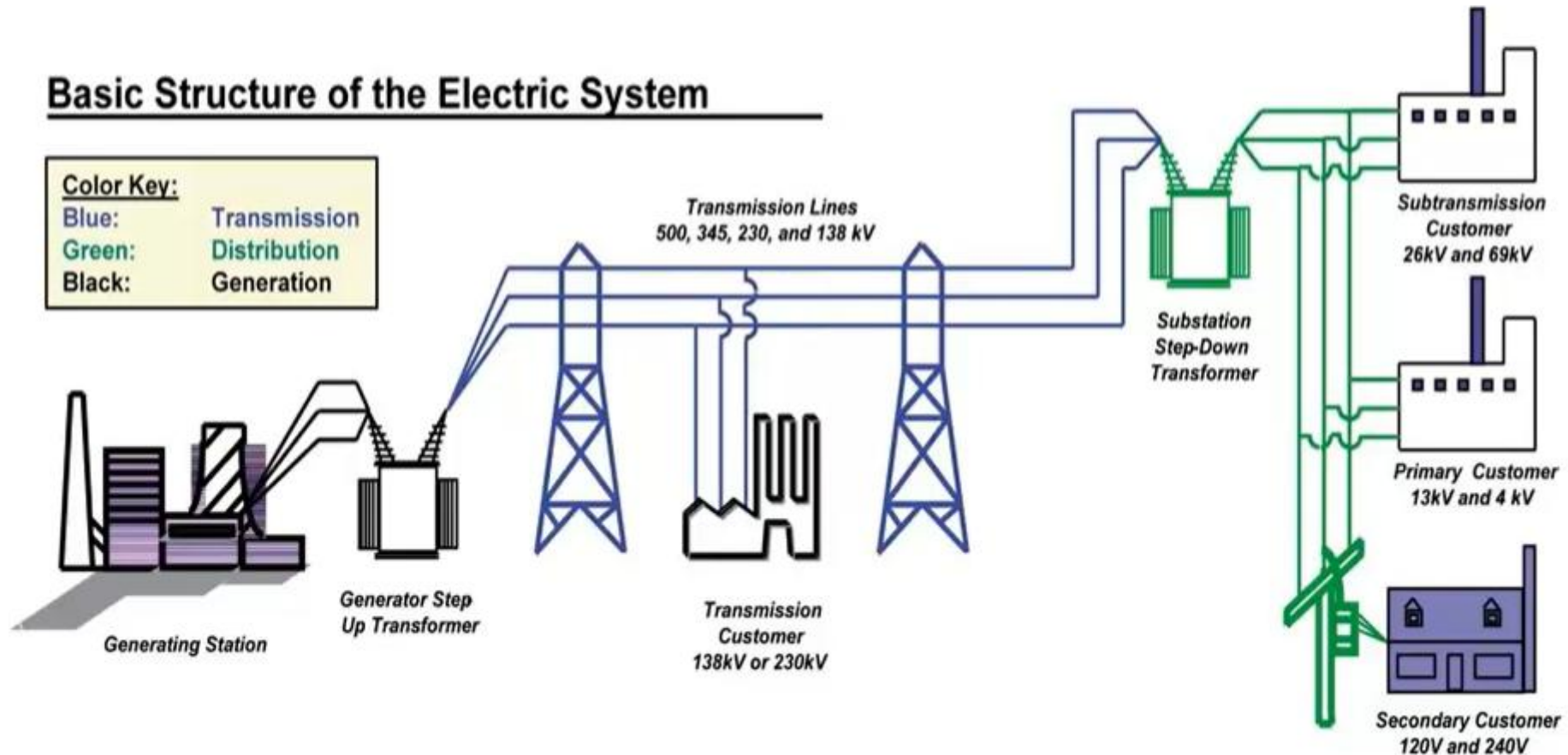
- **High efficiency**
- **Fast response**
- **Compact size**

Example

Electrical Power System

(Generation, Transmission, Distribution)

Basic Structure of the Electric System

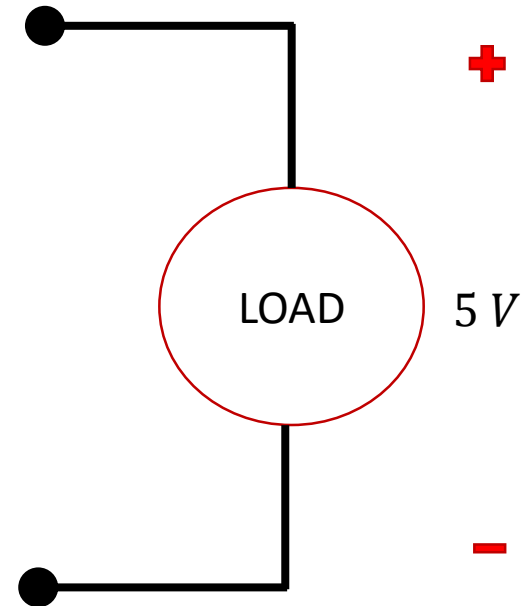
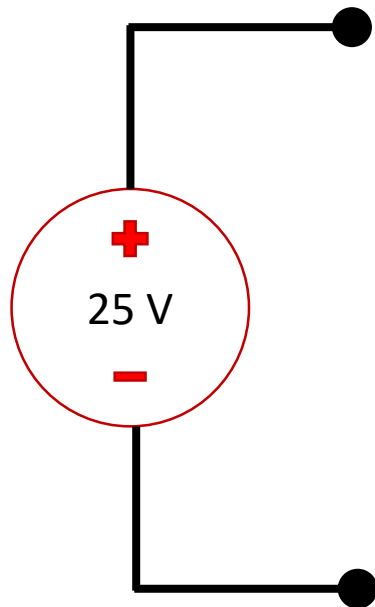


Example

Let us consider a mobile charger of 5 W , 5 V , 1 A . Now let us assume that we have a power bank or a DC voltage source of 25 V . How do we get 5 V to charge our phone?

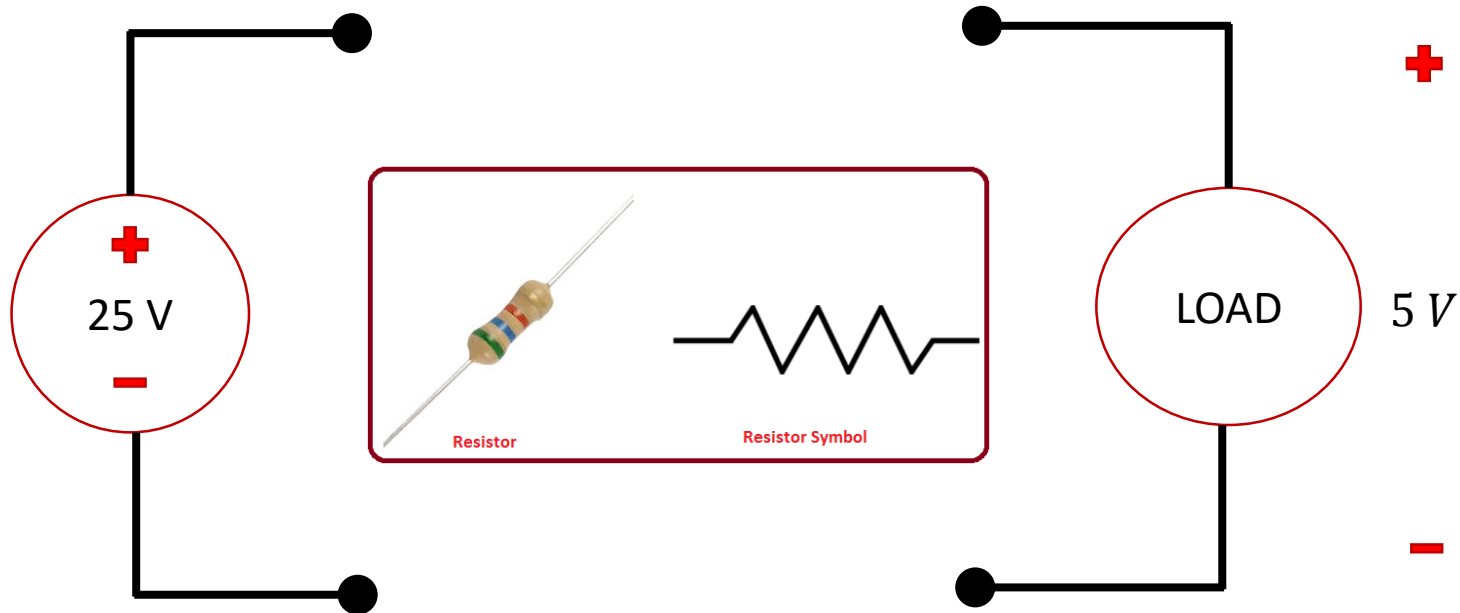
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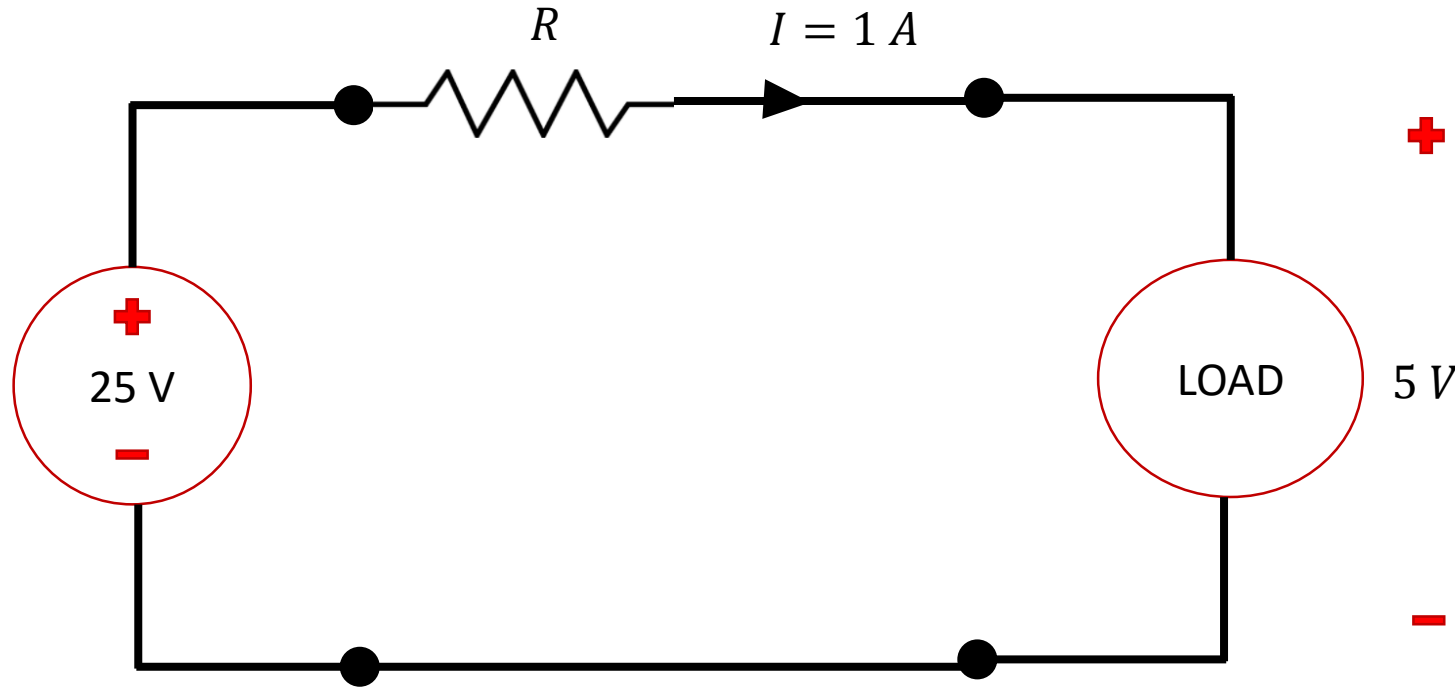
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Example

Let us consider a mobile charger of 5 W , 5 V , 1 A . Now let us assume that we have a power bank or a DC voltage source of 25 V . How do we get 5 V to charge our phone?



$$V_L = 5\text{ V}$$

$$I_L = 1\text{ A}$$

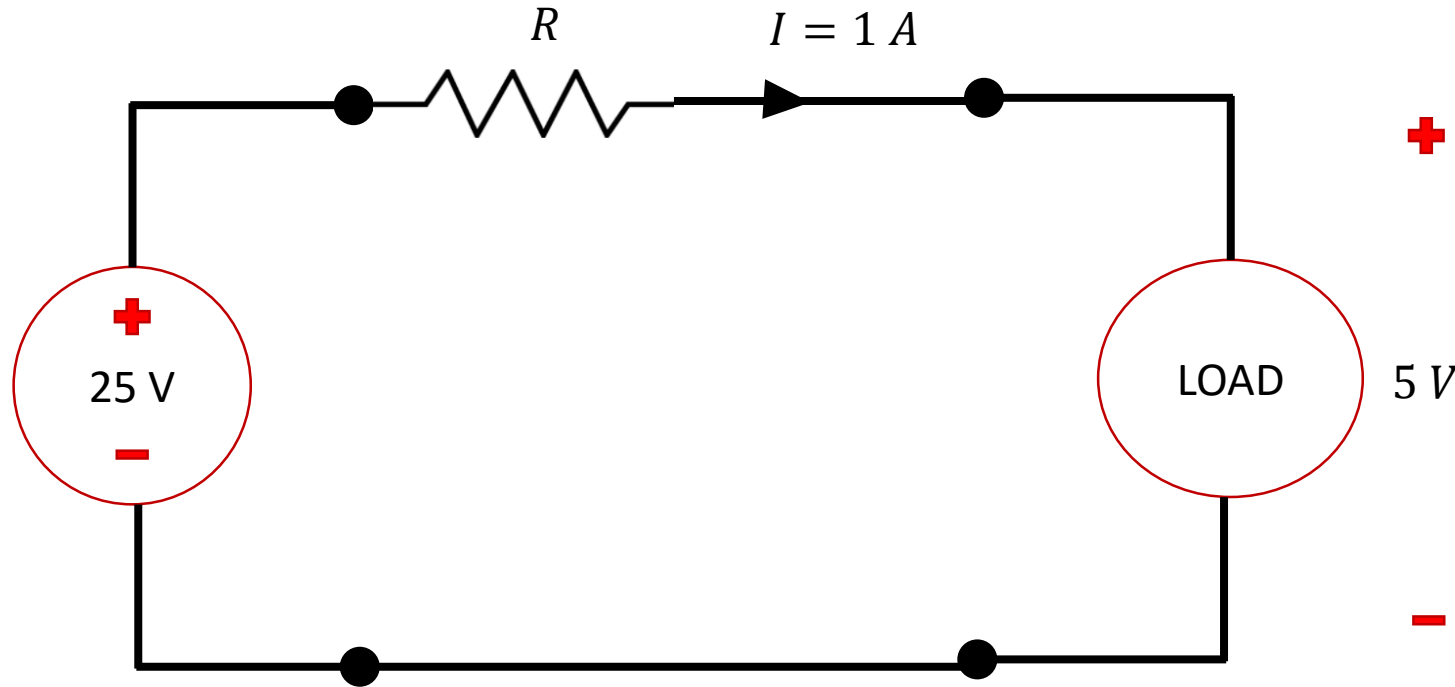
$$R_L = 5\ \Omega$$

$$I_L = 1\text{ A} = \frac{25}{5 + R}$$

$$R = 20\ \Omega$$

Example

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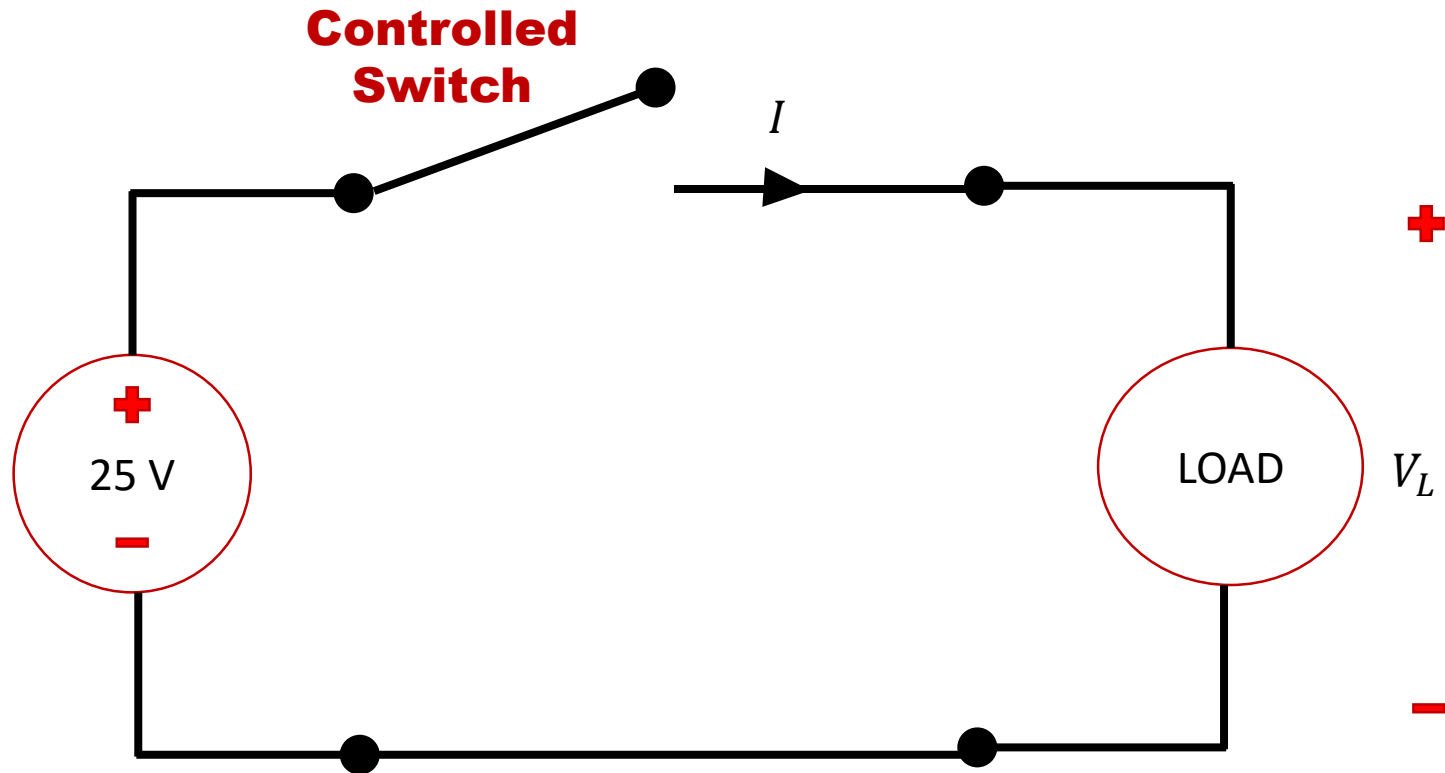
$$P_{in} = 25\text{ W}$$

$$P_{out} = 5\text{ W}$$

$$P_{loss} = I^2 R = 1 * 20 = 20\text{ W}$$

Example

Let us consider a mobile charger of 5 W , 5 V , 1 A . Now let us assume that we have a power bank or a DC voltage source of 25 V . How do we get 5 V to charge our phone?



Case 1: SW is ON

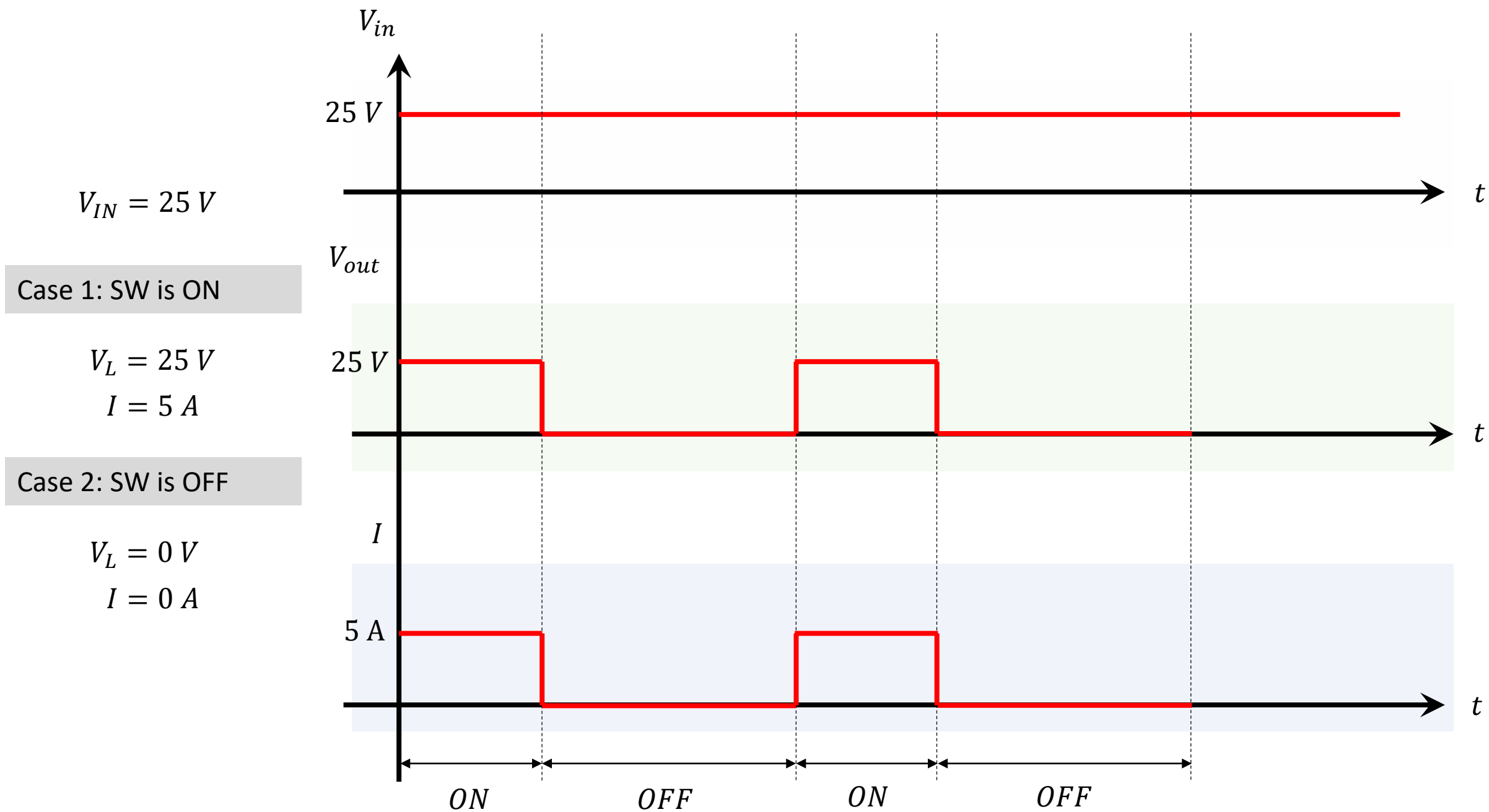
$$V_L = 25\text{ V}$$

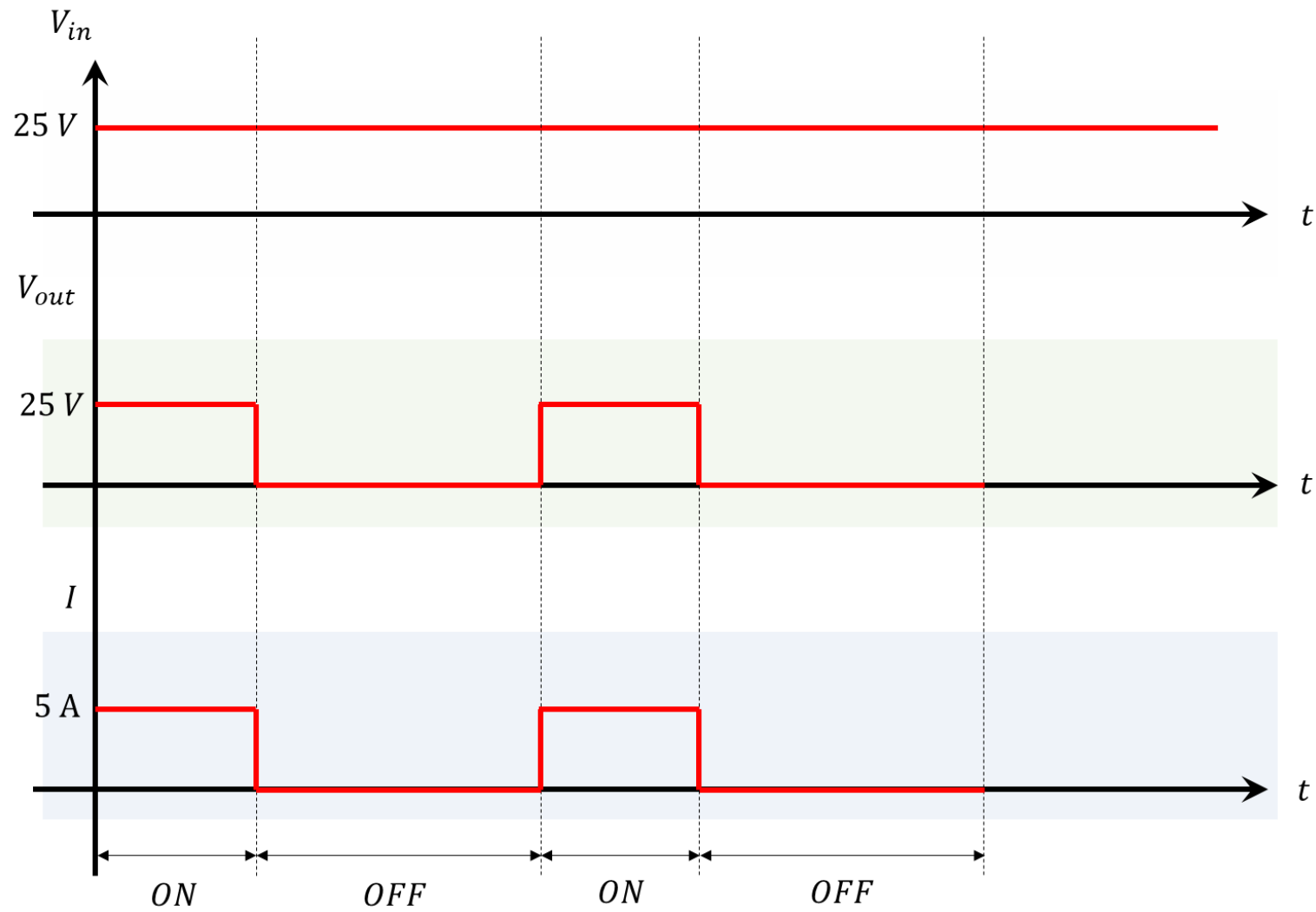
$$I = 5\text{ A}$$

Case 2: SW is OFF

$$V_L = 0\text{ V}$$

$$I = 0\text{ A}$$





$$V_{avg} = \frac{1}{T} \int_0^T v(t) dt$$

$$= \frac{1}{T} \int_0^{T_{on}} 25 dt$$

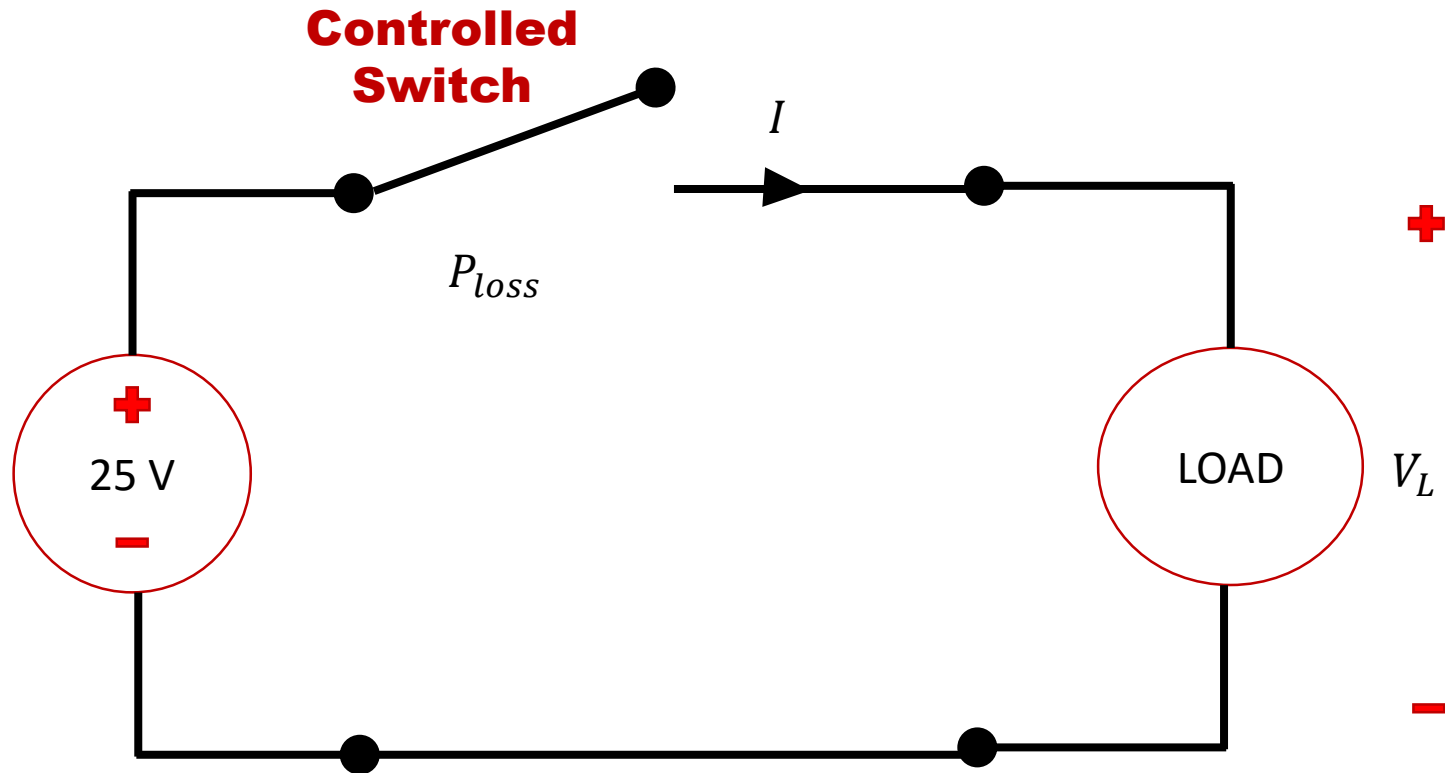
$$= 25 * \frac{T_{on}}{T}$$

$$5 = 25 * \frac{T_{on}}{T}$$

$$\frac{T_{on}}{T} = \frac{1}{5} = \text{duty cycle}$$

Example

Let us consider a mobile charger of 5 W , 5 V , 1 A . Now let us assume that we have a power bank or a DC voltage source of 25 V . How do we get 5 V to charge our phone?



Case 1: SW is ON

$$V_{sw} = 0\text{ V}$$

$$I = 5\text{ A}$$

$$P_{loss} = 0$$

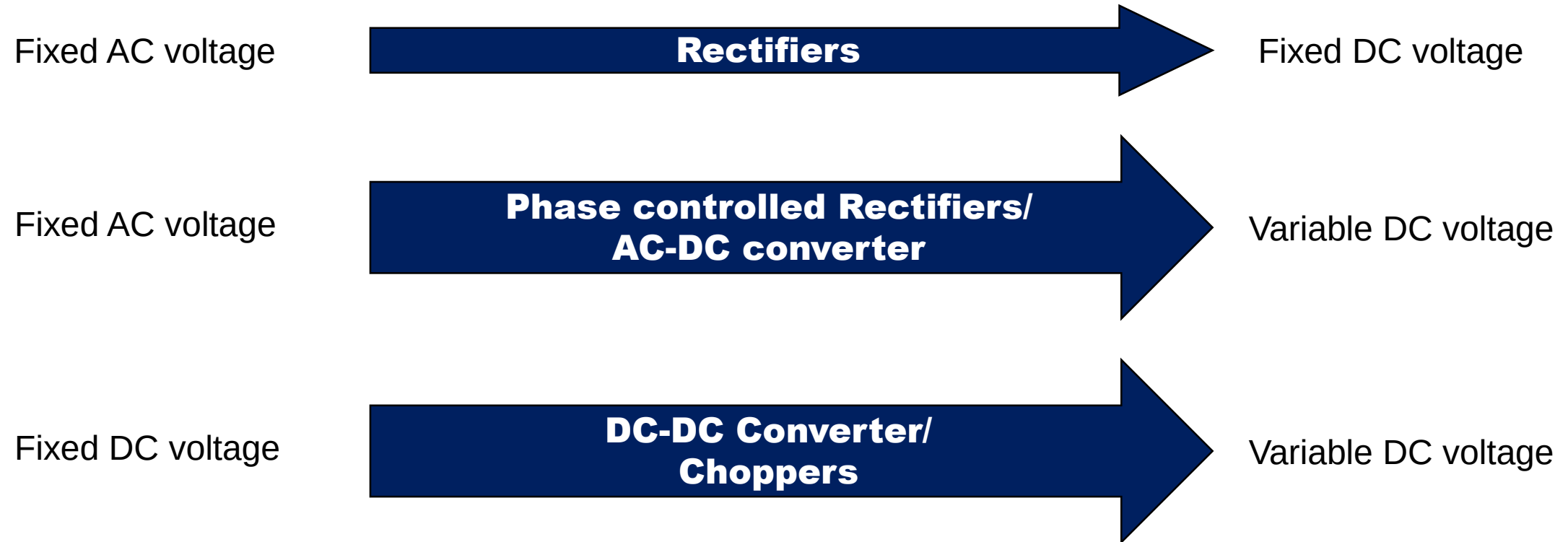
Case 2: SW is OFF

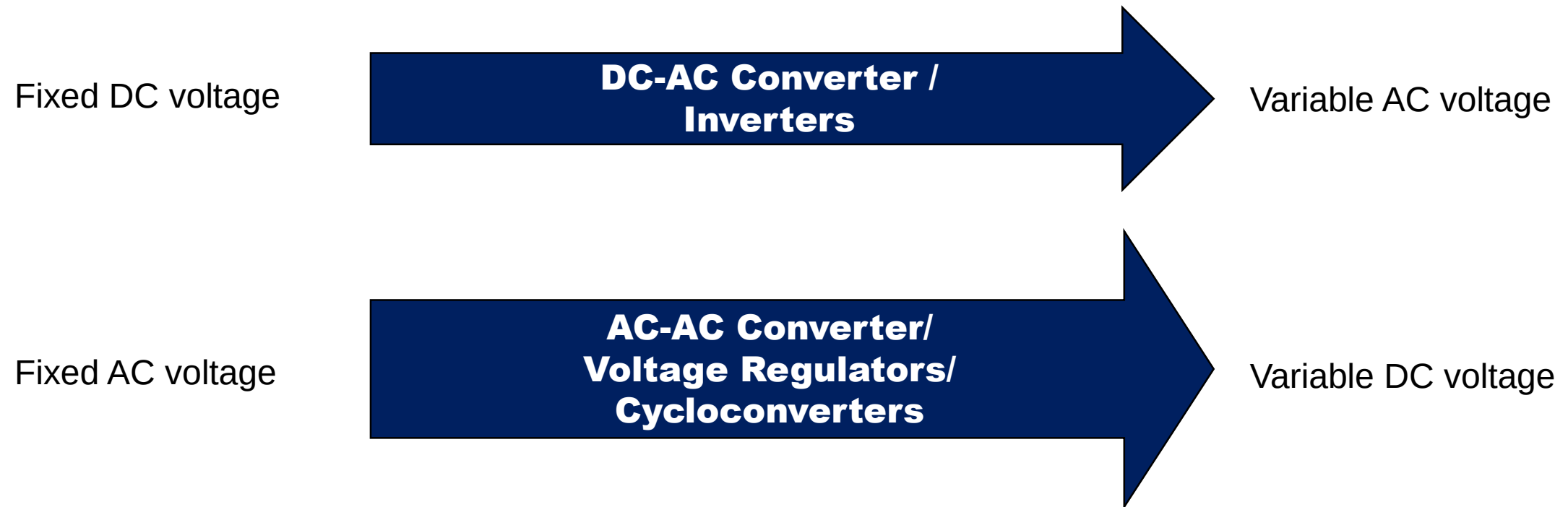
$$V_L = 25\text{ V}$$

$$I = 0\text{ A}$$

$$P_{loss} = 0$$

In practical switch there will be some power loss





Power Diode

- Among all the static switching devices used in power electronics (PE), the power diode is the simplest. It is a two terminal device, and with terminal A known as the anode and terminal K known as the cathode.
- If terminal A experiences a higher potential compared to terminal K, the device is said to be forward biased and a forward current (I_F) will flow through the device. This causes a small voltage drop across the device (<1 V), which under ideal conditions is usually ignored.
- By contrast, when a diode is reverse biased, it does not conduct and the diode then experiences a small current flowing in the reverse direction called the leakage current.
- Both forward voltage drop and leakage current are ignored in an ideal diode.
- In PE applications a diode is usually considered to be an ideal static switch.

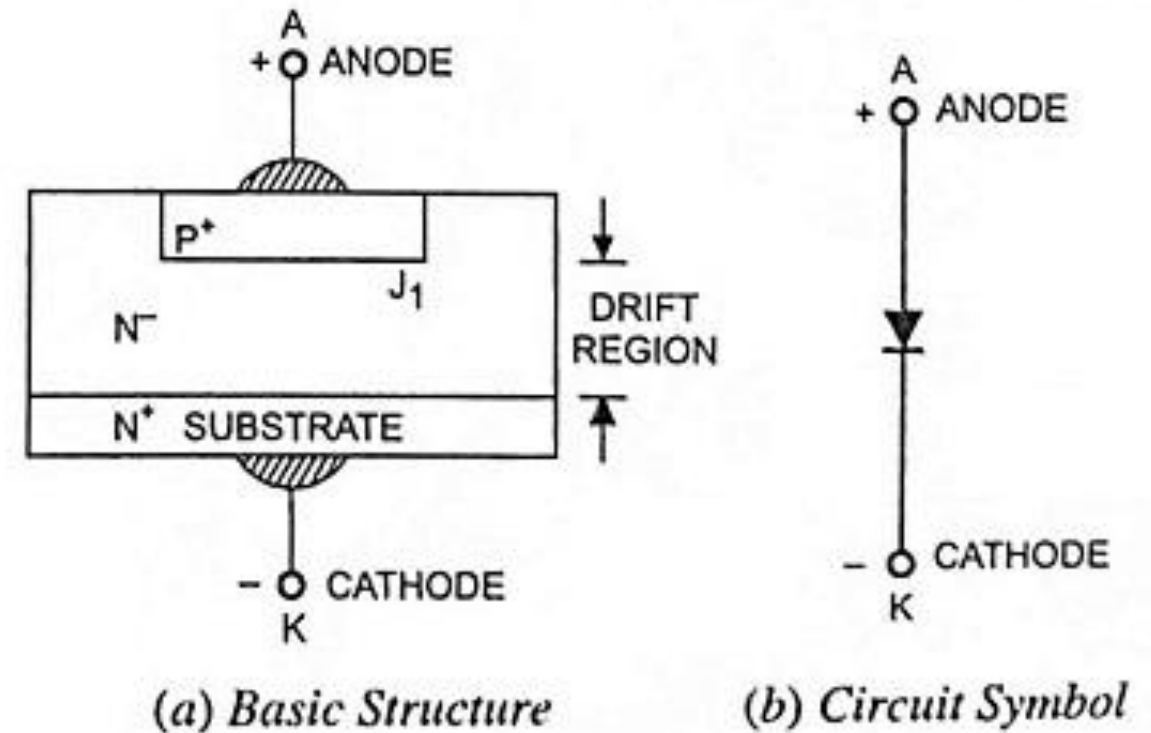
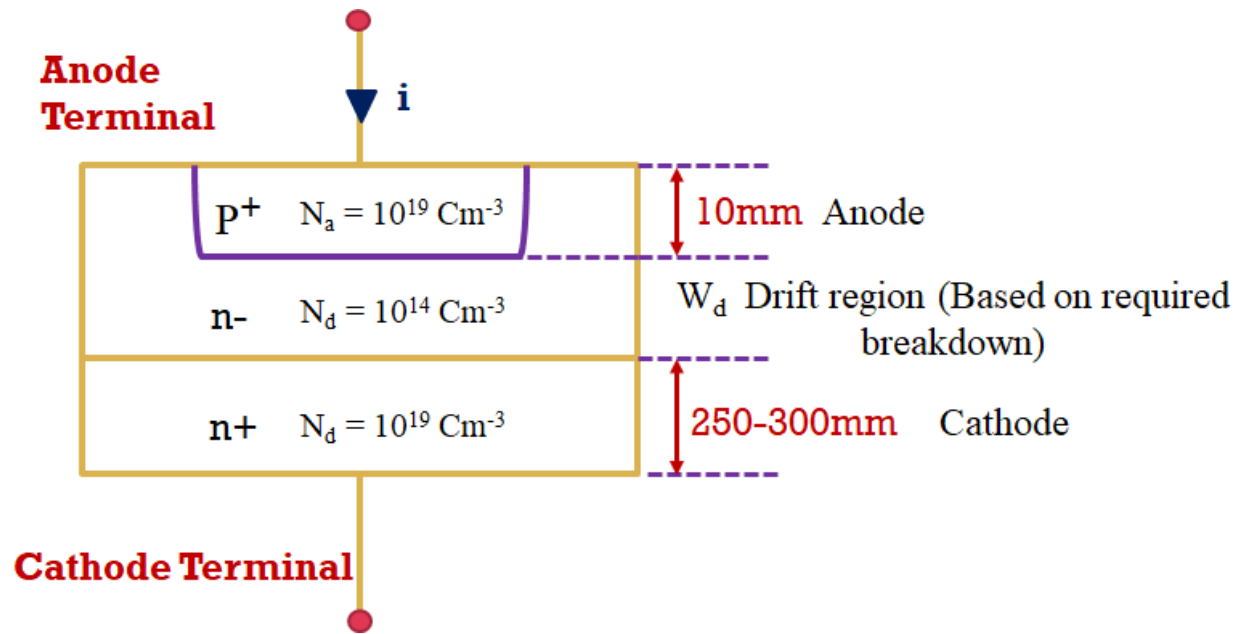
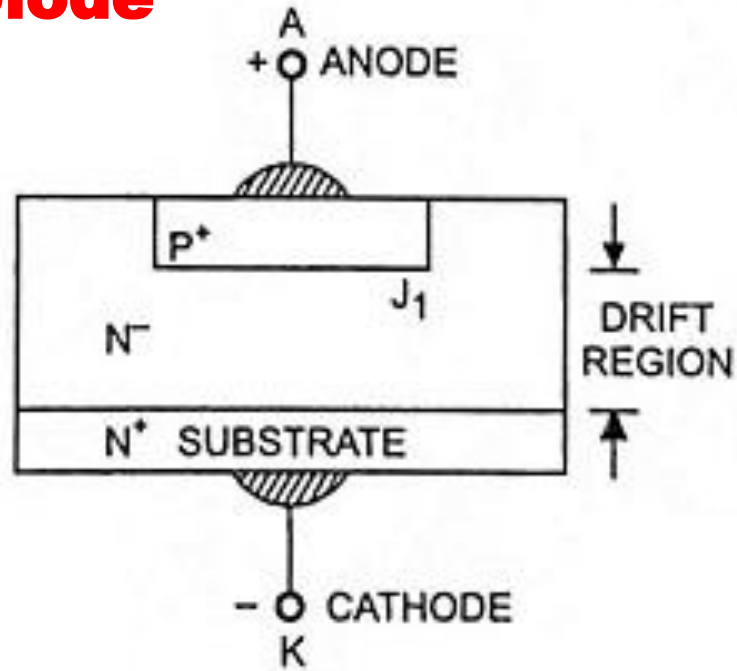


Fig. 8.51

Power Diode



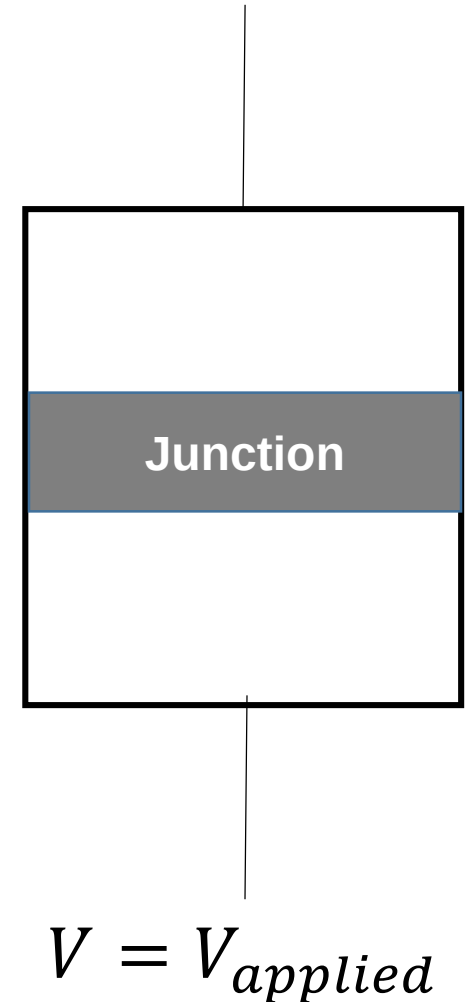
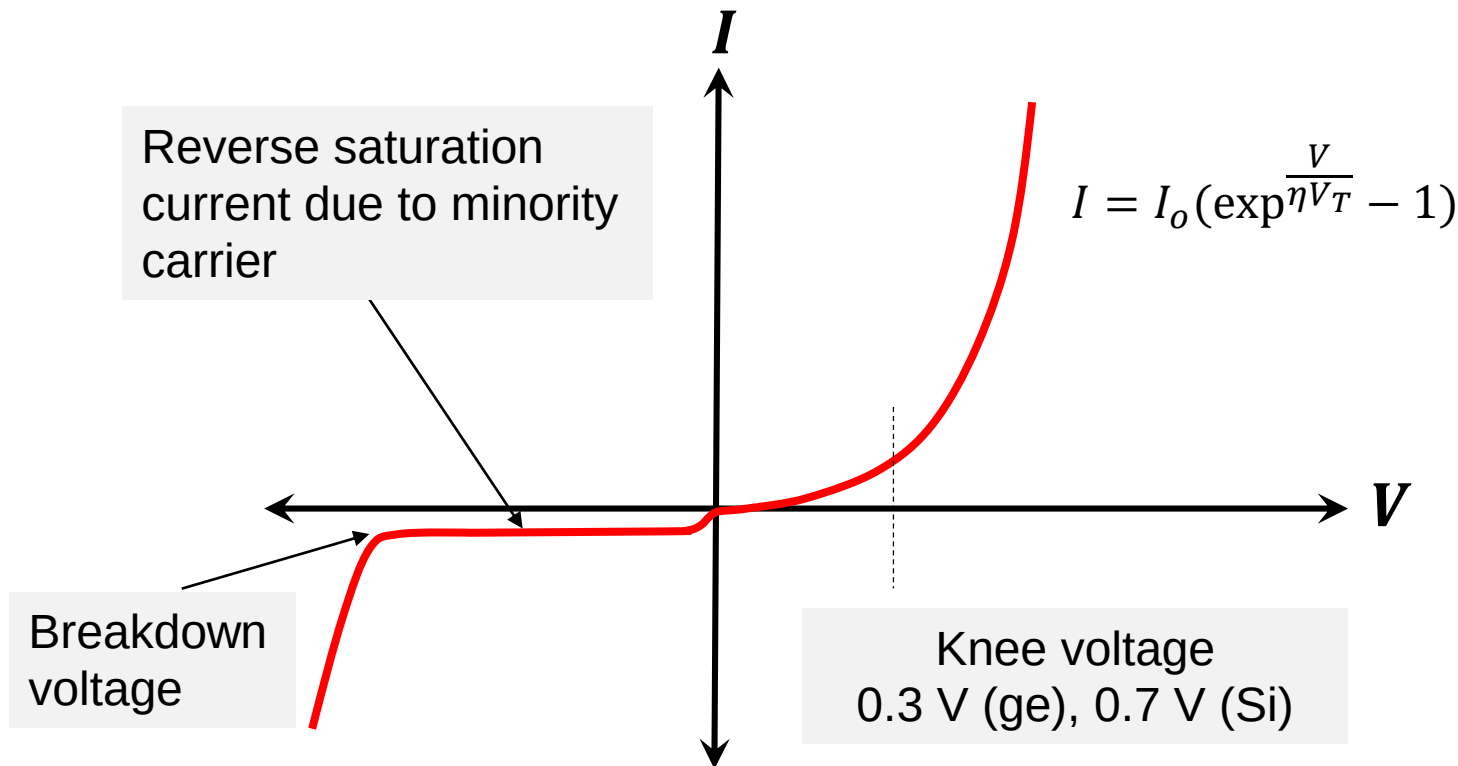
- Power diode is a p-n junction diode with higher breakdown voltage
- The increase in breakdown voltage is due to the addition of n^- drift region
- The thickness of the n^- drift region defines the breakdown voltage
- The n^- drift region also adds significant ohmic resistance when the diode conducts in the forward bias leading to large power dissipation. Proper cooling arrangement become necessary
- This lead to linear forward bias characteristics in power diode

Signal vs Power Diode IV characteristics

- Signal diodes are designed for low power and high speed switching applications

$$I = I_o(\exp^{\frac{V}{\eta V_T}} - 1)$$

In signal diodes, series resistance due to p and n region is neglected.



Signal vs Power Diode IV characteristics

- Power diodes are designed for high power applications and handles much larger current
- High series resistance due to larger junction area and thicker layers.

$$V = V_a - IR_s$$

$$I = I_o \left(\exp \frac{V_a - IR_s}{\eta V_T} - 1 \right) \approx I_o \exp \frac{V_a - IR_s}{\eta V_T}$$

$$\ln(I) = \ln \left(I_o \exp \frac{V_a - IR_s}{\eta V_T} \right)$$

$$\ln(I) = \ln(I_o) + \frac{V_a - IR_s}{\eta V_T}$$

$$\frac{V_a - IR_s}{\eta V_T} = \ln \left(\frac{I}{I_o} \right)$$

$$V_a = IR_s + \eta V_T \ln \left(\frac{I}{I_o} \right)$$

$$V_a = IR_s + \eta V_T \ln \left(\frac{I}{I_o} \right)$$

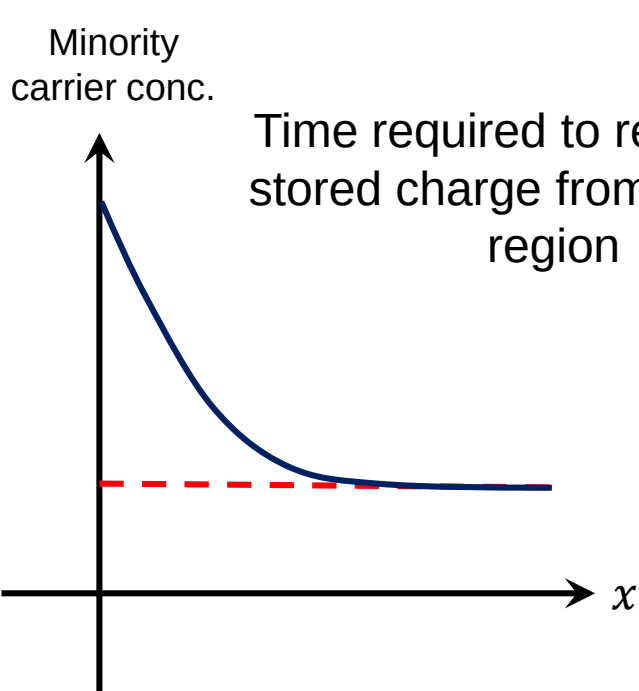
When I is high,

$$IR_s \gg \eta V_T \ln \left(\frac{I}{I_o} \right)$$

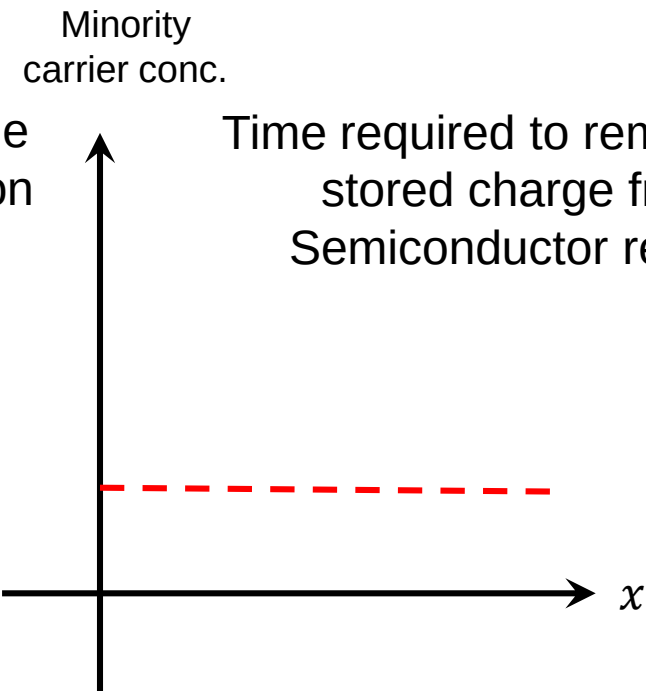
$$V_a \approx IR_s$$

Linear Behavior

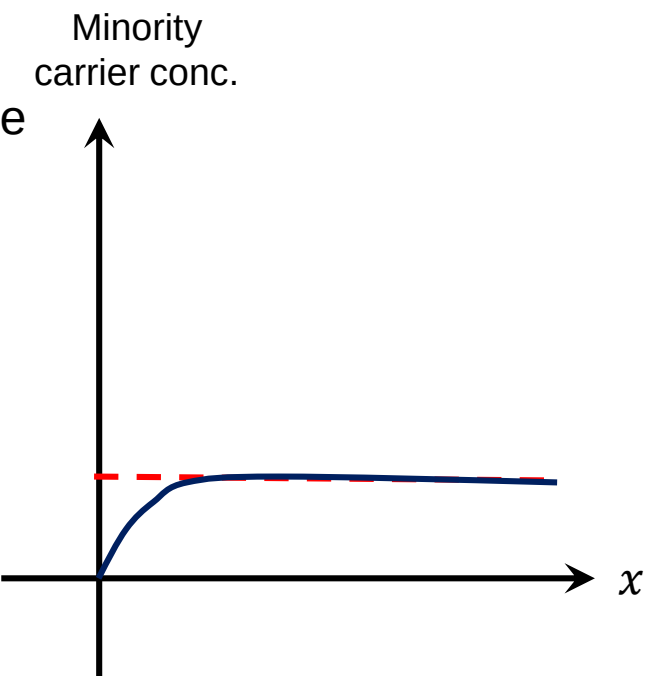
Power Diode reverse recovery characteristics



Forward Bias

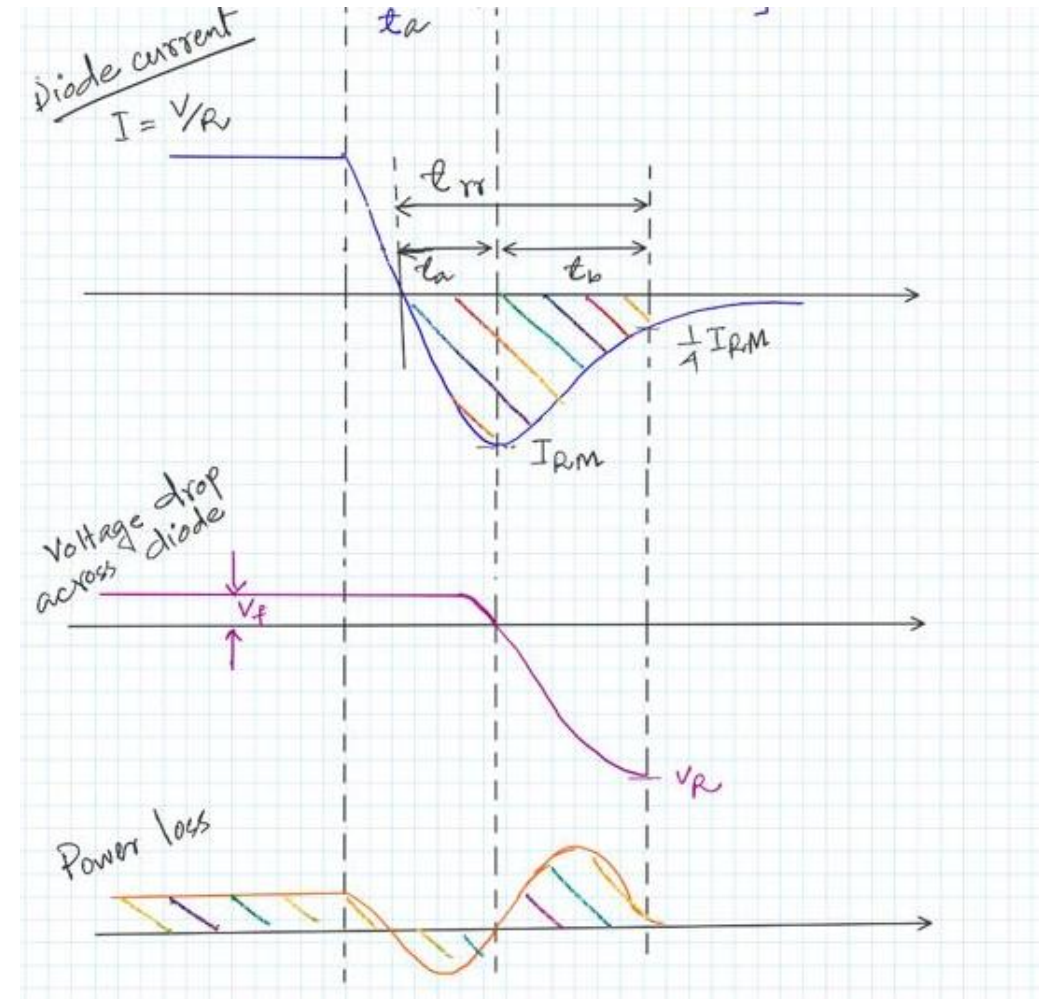
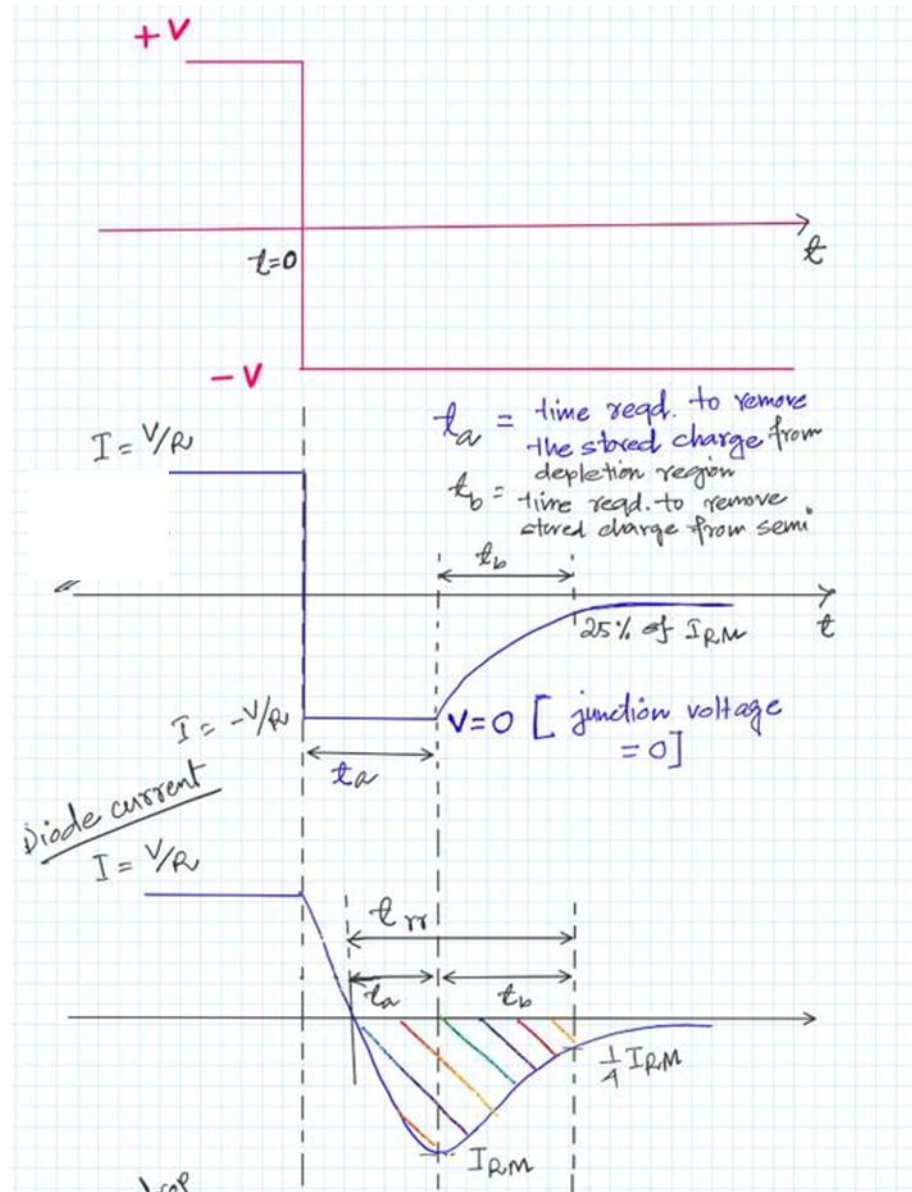


Thermal Equilibrium



Reverse Bias

Reverse Recovery Power Diode IV characteristics



Reverse Recovery Power Diode IV characteristics

- **Reverse recovery time:** The reverse recovery time (t_{rr}) is defined as the time between the instant the forward diode current becomes zero and the instant reverse recovery current decays to 25% of its reverse peak value I_{RM}
- The reverse recovery time (t_{rr}) is composed of two segments of time t_a and t_b

$$t_{rr} = t_a + t_b$$

- t_a is the time between zero crossing of the forward current and peak reverse current I_{RM} . During t_a , stored charge from depletion region is removed
- t_b is measured from the instant of peak reverse current I_{RM} to the instant when 25% of the peak reverse current I_{RM} is reached. During t_b , stored charge from semiconductor region is removed
- The ratio $\frac{t_b}{t_a}$ is called softness factor or S-factor
- A diode with S-factor equal to 1 is called soft recovery diode and a diode with S-factor less than 1 is called snappy recovery or fast recovery diode.

Reverse Recovery Power Diode IV characteristics

$$I_{RM} = t_a \frac{di}{dt} \rightarrow (1)$$

$$Q_R = \frac{1}{2} I_{RM} t_{rr} \rightarrow (2)$$

Assuming $t_a \gg t_b$, $t_{rr} \approx t_a$, (1) can be re-written as,

$$I_{RM} = t_{rr} \frac{di}{dt} \rightarrow (3)$$

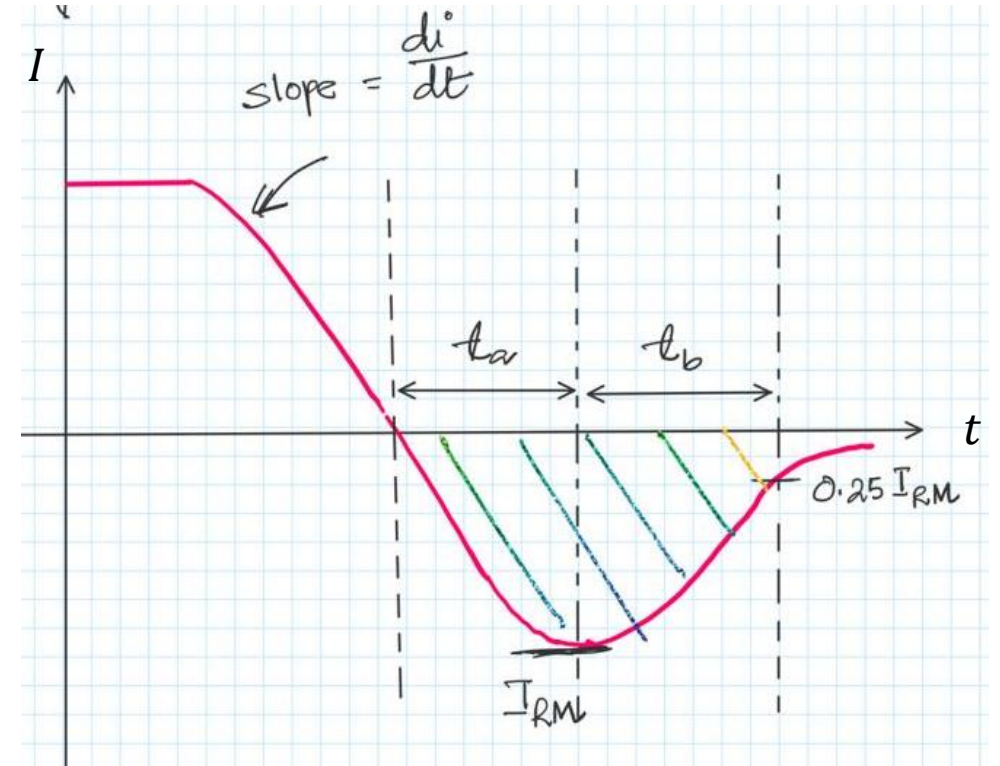
From (2) and (3)

$$Q_R = \frac{1}{2} t_{rr}^2 \frac{di}{dt}$$

$$t_{rr} = \left[\frac{2Q_R}{\frac{di}{dt}} \right]^{\frac{1}{2}}$$

$$Q_R = \frac{1}{2} I_{RM} I_{RM} / \frac{di}{dt}$$

$$I_{RM} = \left[2Q_R \frac{di}{dt} \right]^{\frac{1}{2}}$$



- The reverse recovery time (t_{rr}) and peak inverse current (I_{RM}) depends on storage charge Q_R and rate of change of current $\frac{di}{dt}$
- Q_R also depends on forward current and applied forward voltage

Types of power diodes based on reverse recovery time

1. General Purpose Diode

- High reverse recovery time: $\approx 25 \mu s$
- Current rating: $1 A$ to *Several kA*
- Voltage rating: $5 V$ to $5 kV$
- Primary Applications: Line Frequency (as rectifier)
 - Battery Charging, UPS

2. Fast Recovery Diode

- Reverse recovery time: $< 5 \mu s$
- Current rating: $1 A$ to *Several kA*
- Voltage rating: $5 V$ to $3 kV$ (diffusion technique);
[$400 V$ (epitaxial technique); $t_{rr} = 50 ns$]
- Platinum or gold doping to shorten t_{rr}

3. Schottky Diode

- Metal semiconductor junction
- Current is only due to electrons (majority carrier device)
- No stored charge, no reverse recovery
- Low cut-in voltage and low forward voltage drop
- Current rating: $1 A$ to $300 A$
- Voltage rating: $50 V - 100 V$ (*low*)
- High reverse leakage current
- Higher operating frequency

Numerical

Q1. A silicon power diode carries a forward current of 10 A. When it is suddenly reverse biased, the reverse recovery current waveform is approximately triangular with a peak reverse current of 5 A. The total reverse recovery charge is measured to be $25 \mu\text{C}$.

- a) Calculate the reverse recovery time t_{rr}
- b) If the diode is used in a switching circuit operating at 20 kHz, comment on its suitability.

$$I_{RM} = 5A$$

$$Q_R = 25 * 10^{-6} \text{ C}$$

$$Q_R = \frac{1}{2} I_{RM} t_{rr}$$

$$t_{rr} = \frac{2Q_R}{I_{RM}} = \frac{2 * 25 * 10^{-6}}{5} = 10^{-5} \text{ s} = 10 \mu\text{s}$$

For a switching circuit operating at 20 kHz,

$$T = \frac{1}{f} = \frac{1}{20 * 10^3} = 5 * 10^{-5} = 5 \mu\text{s}$$

Since the t_{rr} is higher than the time period of the circuit, the diode is not suitable.

Numerical

Q.2 A power diode has a reverse recovery time of $4 \mu s$. During reverse recovery, the reverse current rises linearly to a peak value of $8 A$ and then falls linearly to zero.

- Calculate the reverse recovery charge Q_R .
- If the diode is reverse biased at $200 V$, estimate the energy loss per switching due to reverse recovery.

$$t_{rr} = 4 \mu s = 4 * 10^{-6} s$$

$$I_{RM} = 8 A$$

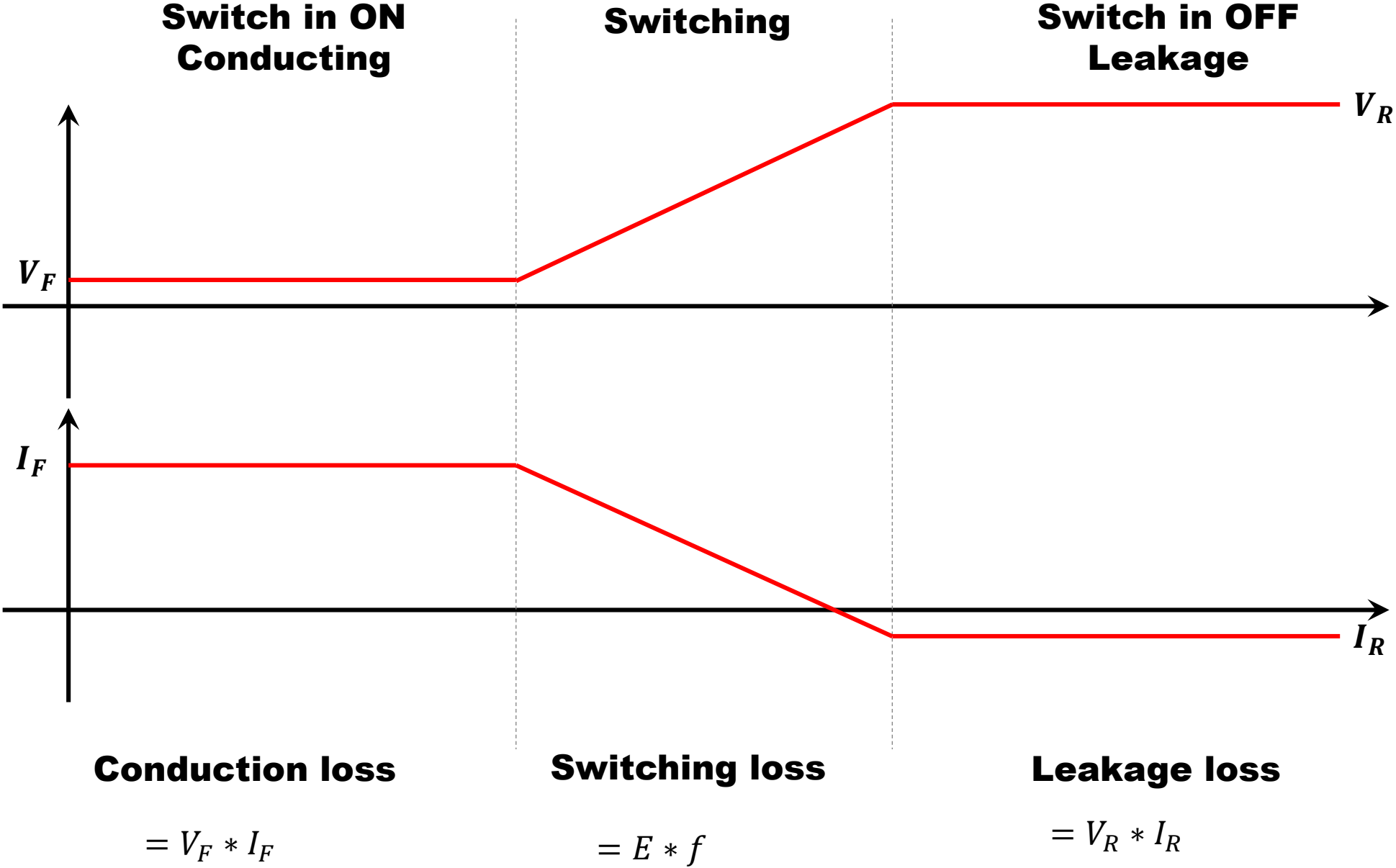
$$Q_R = \frac{1}{2} I_{RM} t_{rr}$$

$$Q_R = \frac{1}{2} * 8 * 4 * 10^{-6} = 16 \mu C$$

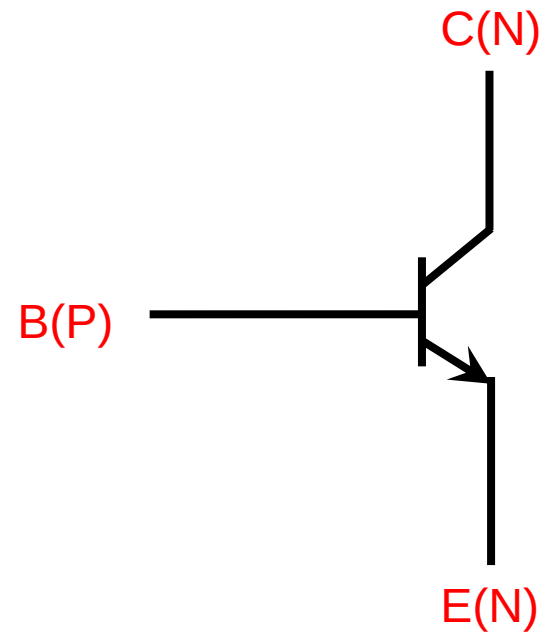
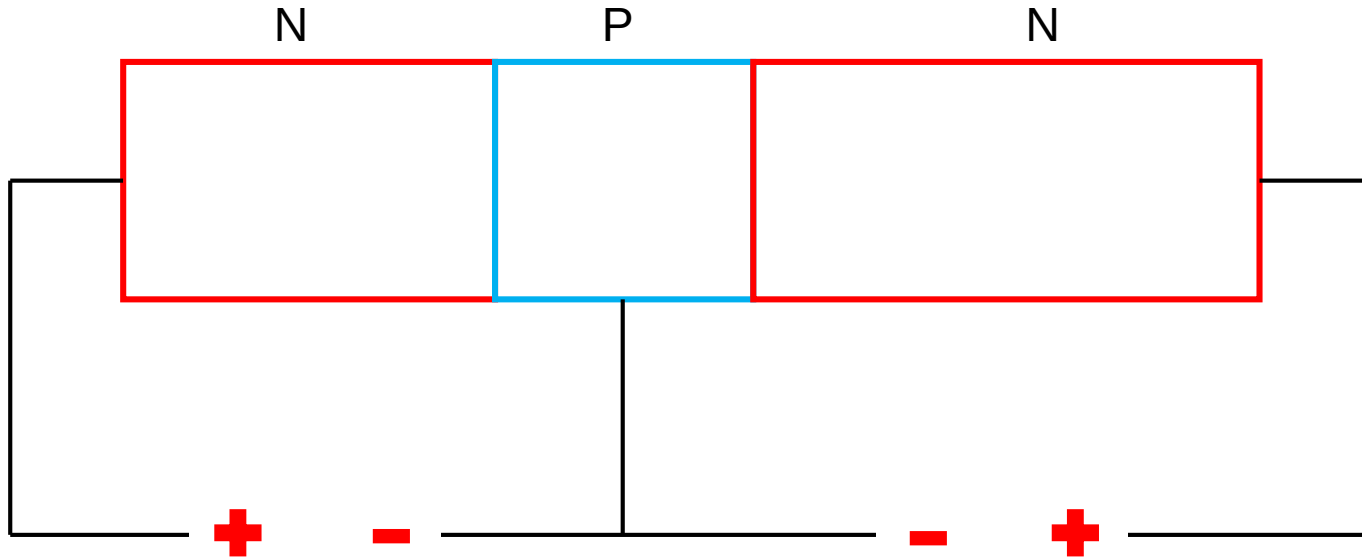
We know power loss = $V * I$

$$\text{Energy loss} = V * I * T = V * Q = 200 * 16 * 10^{-6} = 3.2 * 10^{-3} = 3.2 mJ$$

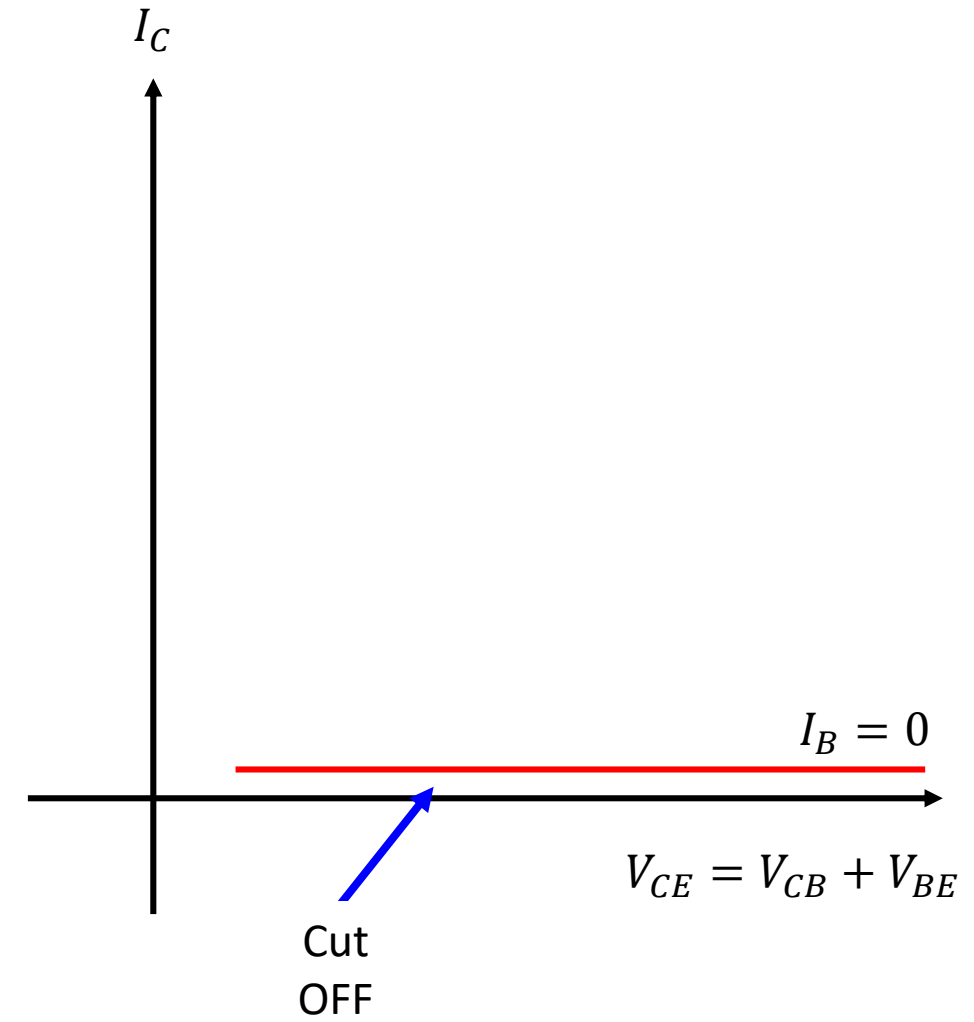
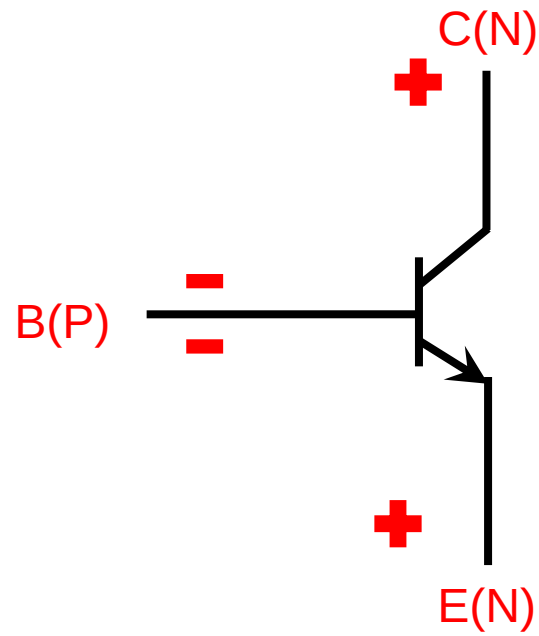
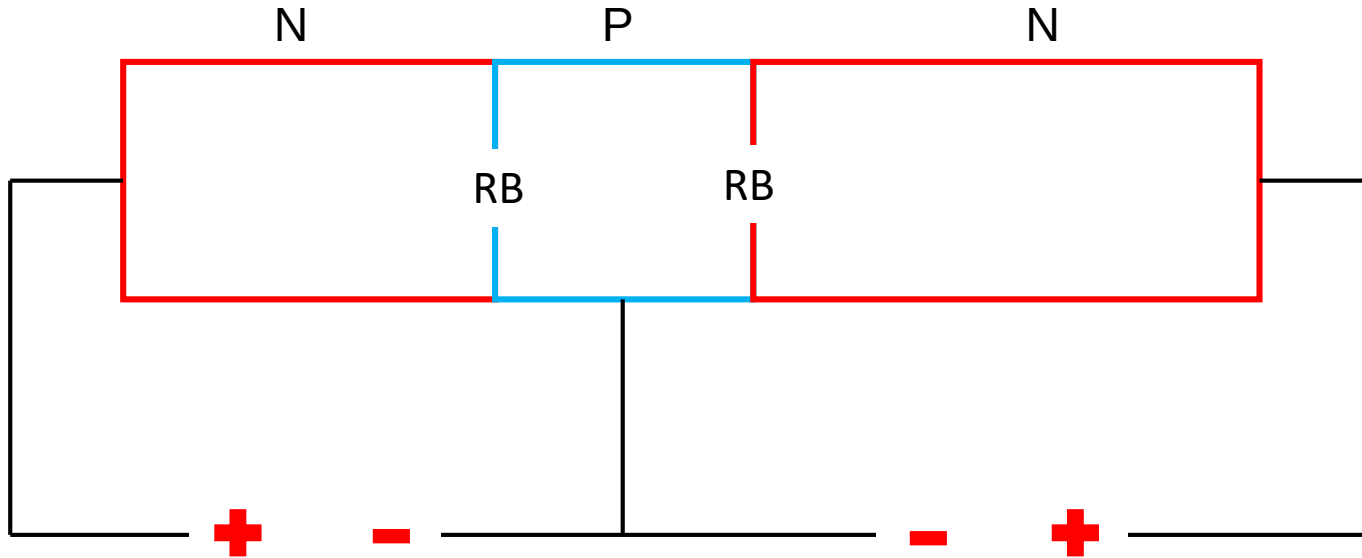
Losses in PE devices



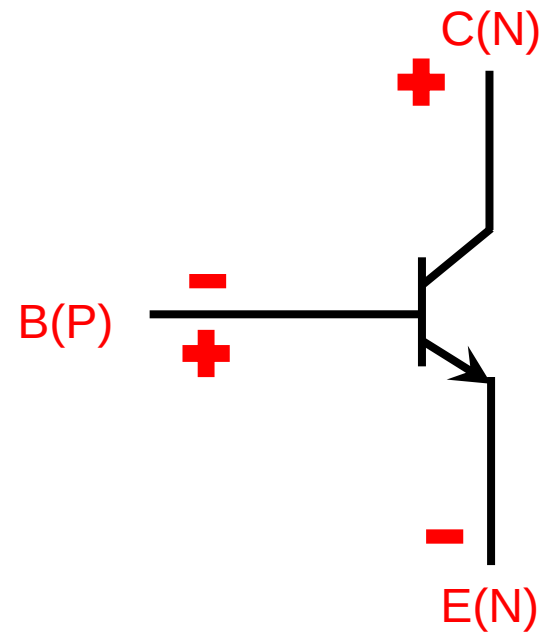
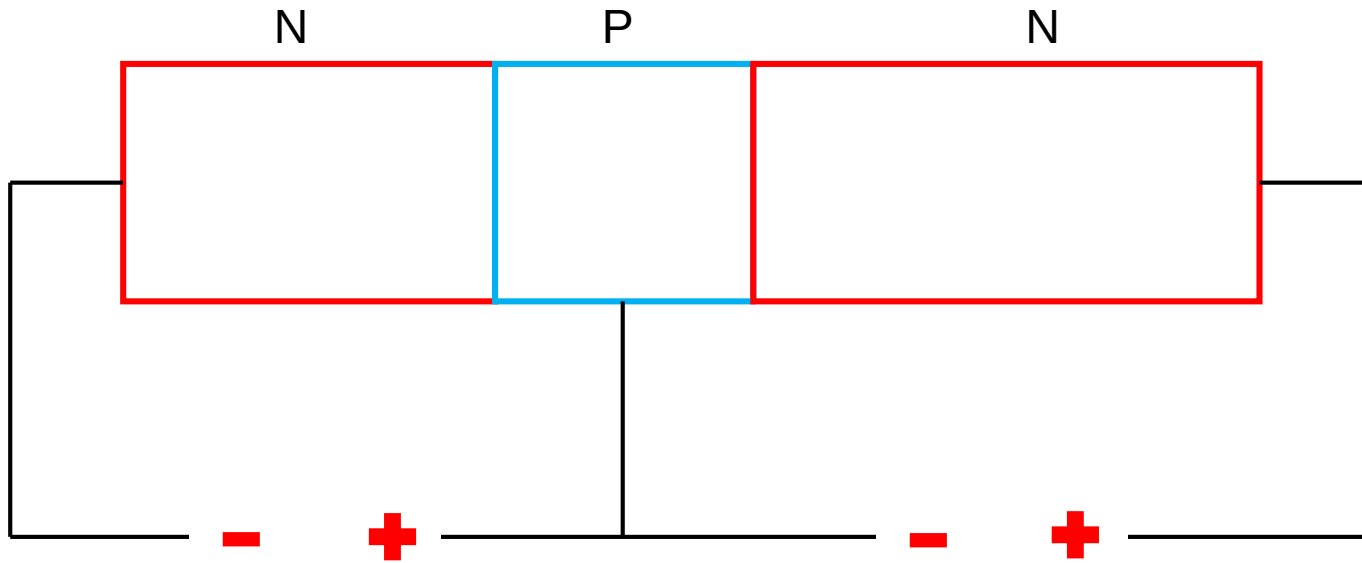
BJT



BJT

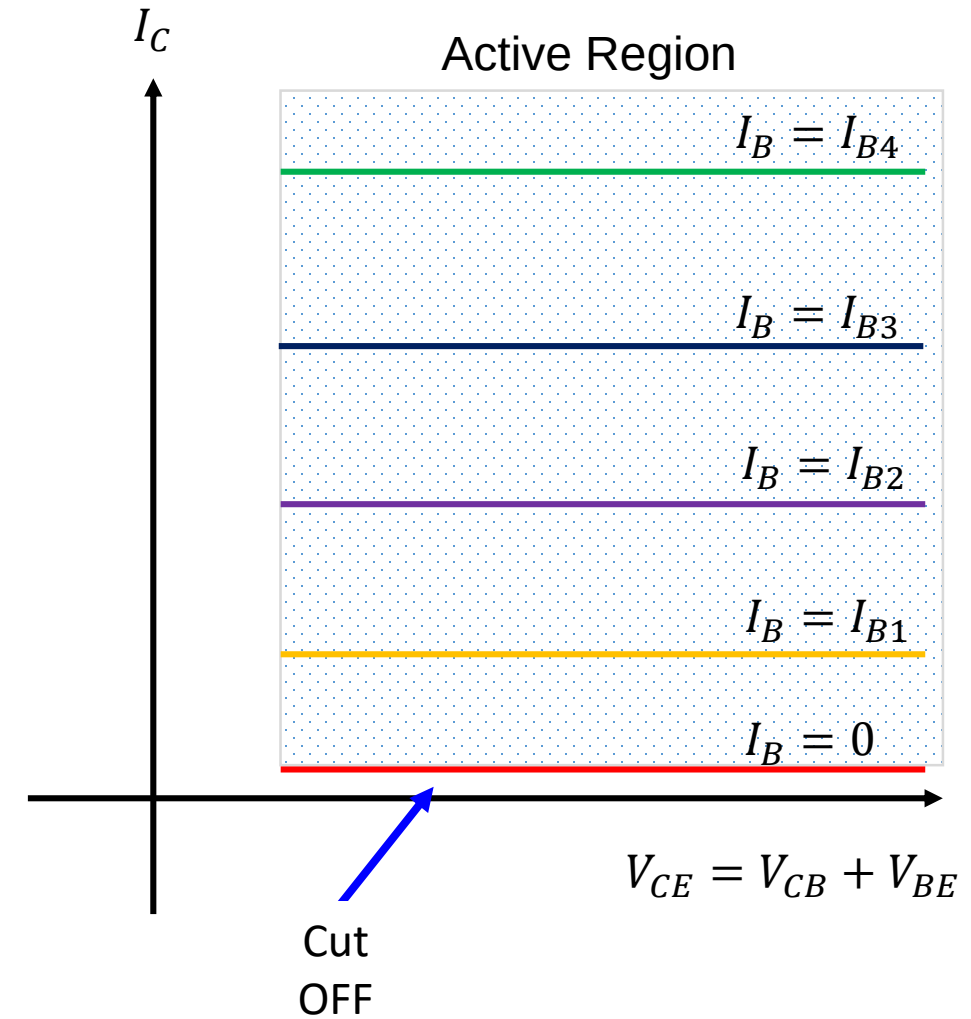
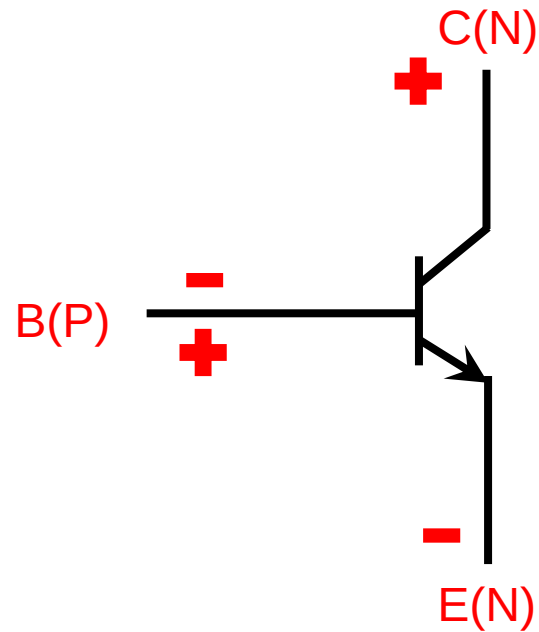
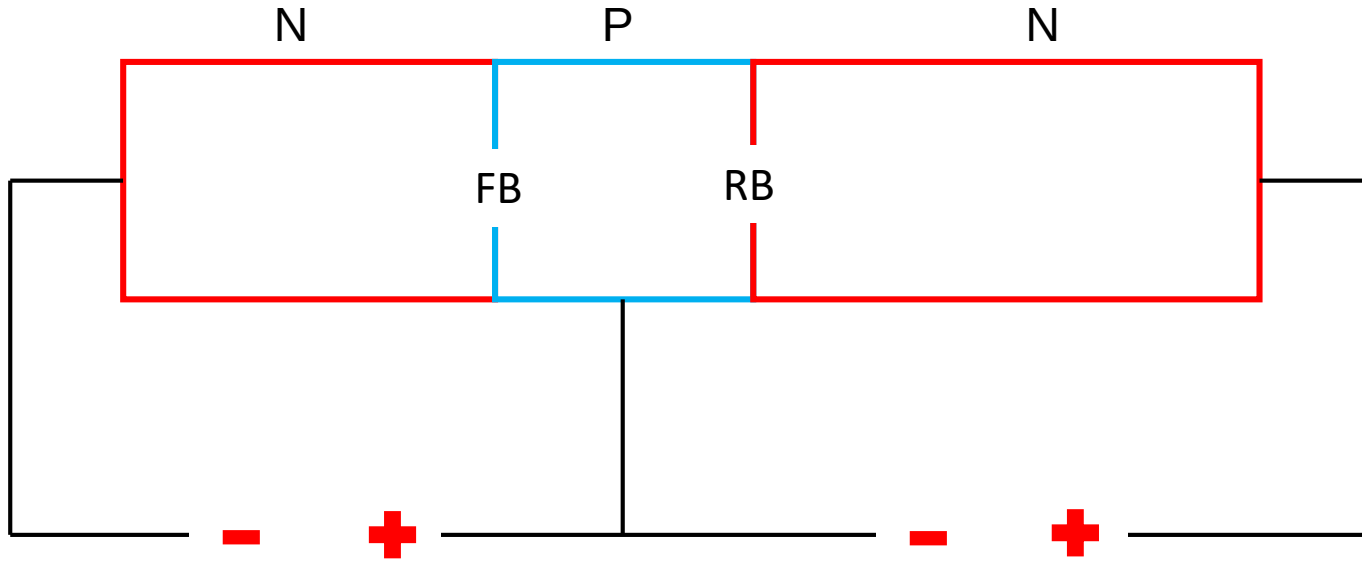


BJT

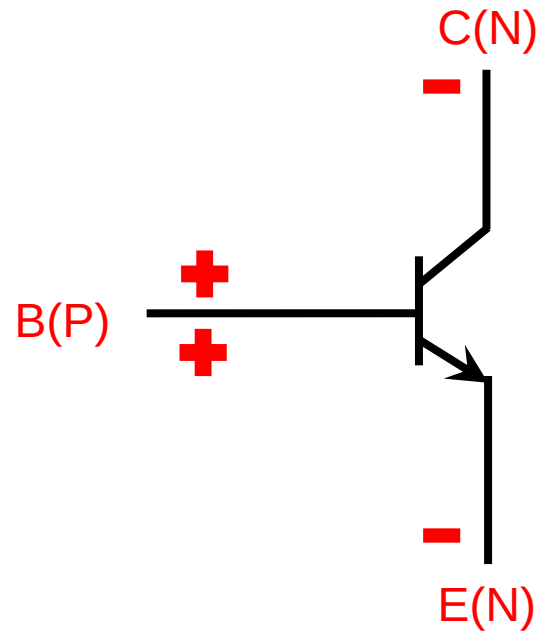
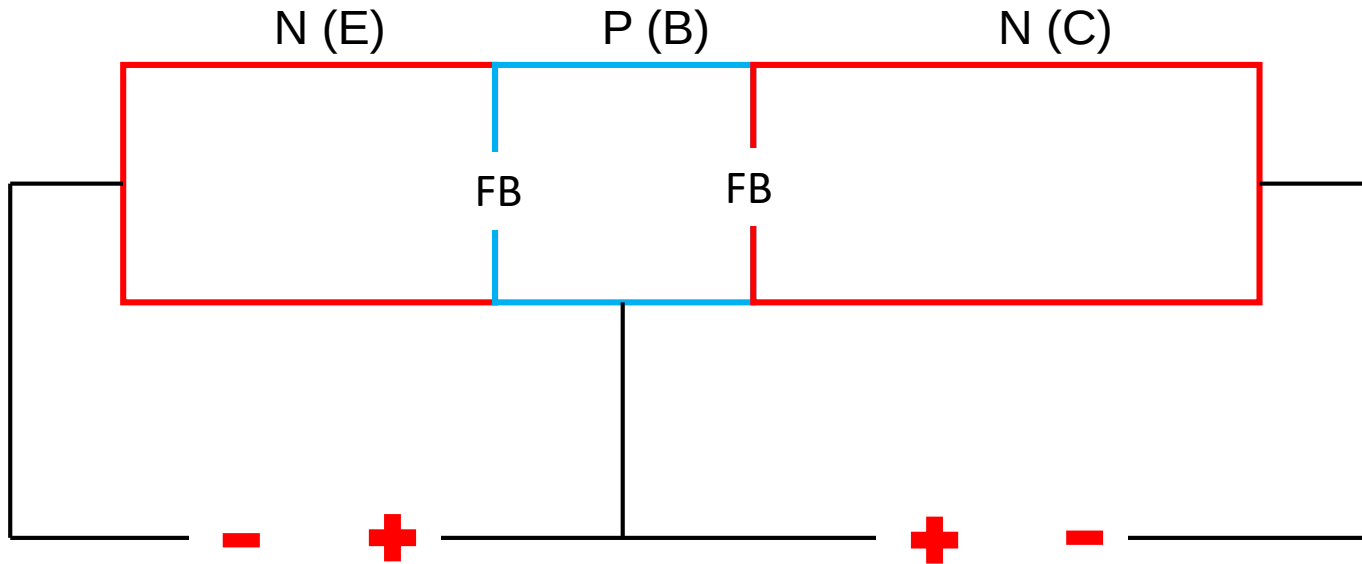


BJT

$$I_C = \beta I_B$$



BJT



$$V_{BE} \geq 0.7; EBJ \rightarrow FB$$

$$V_{BC} \leq 0.4, CBJ \rightarrow RB$$

$$V_{BC} \geq 0.4, CBJ \rightarrow FB$$

$$-V_{BC} \leq -0.4, CBJ \rightarrow FB$$

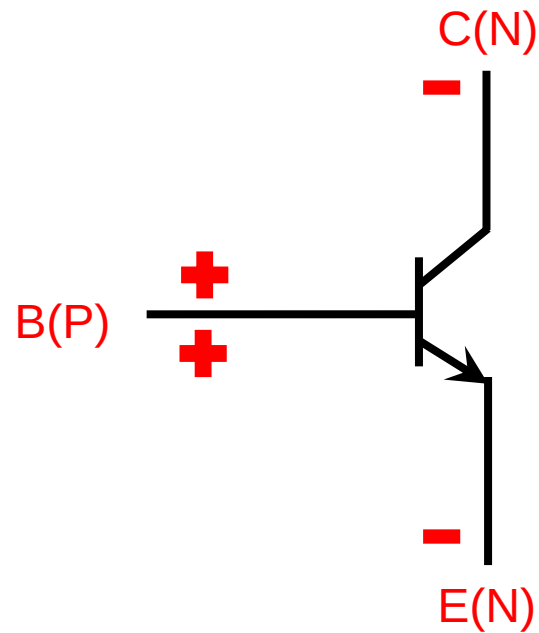
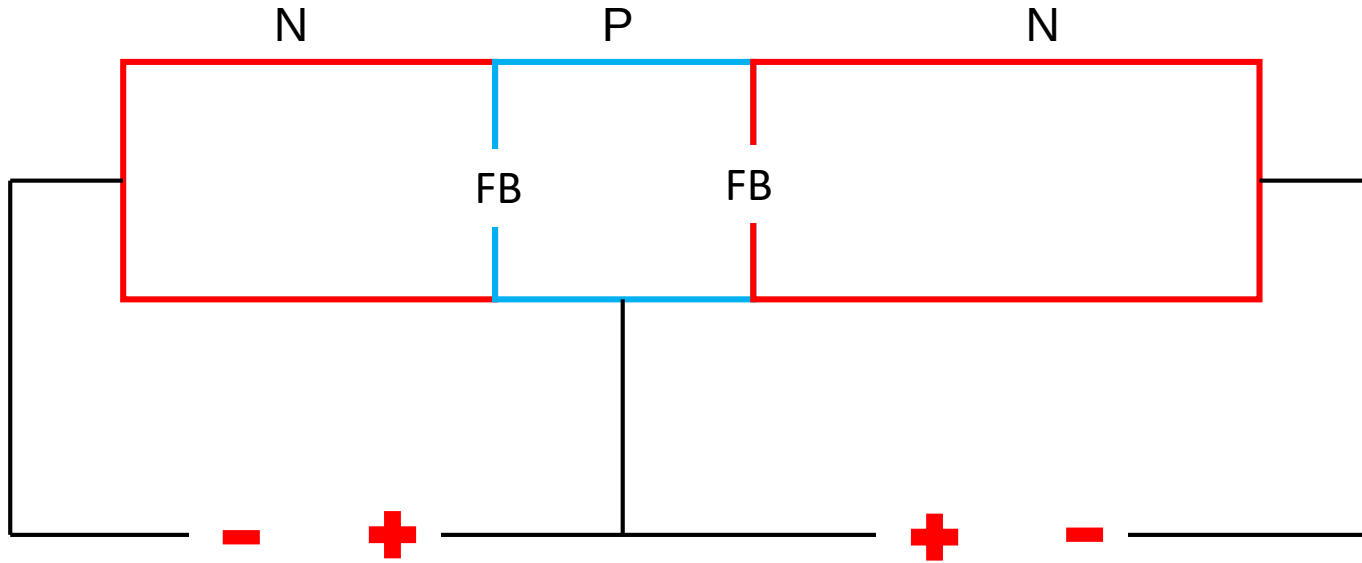
$$V_{CB} \leq -0.4, CBJ \rightarrow FB$$

$$V_{CE} = V_{CB} + V_{BE}$$

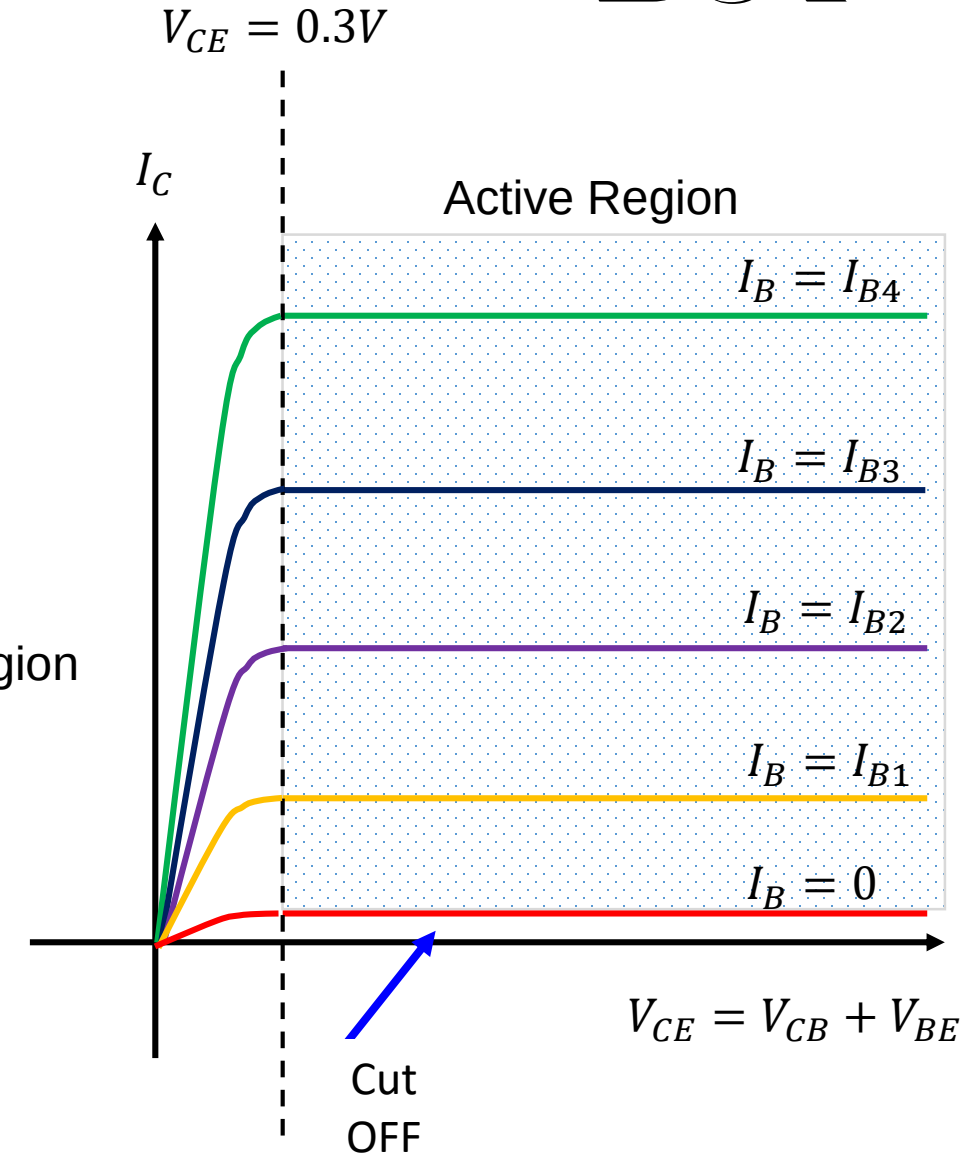
$$V_{CE,sat} \leq -0.4 + 0.7$$

$$V_{CE,sat} \leq 0.3$$

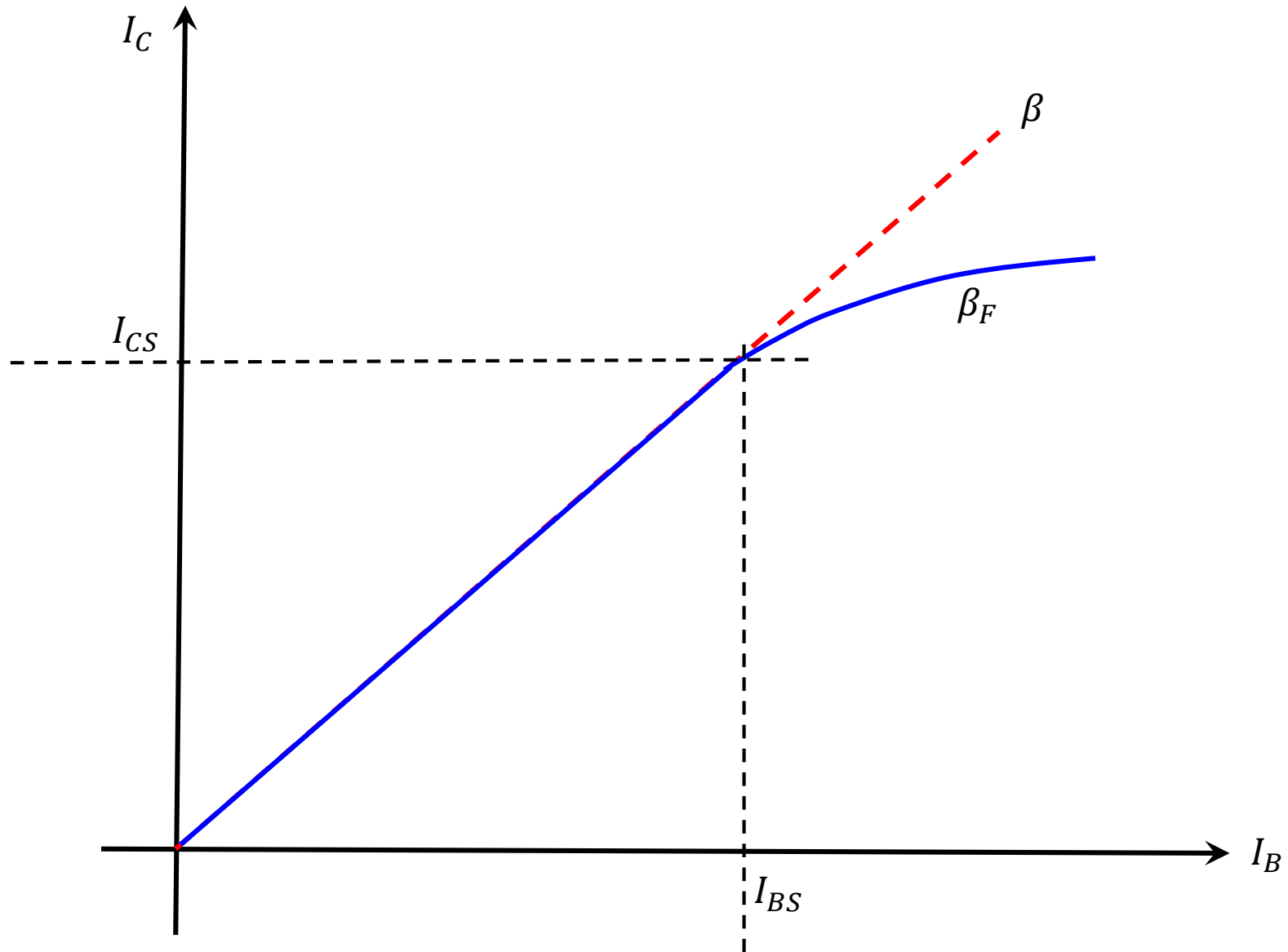
BJT



Saturation Region



BJT



BJT

$$-V_{BB} + I_B R_B + V_{BE} = 0$$

$$I_B = \frac{V_{BB} - V_{BE}}{R_B}$$

$$-V_{CC} + I_C R_C + V_{CE} = 0$$

$$I_C = \frac{V_{CC} - V_{CE}}{R_C}$$

$$I_{CS} = \frac{V_{CC} - V_{CES}}{R_C} \quad I_{BS} = \frac{I_{CS}}{\beta}$$

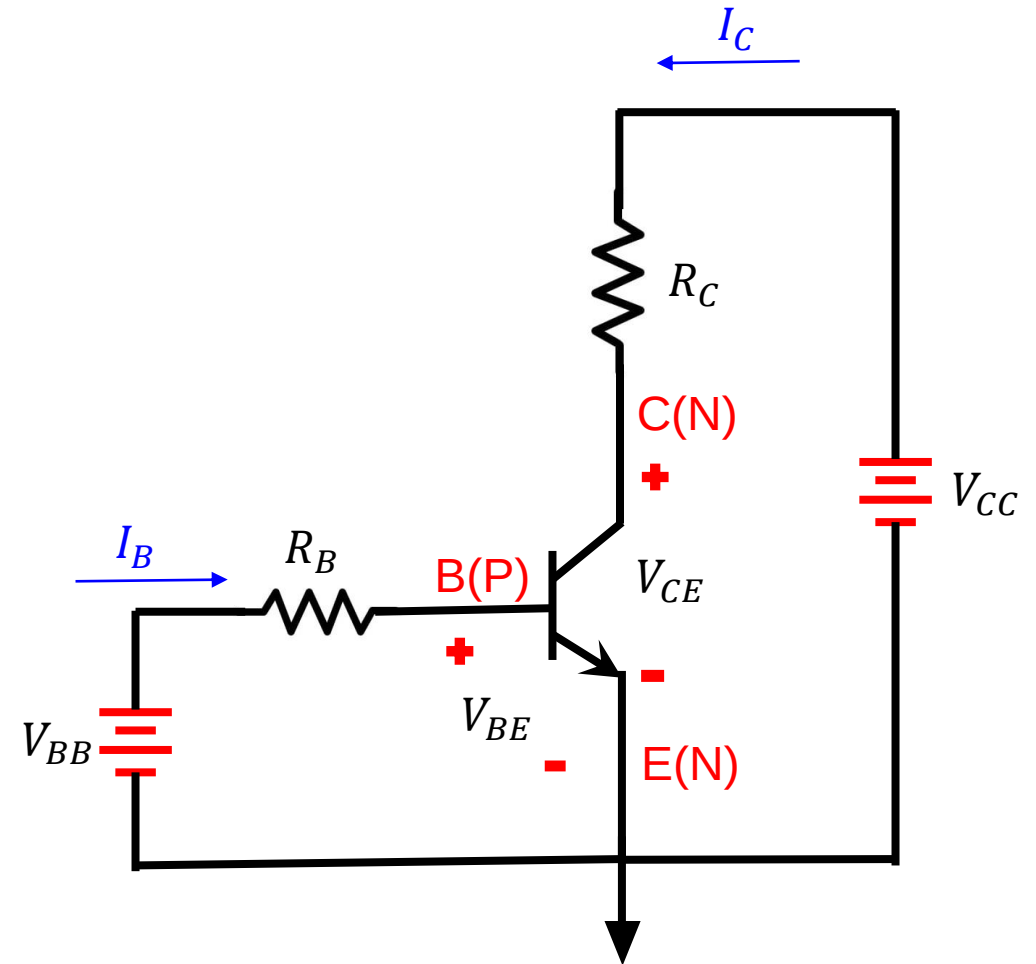
- The minimum base current required to drive the transistor into saturation is denoted as I_{BS}

- The ratio of I_B to I_{BS} is defined as overdrive factor (ODF).

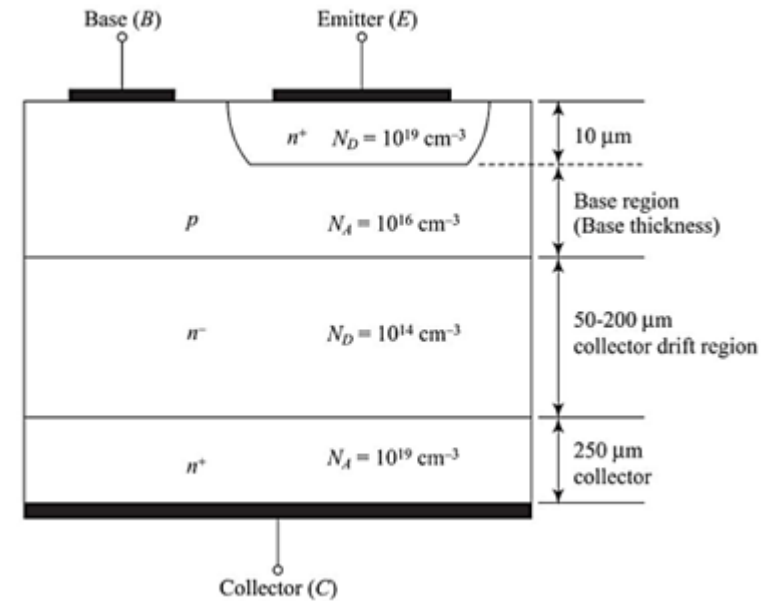
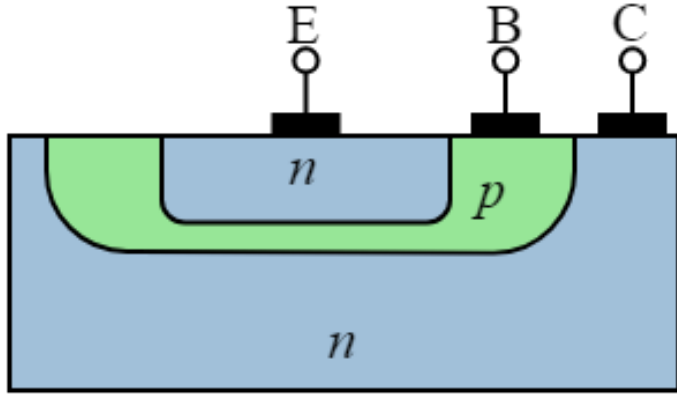
$$ODF = \frac{I_B}{I_{BS}}$$

- The ratio of I_{CS} to I_B is defined as forced current gain β_F

$$\beta_F = I_{CS}/I_B$$

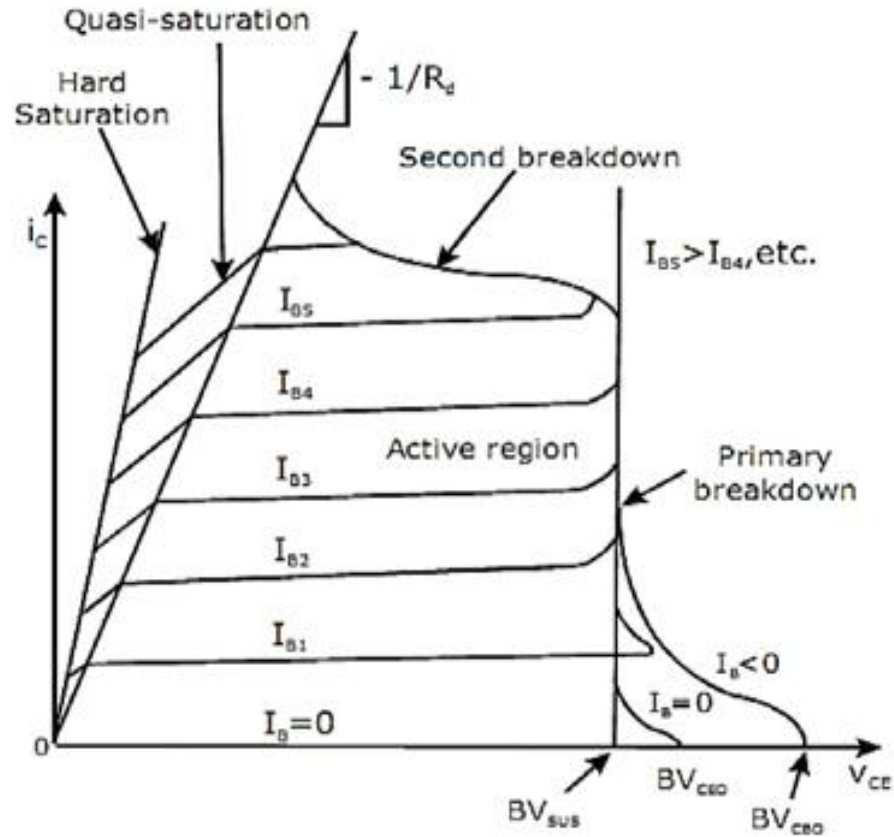


SIGNAL VS POWER BJT



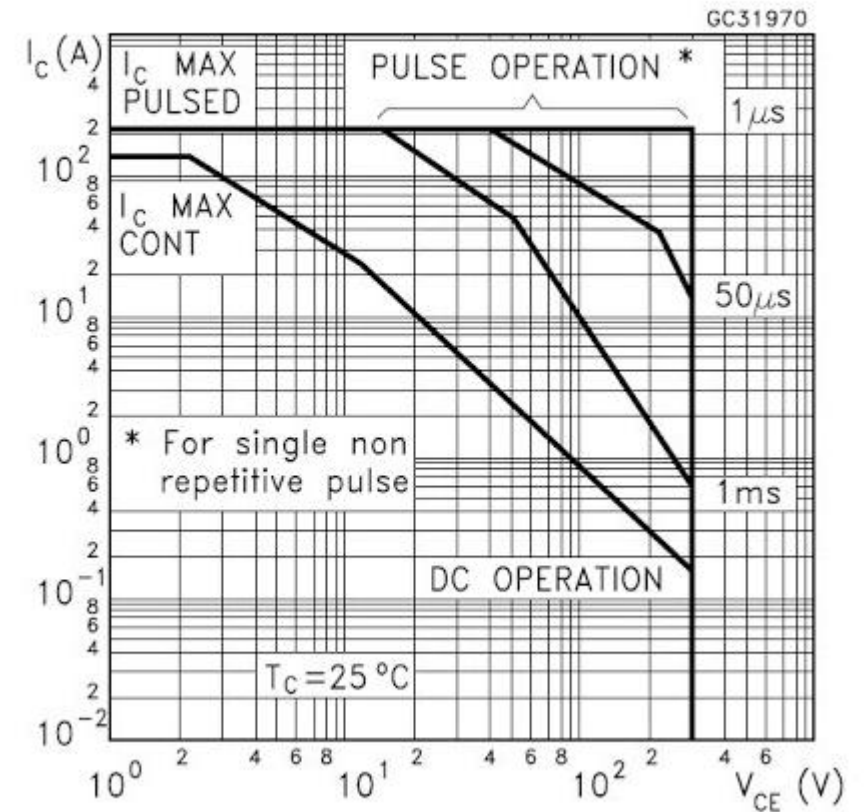
- A power transistor is a vertically oriented four layer structure of alternating p-type and n type. This is maximizing the cross-section area results in current rating of BJT, minimize the on-state resistance, and thus reduce the power losses.
- The doping of emitter layer and collector layer is quite large typically 10¹⁹ /cm³
- A special layer called the collector drift region (n⁻) has a light doping level of 10¹⁴/cm³ .
- The thickness of the drift region determines the breakdown voltage of the transistor

POWER BJT STEADY STATE CHARACTERISTICS



The power transistor has steady state characteristics almost similar to signal level transistors except that the V-I characteristics has a region of quasi saturation

Safe Operating Area



The **Safe Operating Area (SOA)** of a transistor defines the voltage and current conditions under which the device can operate reliably without failure

POWER BJT SOA

It is usually represented as a plot with: Collector-emitter voltage (V_{CE}) on the x-axis and Collector current (I_C) on the y-axis

SOA considers multiple factors, including:

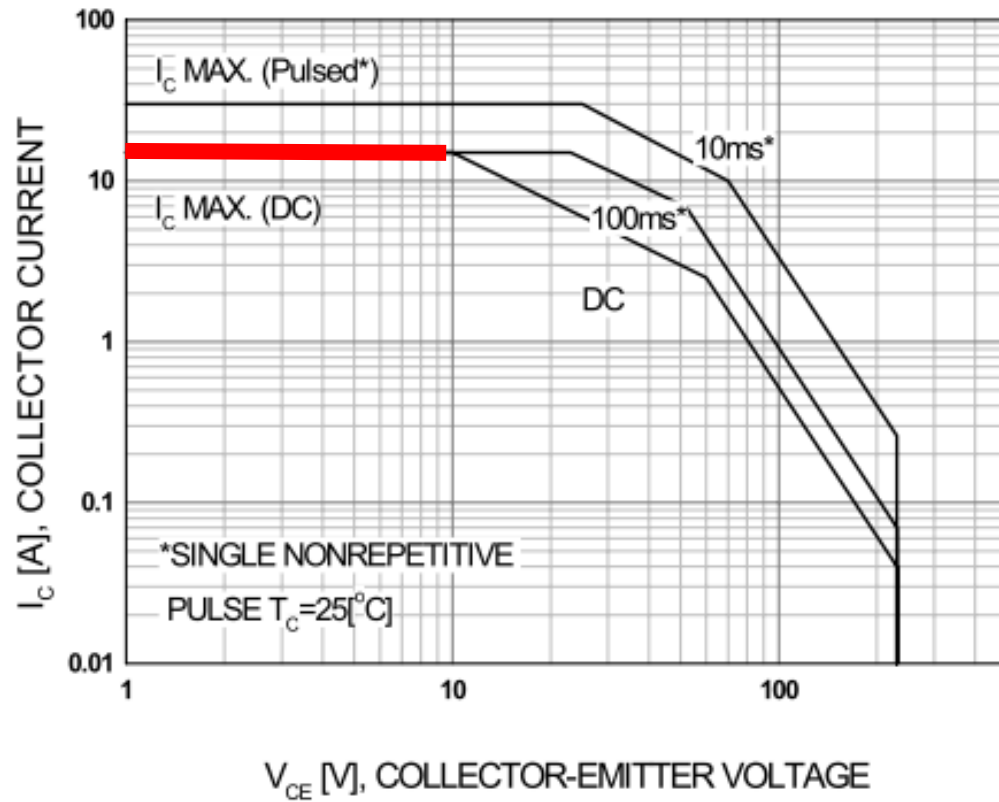
- Maximum current handling capability – Avoids excessive current that can damage the transistor.
- Maximum voltage withstand capability – Ensures the transistor does not experience breakdown.
- Power dissipation limit – Prevents overheating.
- Second breakdown limit – Protects against failure due to localized heating effects.
- Thermal constraints – Ensures junction temperature remains within safe limits.

For a Bipolar Junction Transistor (BJT), there are two key safe operating areas:

- ❖ Forward Bias Safe Operating Area (FBSOA)
- ❖ Reverse Bias Safe Operating Area (RBSOA)

POWER BJT FBSOA

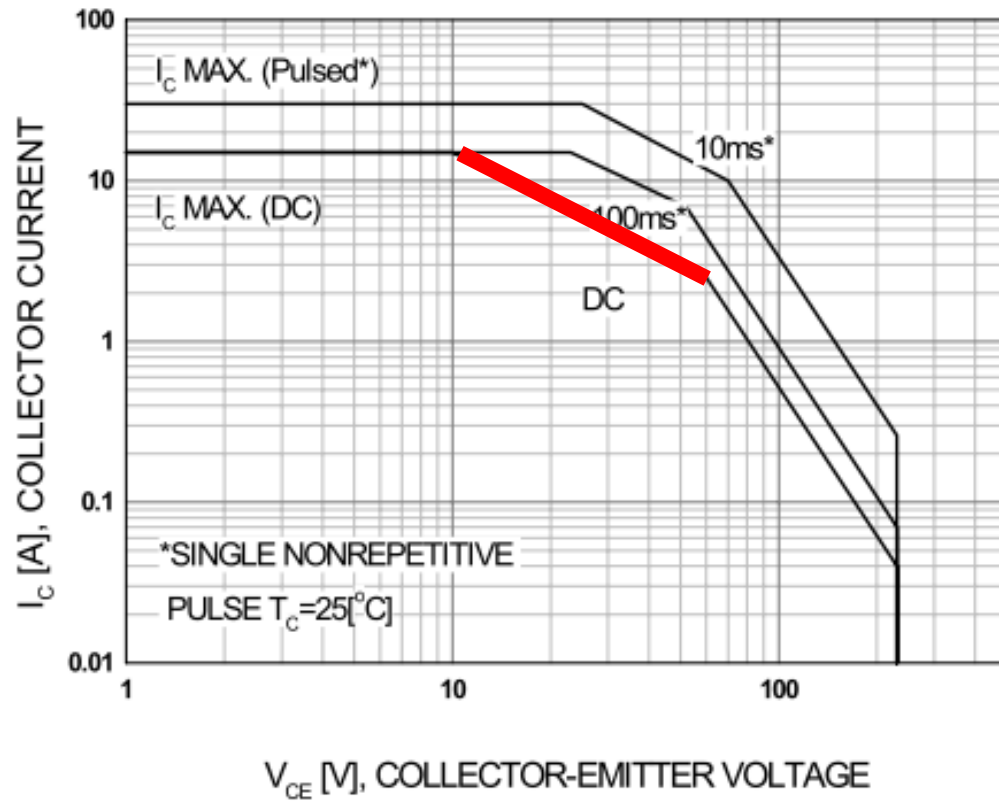
It is usually represented as a plot with: Collector-emitter voltage (V_{CE}) on the x-axis and Collector current (I_C) on the y-axis in **log scale**.



Maximum Current Limit

POWER BJT ~~FBSOA~~

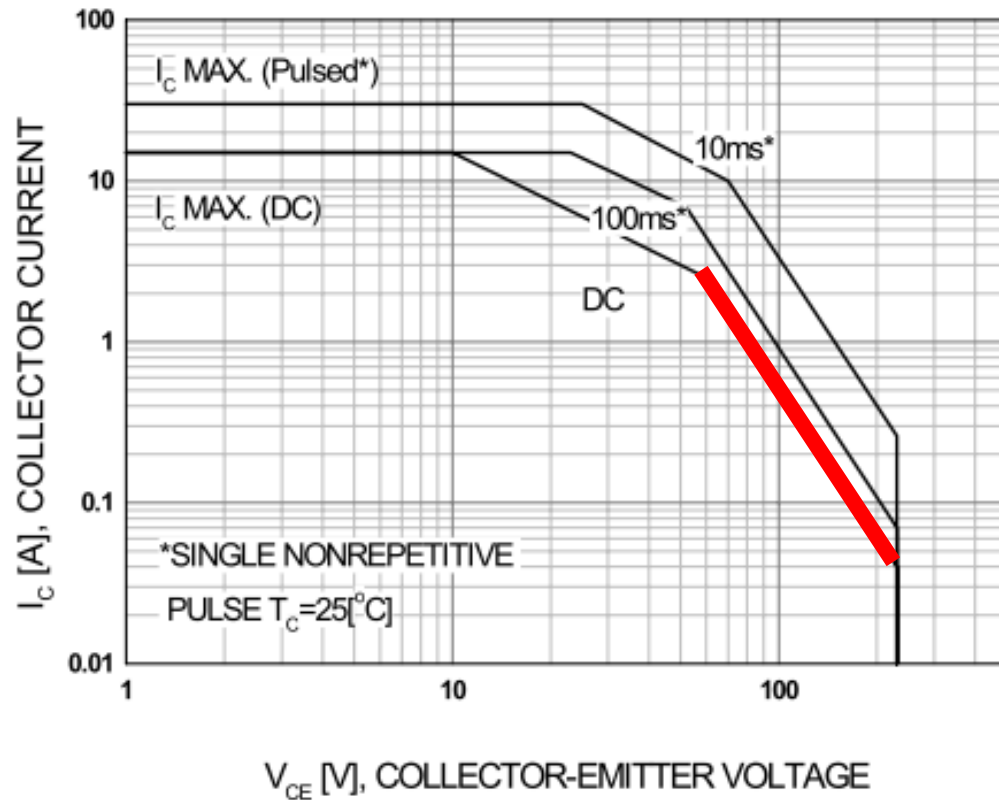
It is usually represented as a plot with: Collector-emitter voltage (V_{CE}) on the x-axis and Collector current (I_C) on the y-axis in **log scale**.



Maximum Power Limit

POWER BJT FBSOA

It is usually represented as a plot with: Collector-emitter voltage (V_{CE}) on the x-axis and Collector current (I_C) on the y-axis in **log scale**.



Secondary Breakdown Limit

- Secondary breakdown is a destructive failure mode in power BJTs, occurring due to localized overheating when the device operates in the active or saturation region with high voltage and current simultaneously.
- It is caused by non-uniform current distribution, leading to excessive power dissipation in small regions, resulting in thermal runaway and permanent damage.

Impact on Device Performance:

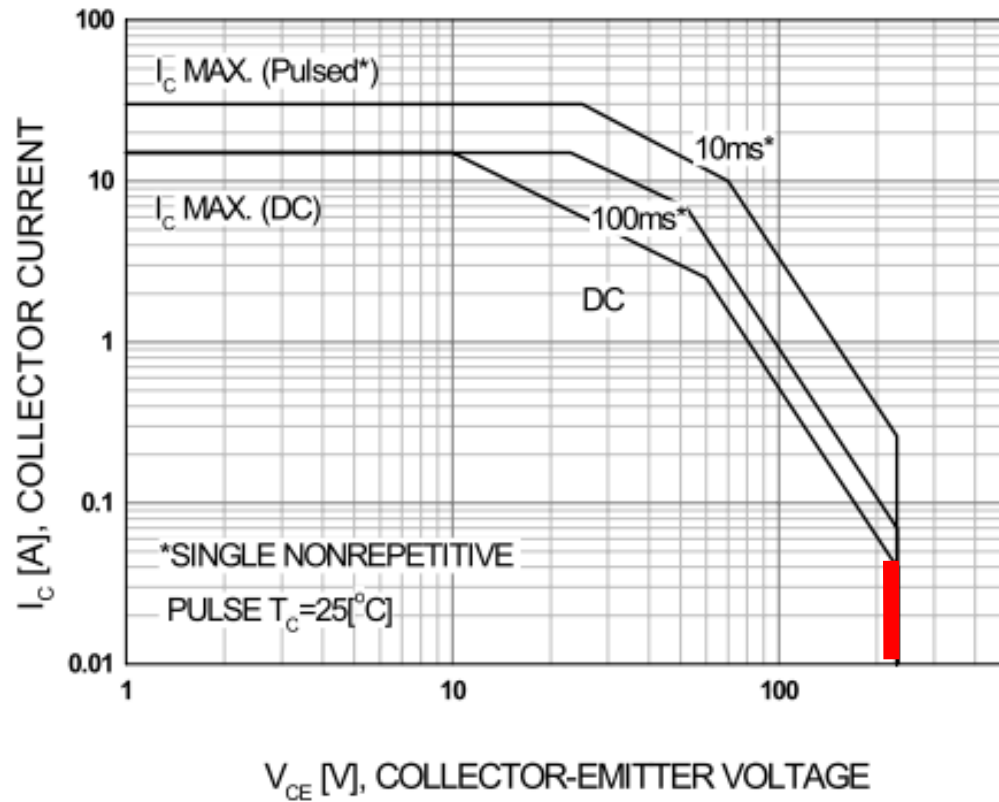
- Reduces the safe operating area (SOA) of the BJT.
- Limits the maximum voltage and current handling capability.
- Increases the risk of irreversible damage if proper heat dissipation is not ensured.

Mitigation: Proper heat sinking, derating, and using current-limiting circuits can help prevent secondary breakdown.

Power MOSFET doesn't exhibit secondary breakdown

POWER BJT FBSOA

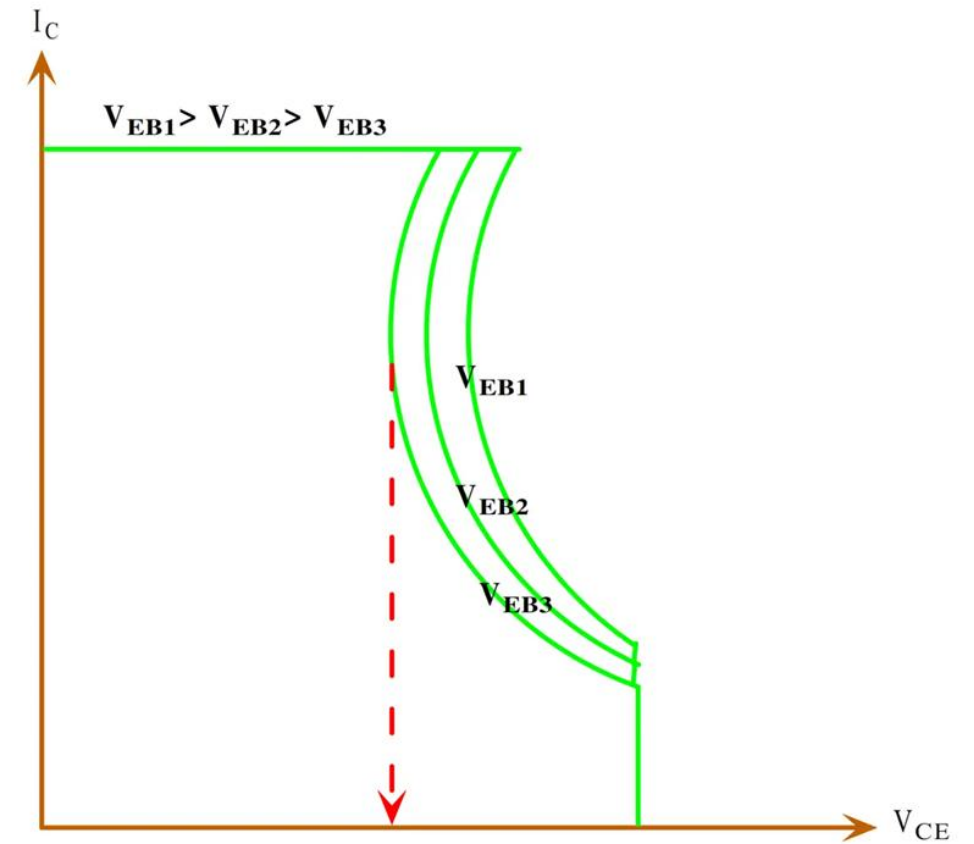
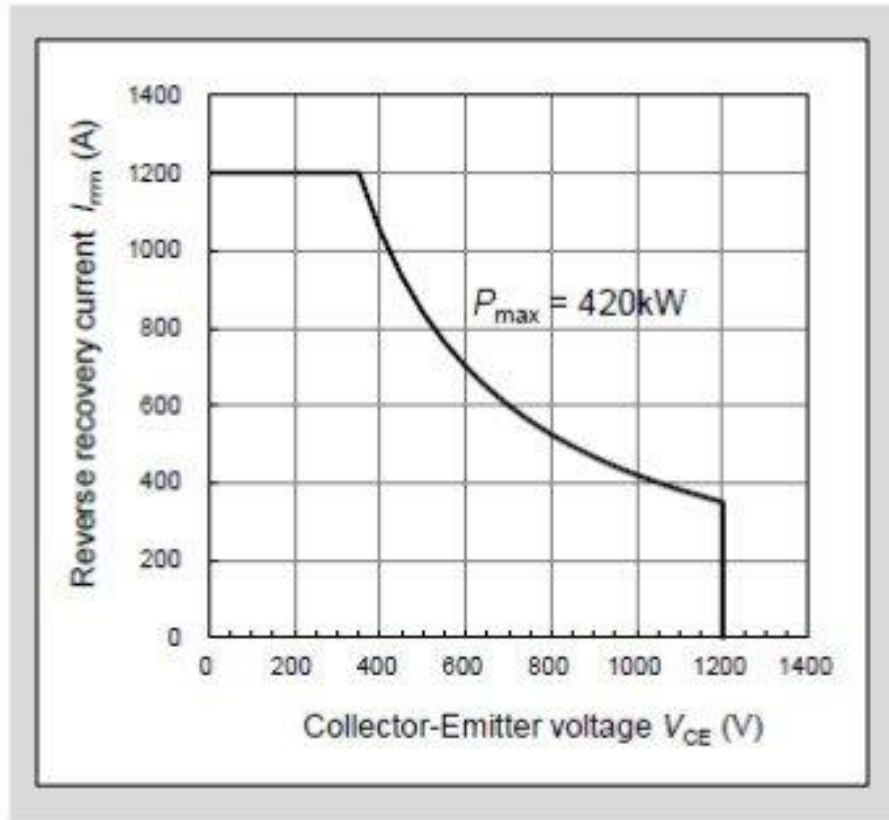
It is usually represented as a plot with: Collector-emitter voltage (V_{CE}) on the x-axis and Collector current (I_C) on the y-axis in **log scale**.



Maximum voltage Limit

POWER BJT RBSOA

It is usually represented as a plot with: Collector-emitter voltage (V_{CE}) on the x-axis and Reverse recovery current (I_C) on y axis.



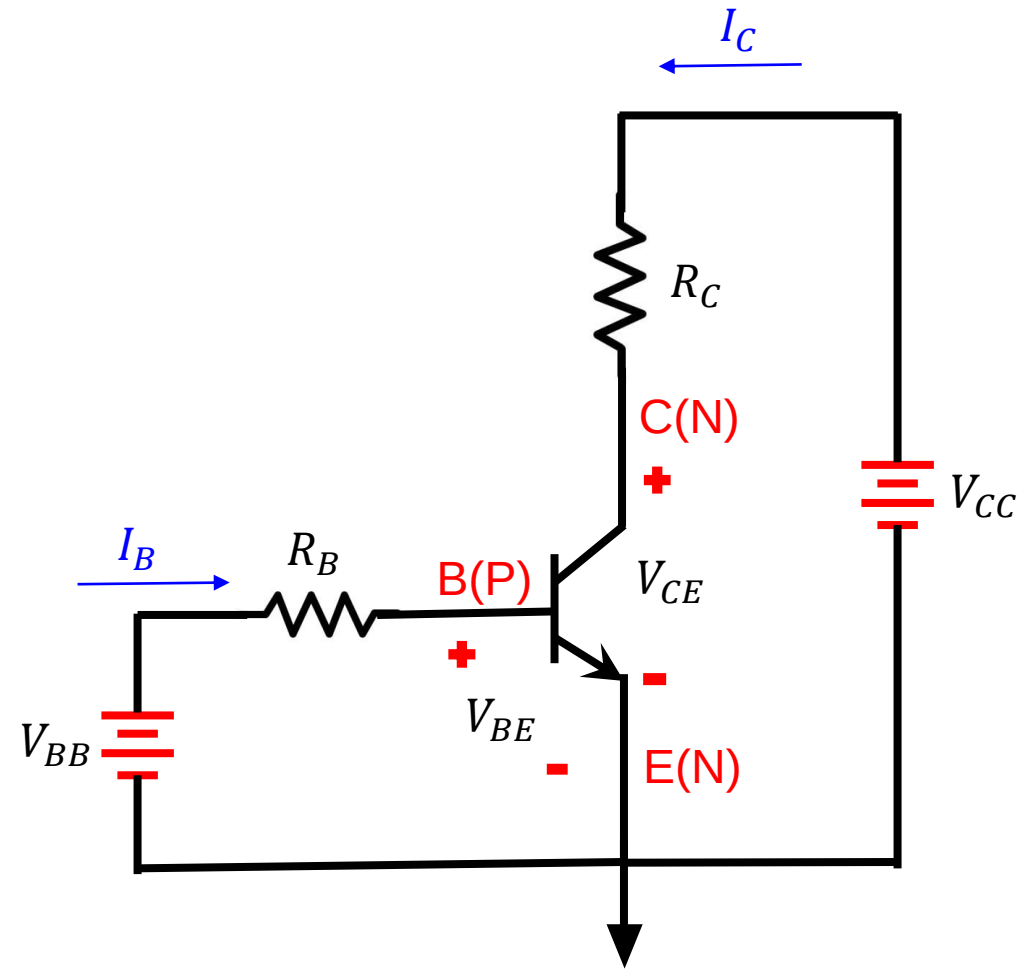
$$-V_{BB} + I_B R_B + V_{BE} = 0$$

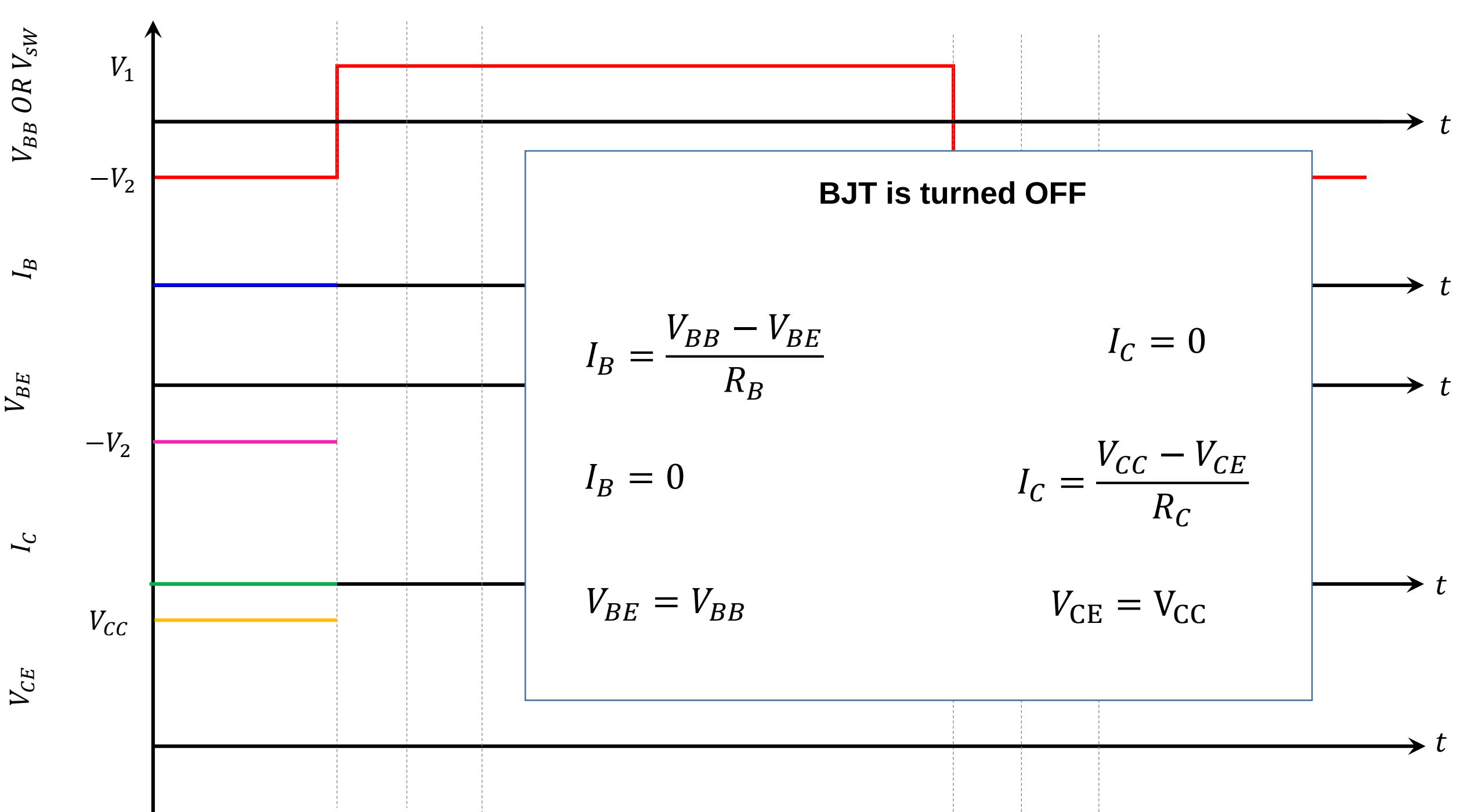
$$I_B = \frac{V_{BB} - V_{BE}}{R_B}$$

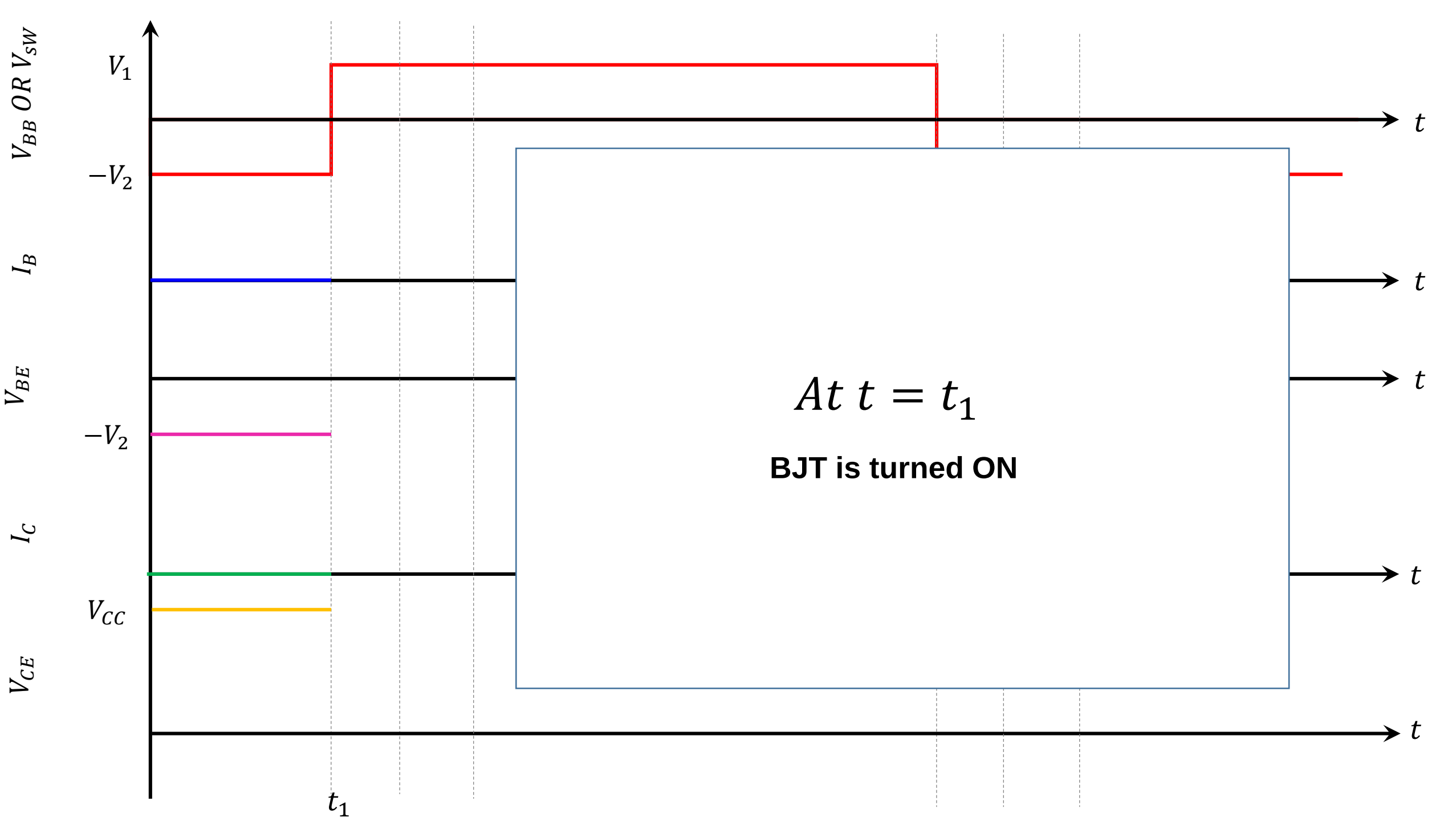
$$-V_{CC} + I_C R_C + V_{CE} = 0$$

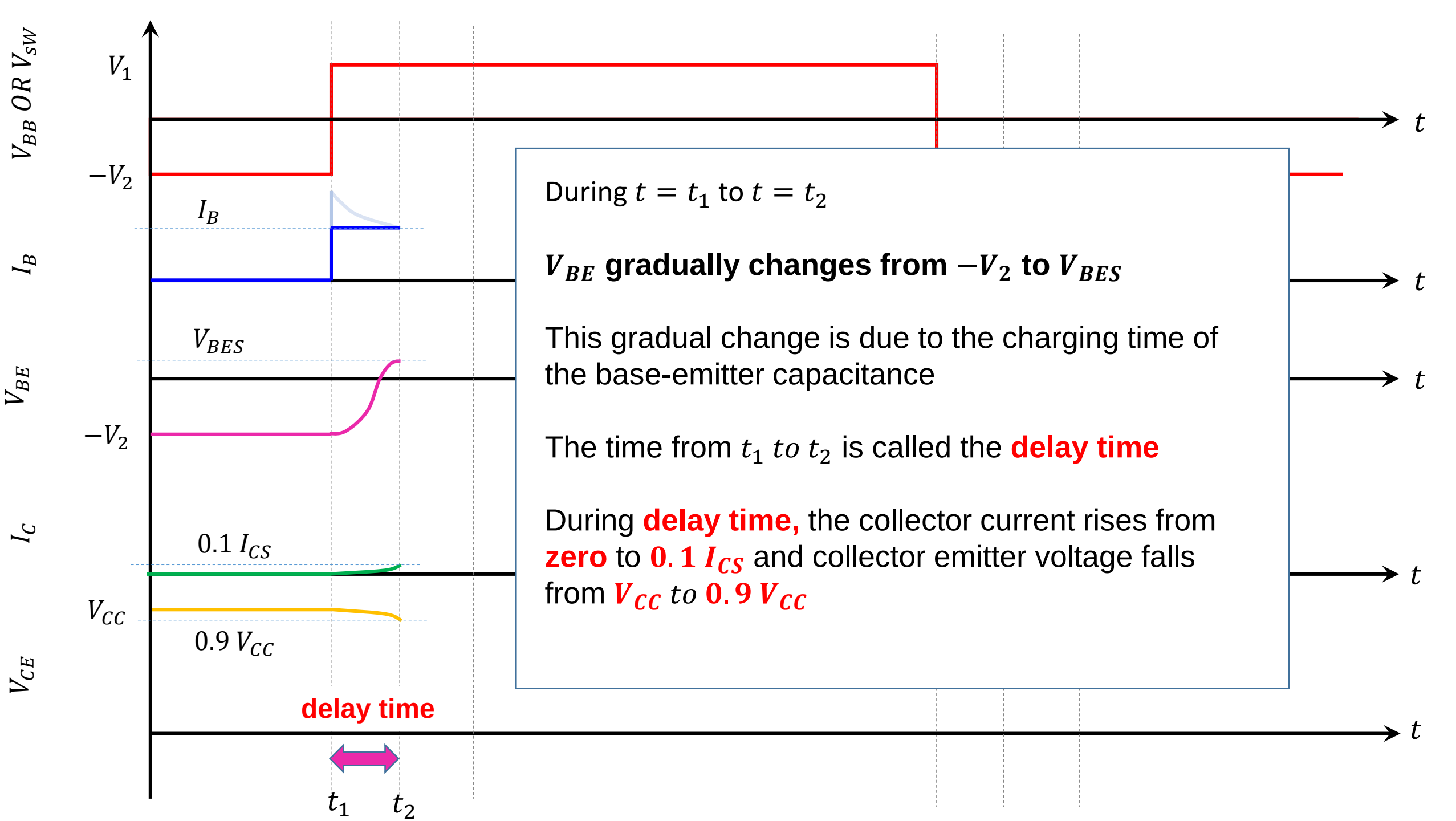
$$I_C = \frac{V_{CC} - V_{CE}}{R_C}$$

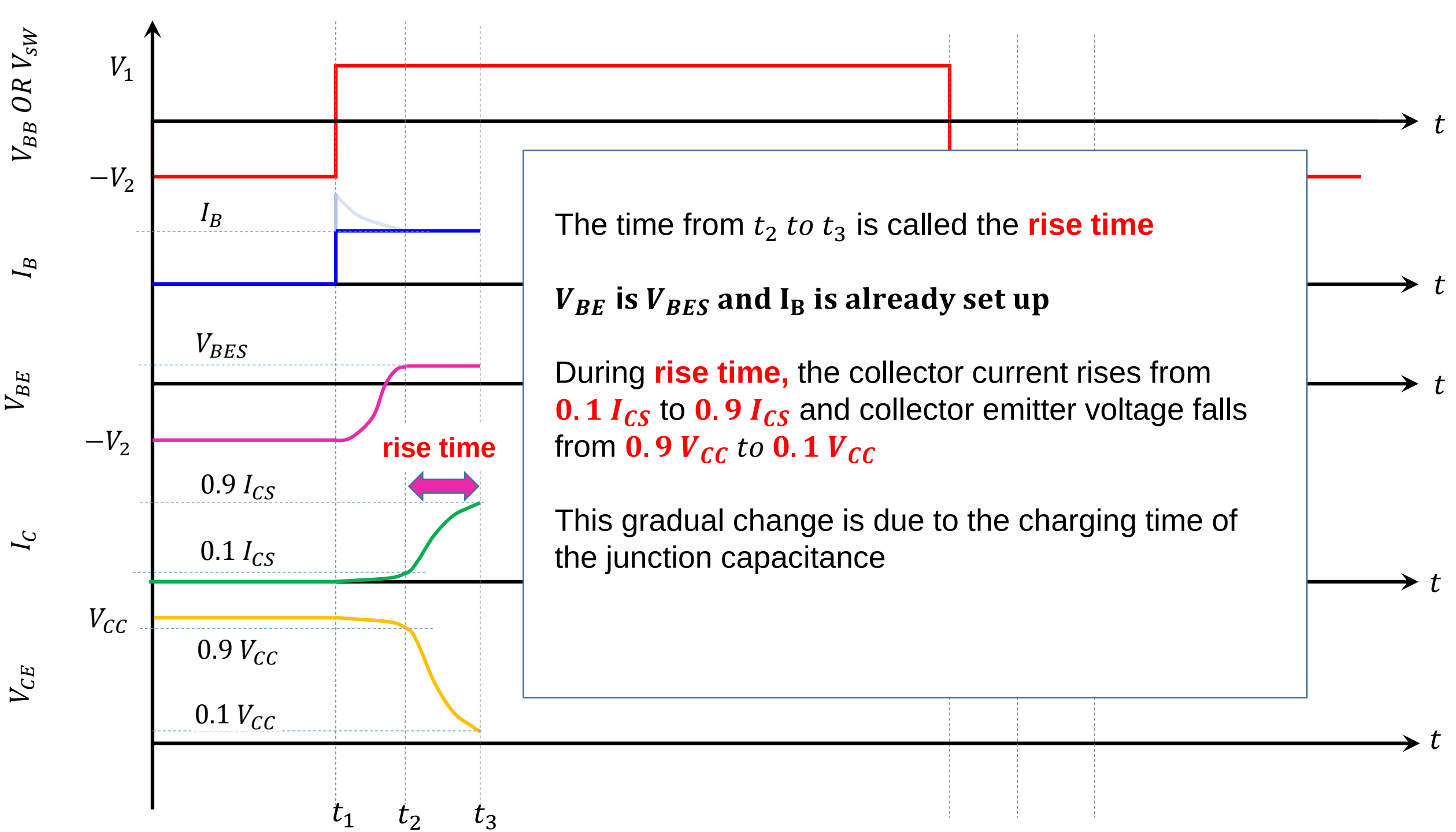
$$I_{CS} = \frac{V_{CC} - V_{CES}}{R_C} \quad I_{BS} = \frac{I_{CS}}{\beta}$$

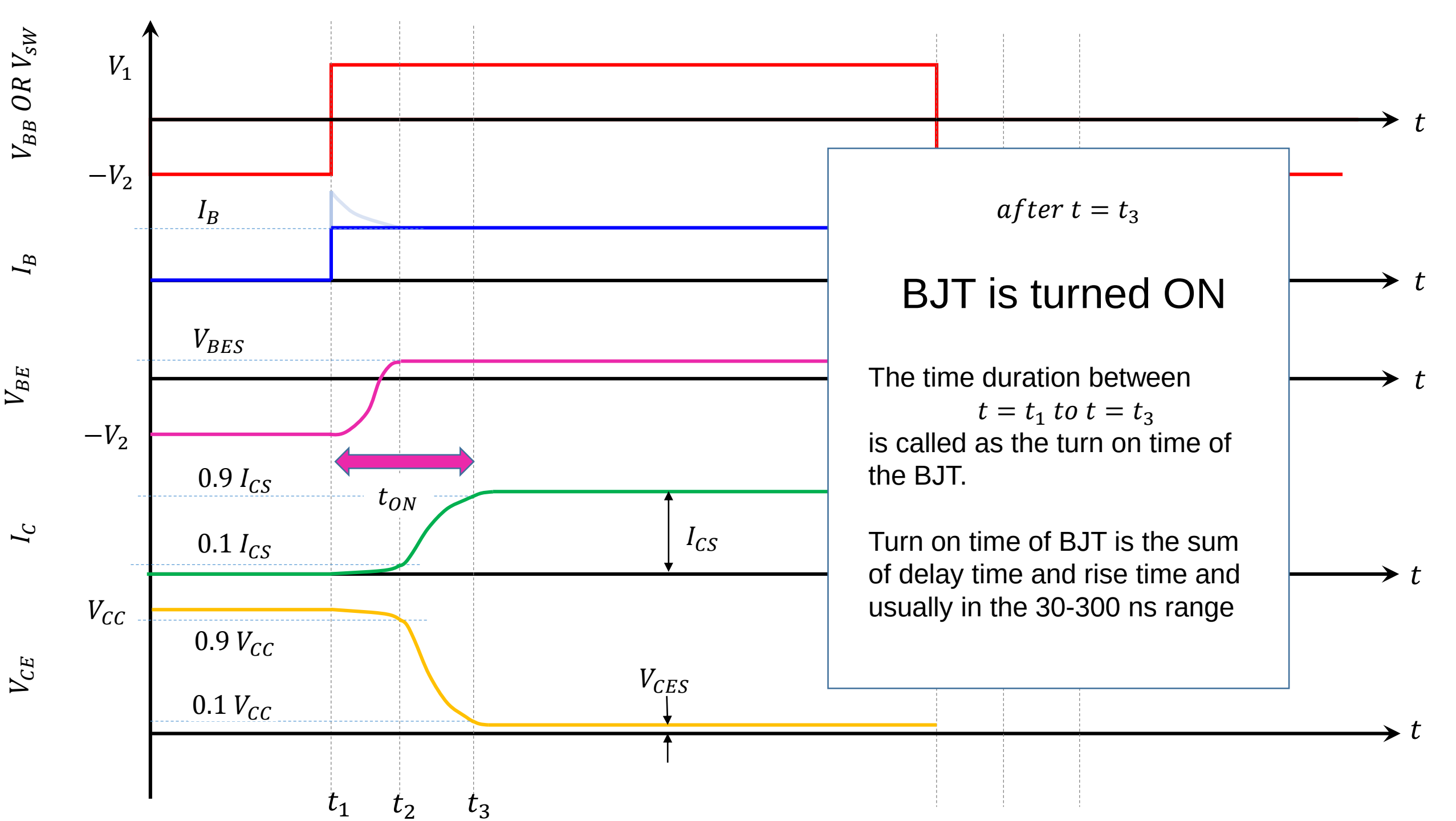


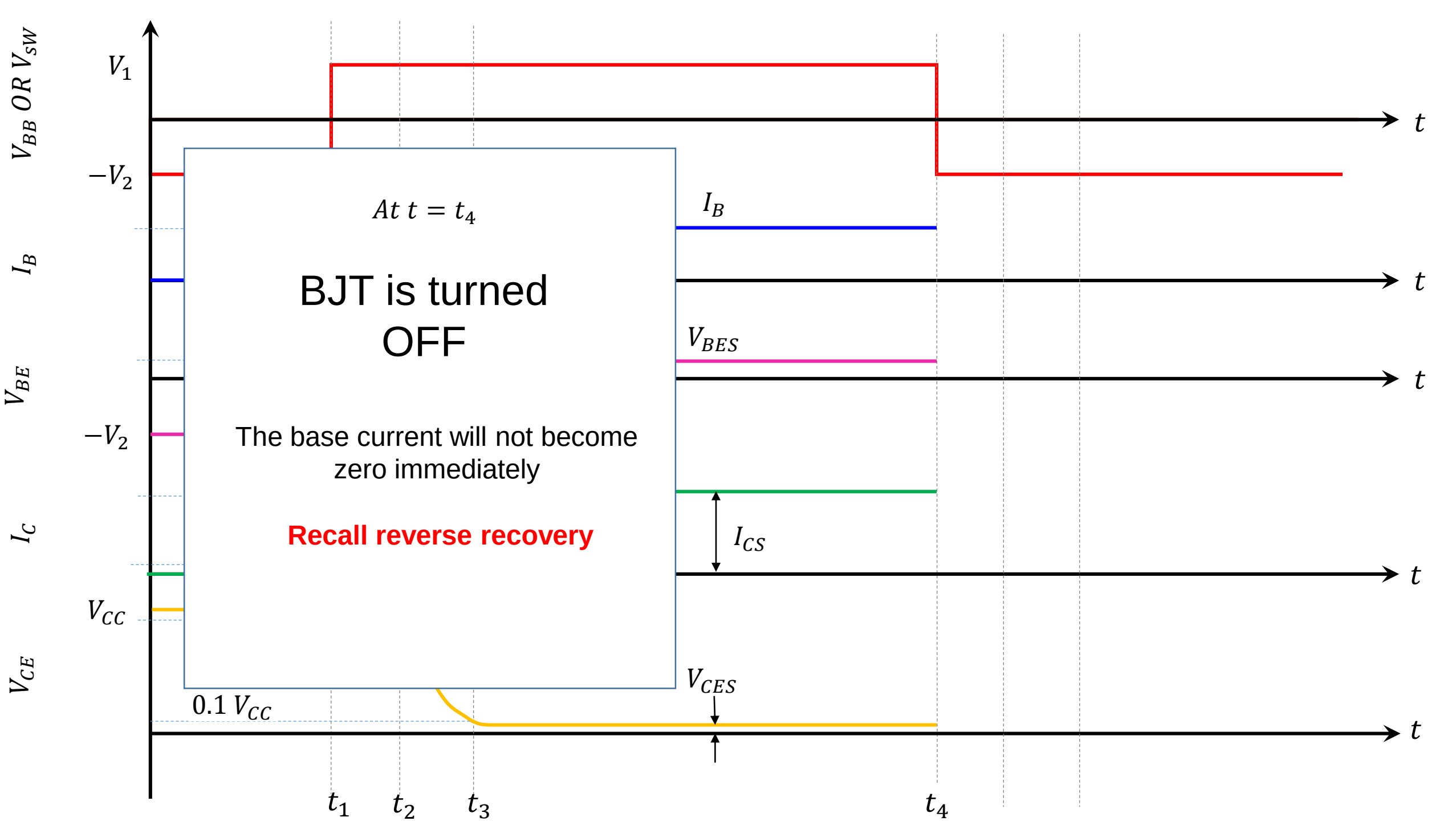


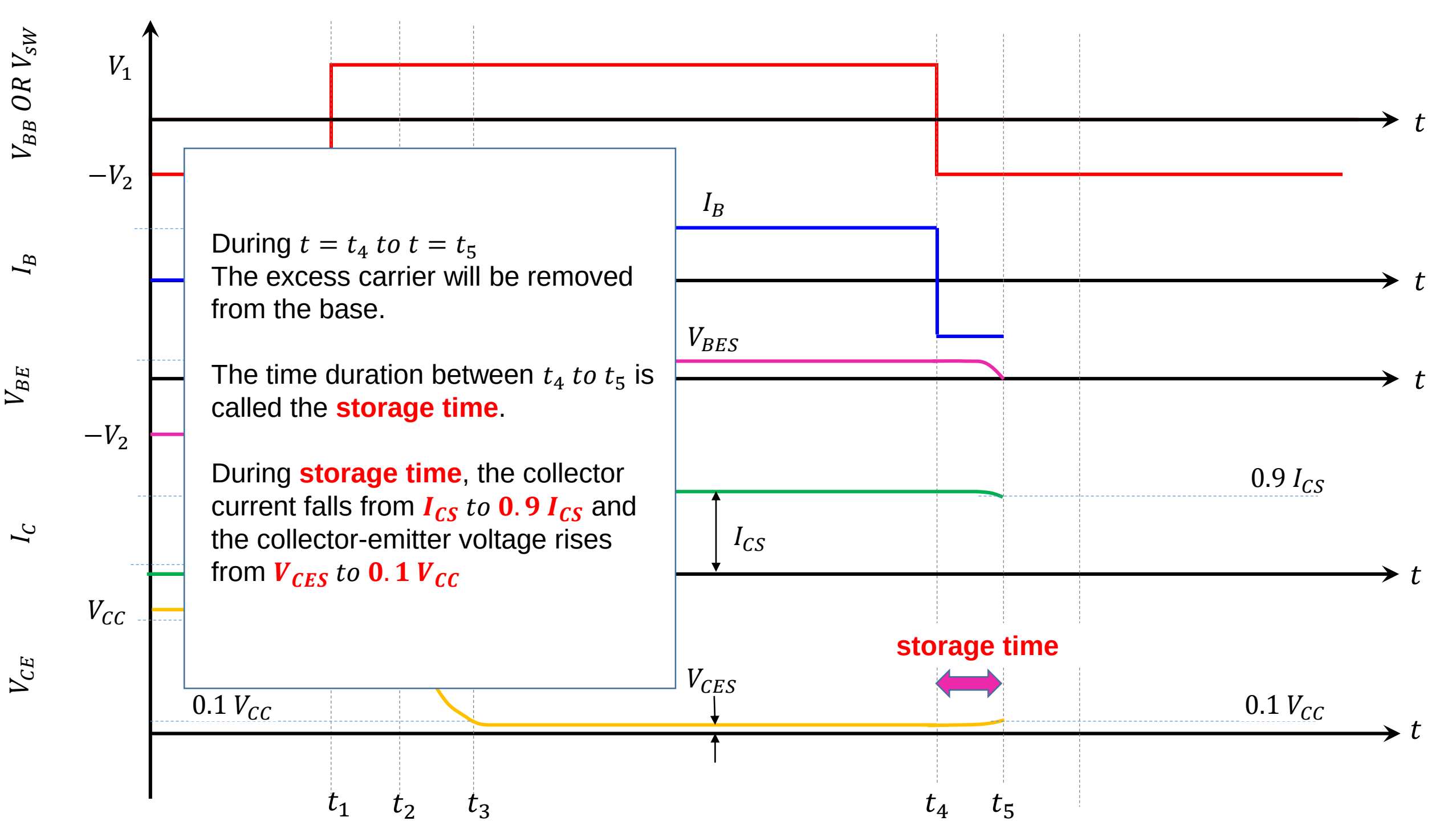


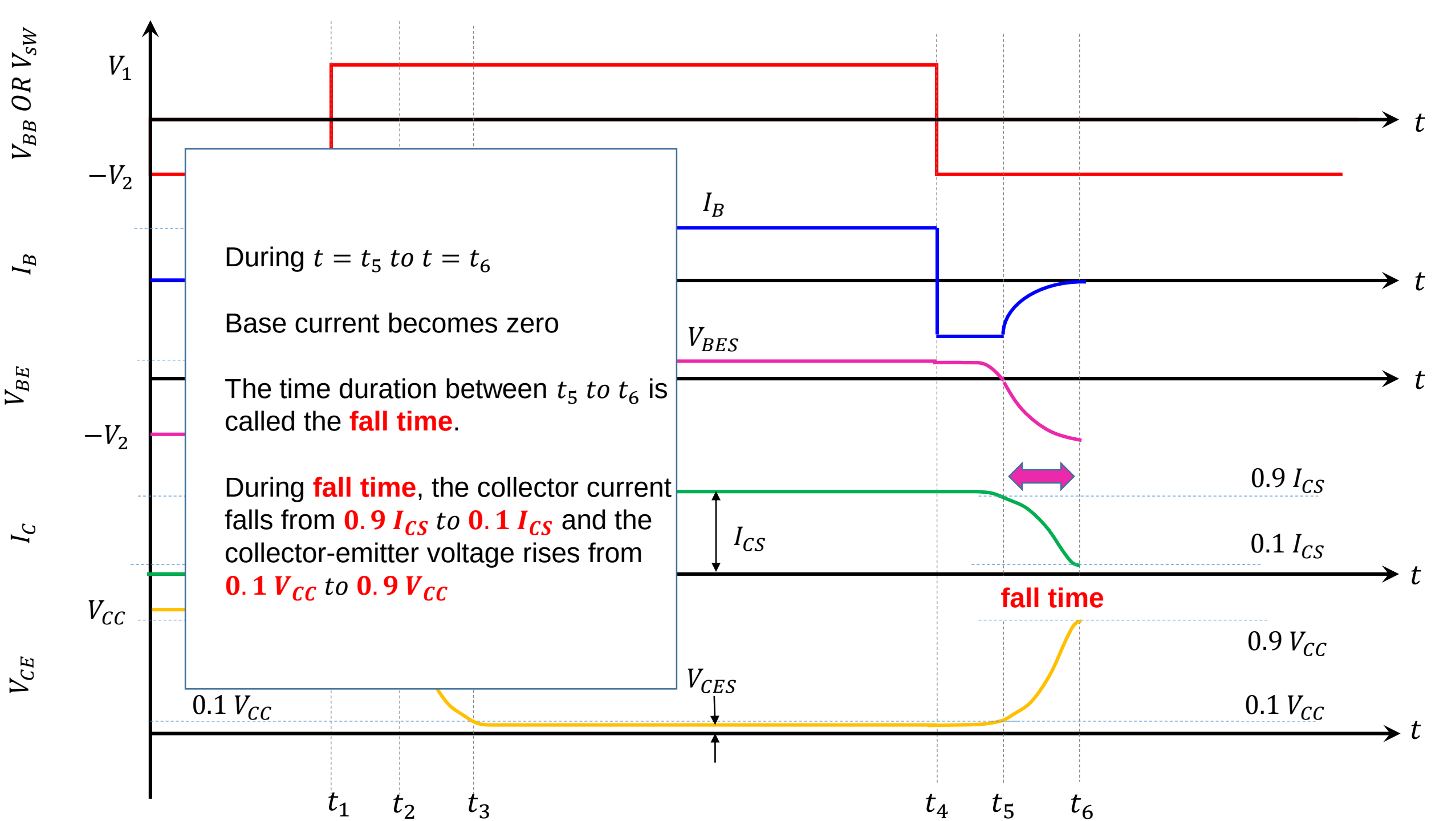


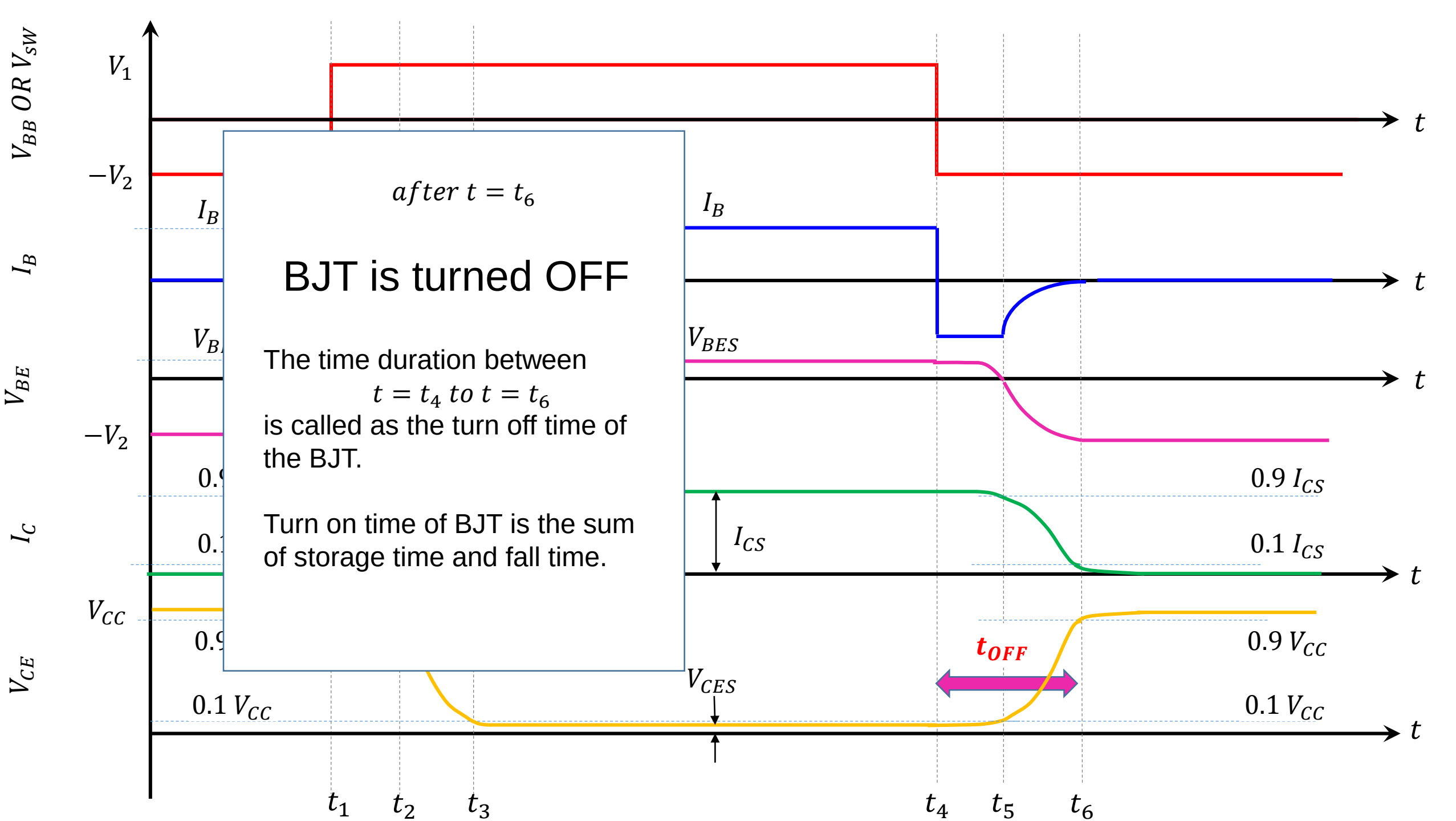


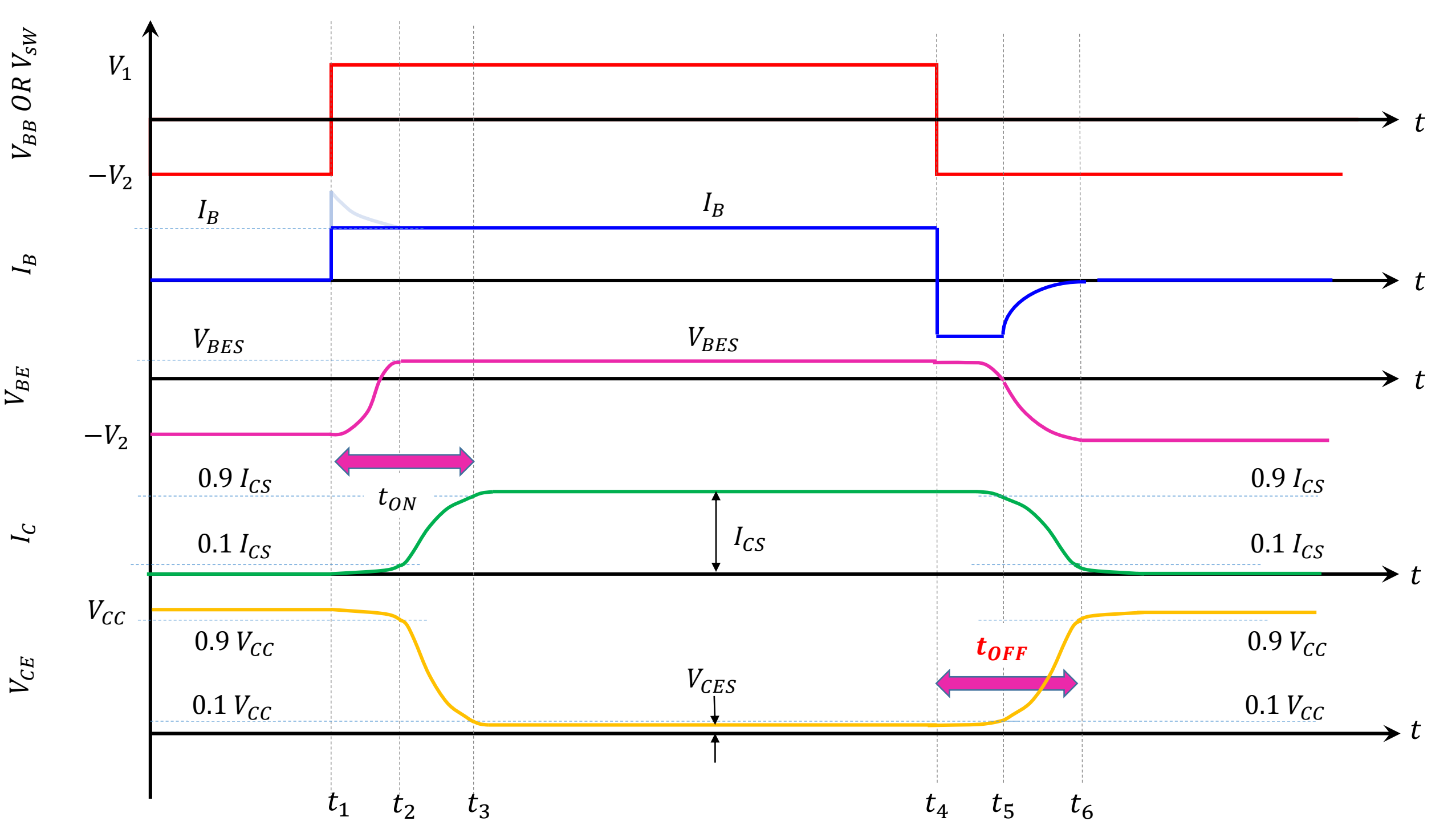




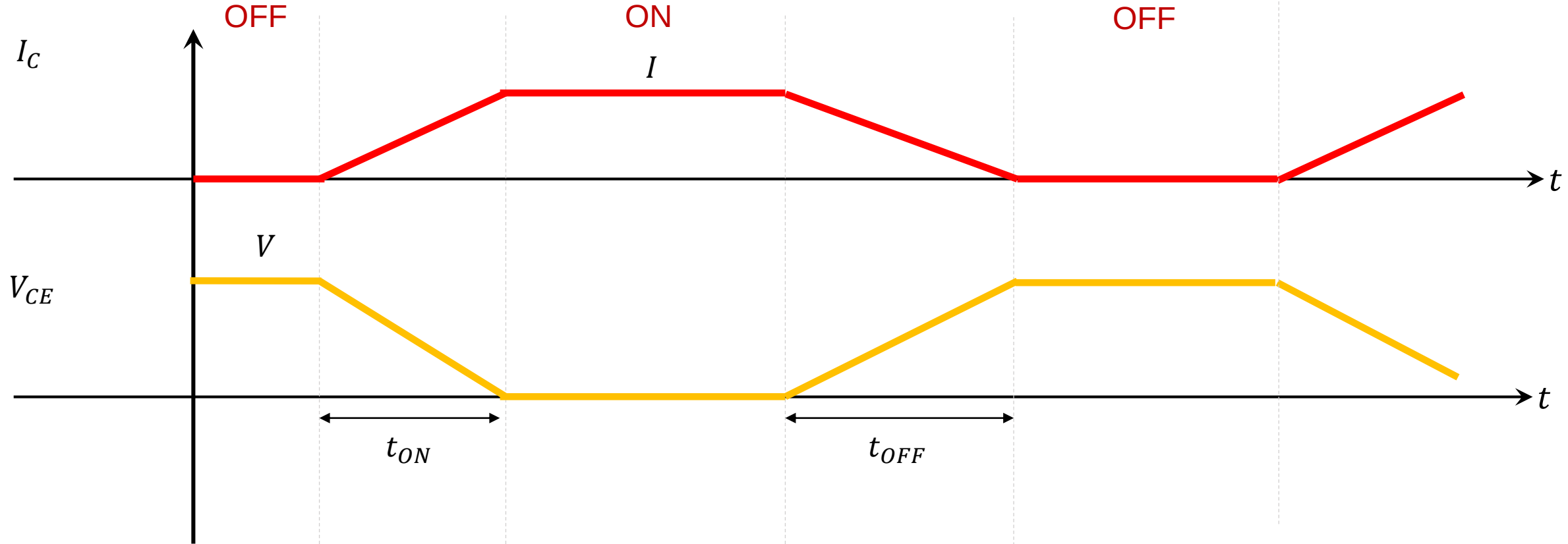








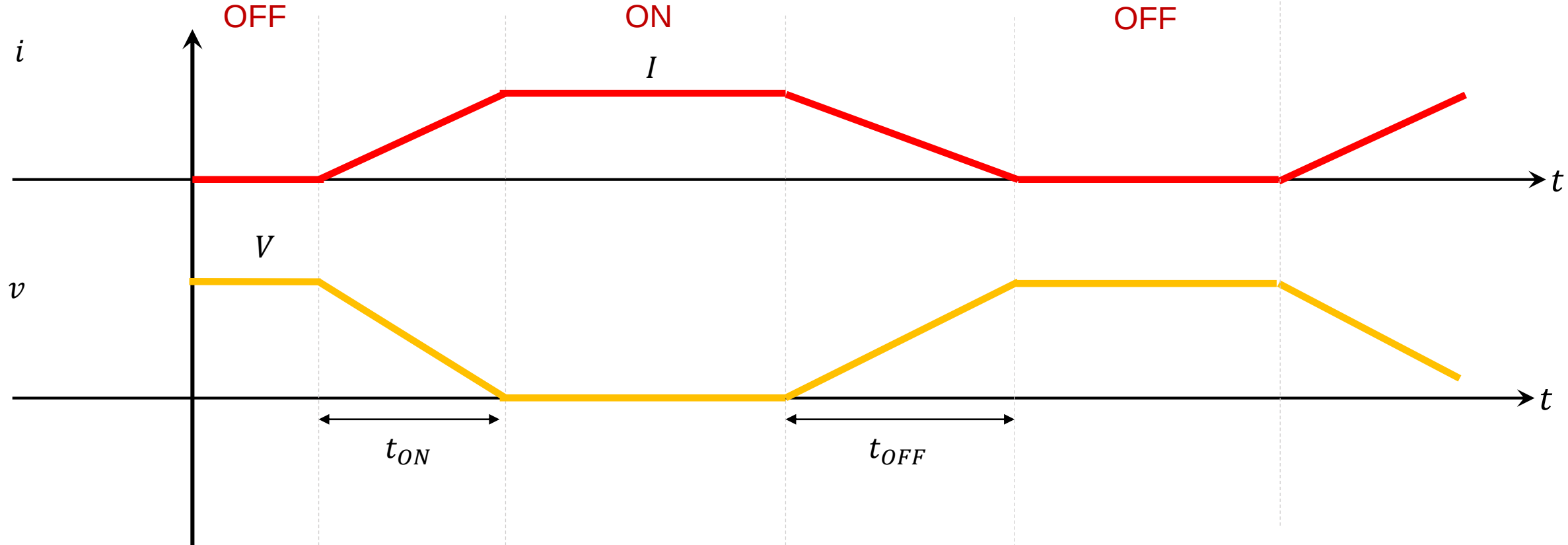
Q.1 A power BJT has its switching waveform as below.



Calculate the switching energy loss during ON state and OFF state. Also calculate the average power loss if the switching frequency is f

Q.1 A power BJT has its switching waveform as below.

$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1} (x - x_1)$$



During turn ON

$$i - 0 = \frac{I - 0}{t_{ON} - 0} (t - 0) \Rightarrow i = \frac{I}{t_{ON}} t$$

$$v - V = \frac{0 - V}{t_{ON} - 0} (t - 0) \Rightarrow v = V \left(1 - \frac{t}{t_{ON}} \right)$$

during turn OFF

$$i = I \left(1 - \frac{t}{t_{OFF}} \right)$$

$$v = \frac{V}{t_{OFF}} t$$

$$E_{on} = \int_0^{t_{ON}} v * i * dt = \int_0^{t_{ON}} V \left(1 - \frac{t}{t_{ON}} \right) * \frac{I}{t_{ON}} t * dt$$

$$E_{on} = \frac{VI}{t_{ON}} \int_0^{t_{ON}} \left(1 - \frac{t}{t_{ON}} \right) * t * dt = \frac{VI}{t_{ON}} \left[\left(\frac{t^2}{2} - \frac{t^3}{3t_{ON}} \right) \right]_0^{t_{ON}}$$

$$E_{on} = \frac{VI}{t_{ON}} \left(\frac{t_{ON}^2}{2} - \frac{t_{ON}^3}{3t_{ON}} \right) = \frac{VI}{t_{ON}} * \frac{t_{ON}^2}{6} = \frac{VI t_{ON}}{6}$$

$$P_{on} = E_{on} * f = \frac{VI t_{ON}}{6} * f$$

E_{off} = homework

Q.1 alternate

During power ON and OFF of a power BJT (or any other switch) supplying a resistive load, the voltage and current variations may be assumed to be linear w.r.t time.

1. Calculate the switching energy loss during ON state and OFF state. Also calculate the average power loss if the switching frequency is f .
2. Show that the maximum power loss during switching ON or OFF happens at $\frac{t_{ON}}{2}$ and $\frac{t_{OFF}}{2}$ respectively.
3. Show that the maximum power loss during switching ON or OFF is $\frac{VI}{4}$
4. Show that the switching energy loss during switching ON or OFF is $\frac{VIT}{6}$

Hint: Maximum power loss during switching ON

$$i = \frac{I}{t_{ON}} t$$

$$v = V \left(1 - \frac{t}{t_{ON}} \right)$$

$$p = iv = \frac{VI}{t_{on}} t \left(1 - \frac{t}{t_{ON}} \right)$$

$$\frac{dP}{dt} = 0 \quad t = \frac{t_{ON}}{2}$$

$$p_{\max} = \frac{VI}{4}$$

Q.1 A power BJT switches with the following parameters:

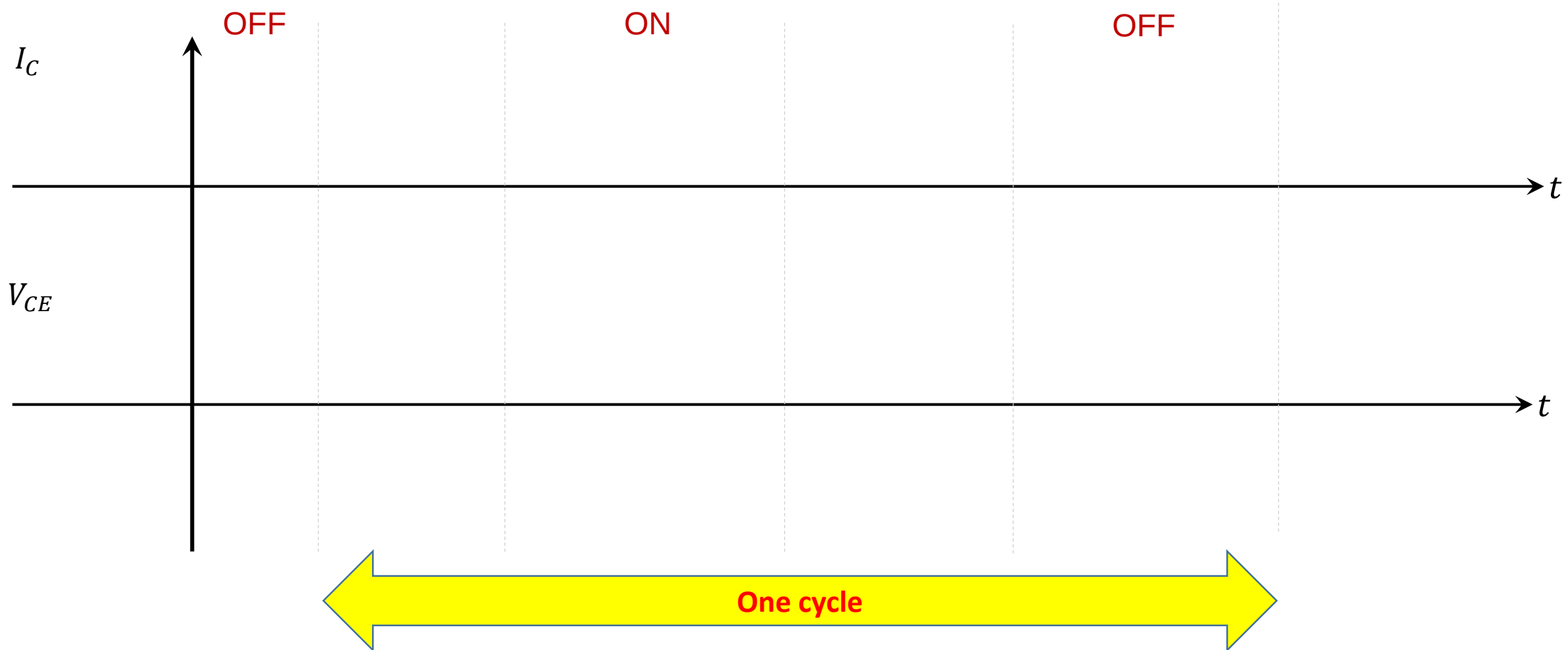
$$V_{CE} = 400\text{ V}, V_{CE(sat)} = 2\text{ V}, I_C = 10\text{ A}, t_{ON} = 2\mu\text{ s}, t_{OFF} = 3\mu\text{ s}, D = 0.5$$

If the switching frequency of the BJT is 20 KHz, calculate the total power loss.

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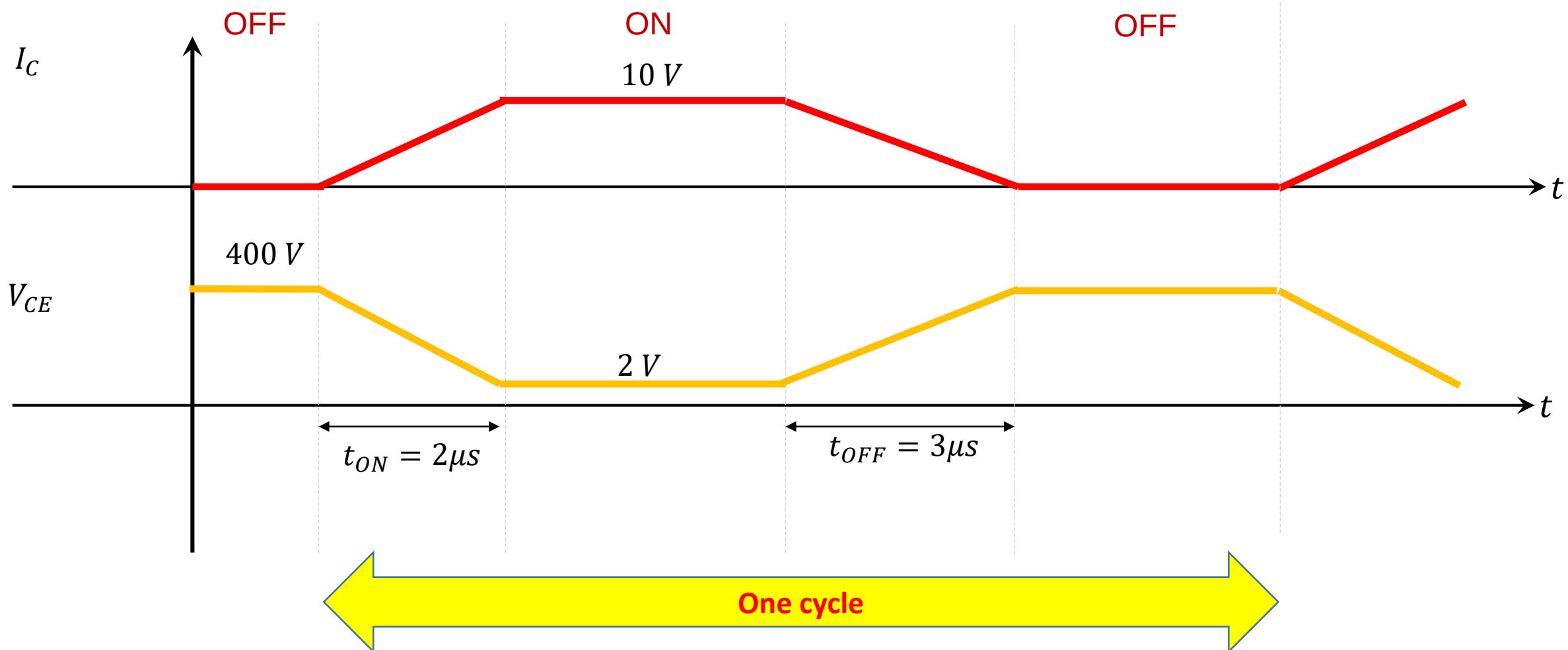
If the switching frequency of the BJT is 20 KHz, calculate the total power loss and efficiency



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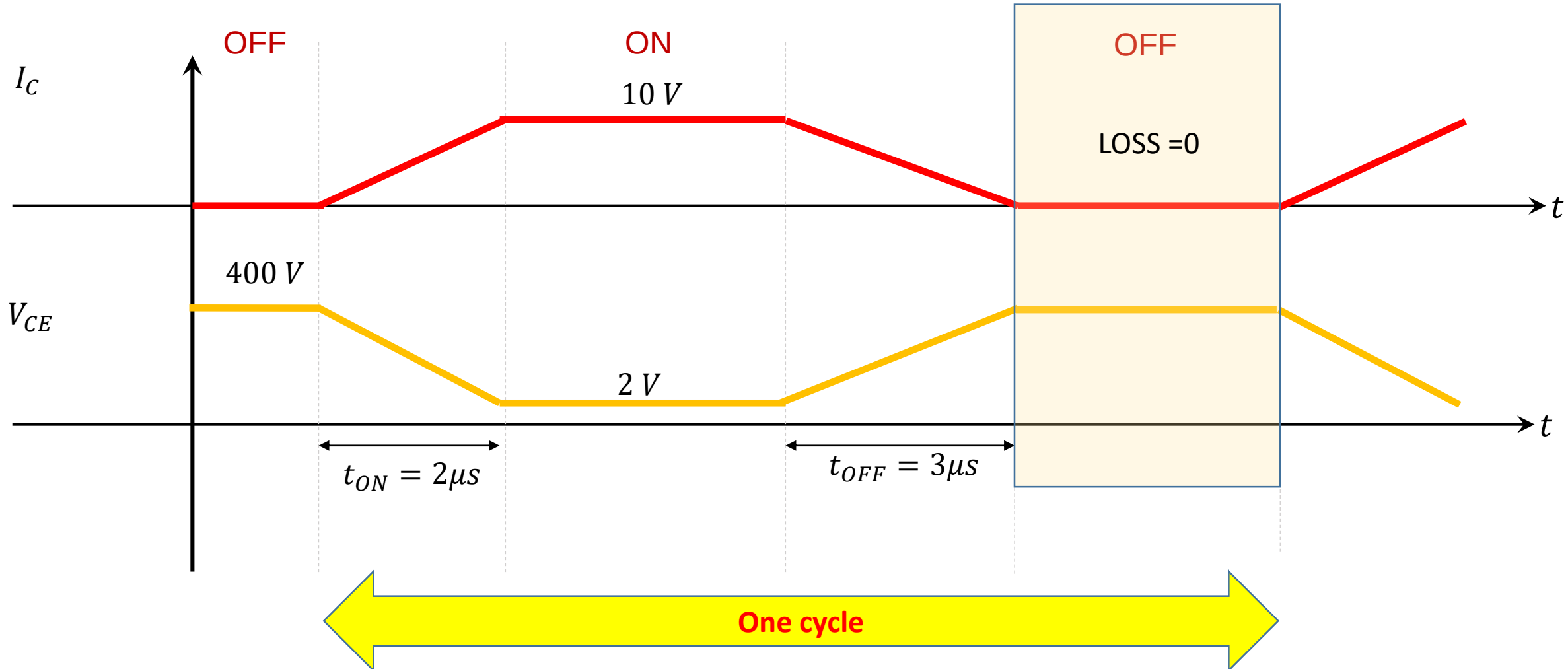
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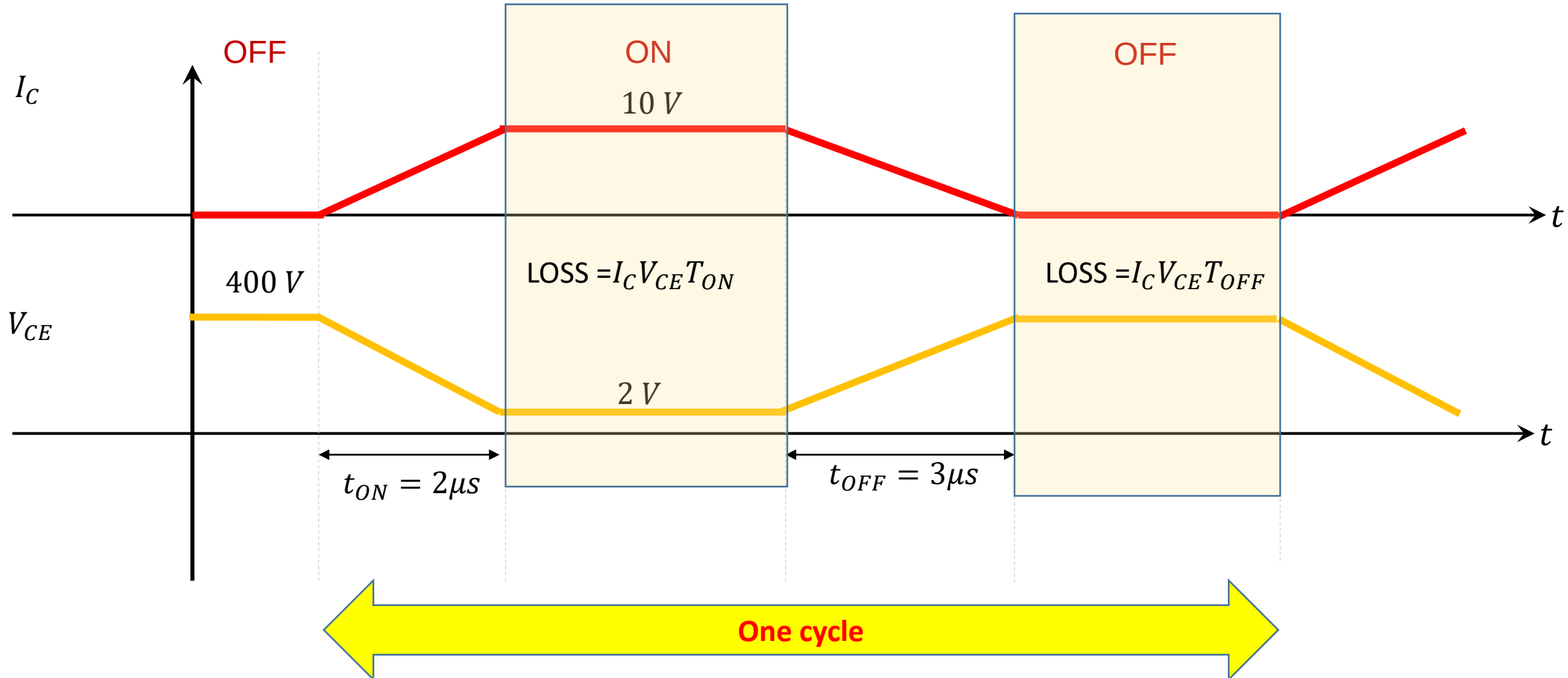
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If the switching frequency of the BJT is 20 KHz, calculate the total power loss and efficiency

Conduction loss in one cycle

$$P'_C = I_C V_{CE} T_{ON} + I_C V_{CE} T_{OFF}$$

Total Conduction loss

$$P_C = P'_C * f = \frac{P'_C}{T} = \frac{(I_C V_{CE} T_{ON} + I_C V_{CE} T_{OFF})}{T}$$

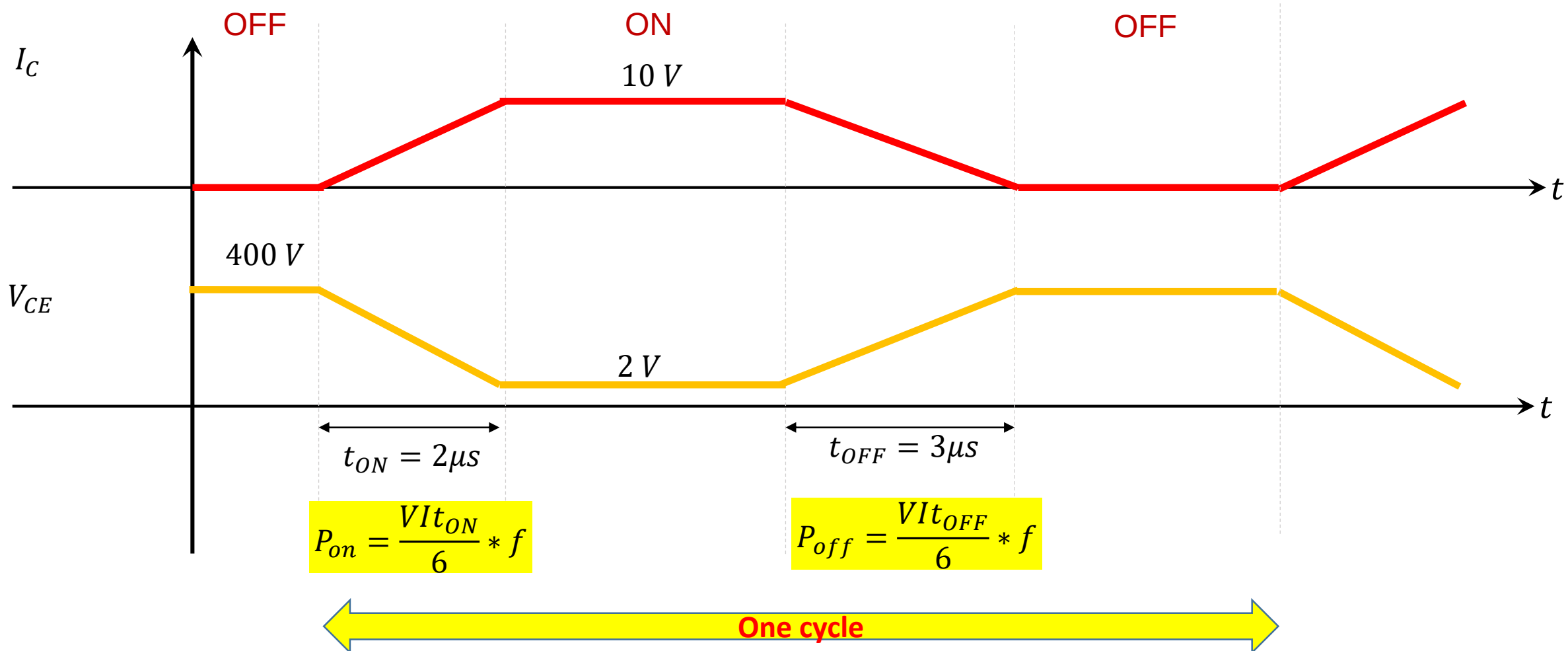
$$P_C = \frac{I_{CS} V_{CES} T_{ON}}{T} = I_{CS} * V_{CES} * D$$

10 W

Q.1 A power BJT switches with the following parameters:

$$V_{CE} = 400\text{ V}, V_{CE(sat)} = 2\text{ V}, I_C = 10\text{ A}, t_{ON} = 2\mu\text{s}, t_{OFF} = 3\mu\text{s}, D = 0.5$$

If the switching frequency of the BJT is 20 KHz, calculate the total power loss and efficiency



Q.1 A power BJT switches with the following parameters:

$$V_{CE} = 400\text{ V}, V_{CE(sat)} = 2\text{ V}, I_C = 10\text{ A}, t_{ON} = 2\mu\text{ s}, t_{OFF} = 3\mu\text{ s}, D = 0.5$$

If the switching frequency of the BJT is 20 KHz, calculate the total power loss and efficiency

Switching loss during turn ON and OFF

$$P_{on} = \frac{V I t_{ON}}{6} * f$$

$$P_{off} = \frac{V I t_{OFF}}{6} * f$$

Total Switching loss

$$P_{sw} = P_{on} + P_{off} = \frac{V_{CE} I_C}{6} (t_{ON} + t_{OFF}) f$$

66.67 W

$$P_{sw} = \frac{400 * 10}{6} * (2 \times 10^{-6} + 3 \times 10^{-6}) * 20 \times 10^3$$

Total Power Loss = 76.67 W

BJT Problem

Q. The BJT is specified to have a range of $\beta = 8 \text{ to } 40$. The load resistance $R_e = 11\Omega$. The dc supply voltage $V_{CC} = 200 \text{ V}$ and the input voltage to the base circuit is $V_{BB} = 10 \text{ V}$. If $V_{CE,sat} = 1.0 \text{ V}$ and $V_{BE,sat} = 1.5 \text{ V}$, Find

- a) The value of R_B that results in saturation with a ODF of 5.
- b) The forced β_F
- c) The power loss in the transistor

BJT Problem

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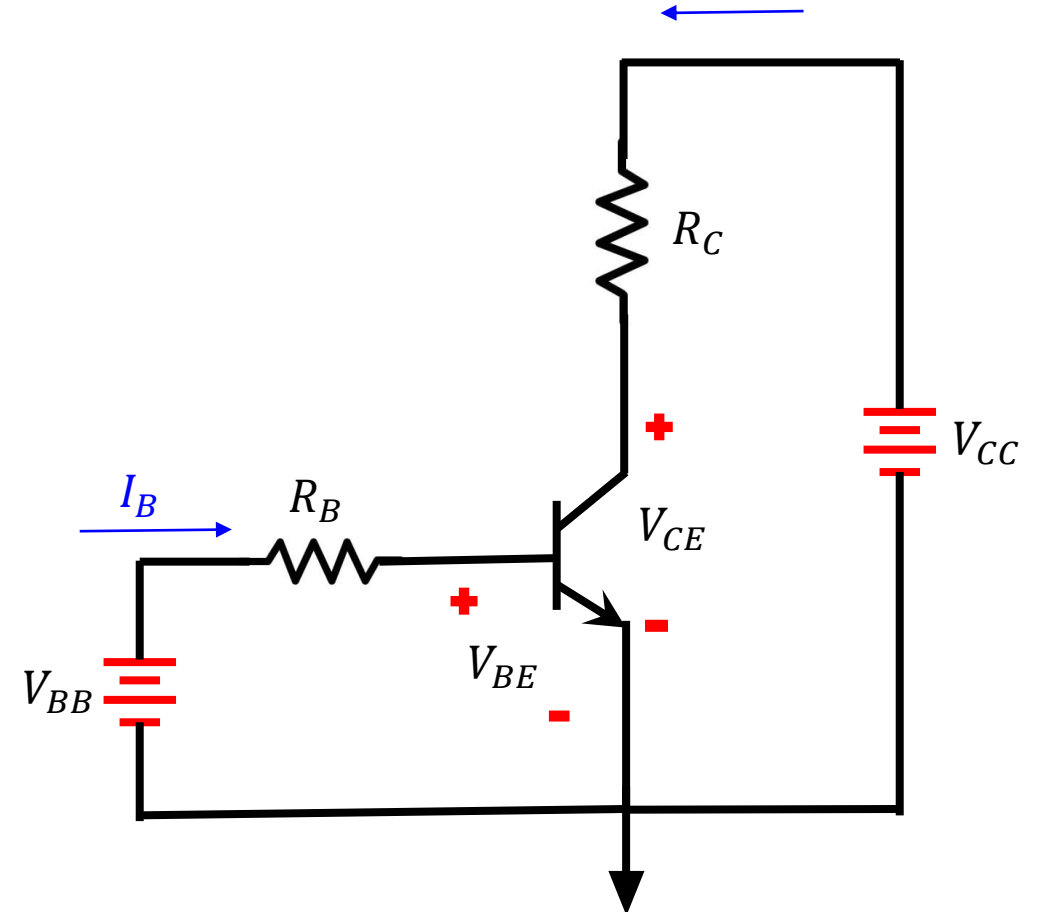
$$ODF = \frac{I_B}{I_{BS}} = 5$$

$$I_B = \frac{V_{BB} - V_{BE,sat}}{R_B} = \frac{10 - 1.5}{R_B}$$

$$I_{BS} = \frac{I_{CS}}{\beta}$$

$$I_{CS} = \frac{V_{CC} - V_{CE,sat}}{R_C} = \frac{200 - 1}{11} = 18.1 \text{ A}$$

$$I_{BS} = \frac{18.1}{\beta}$$



BJT Problem

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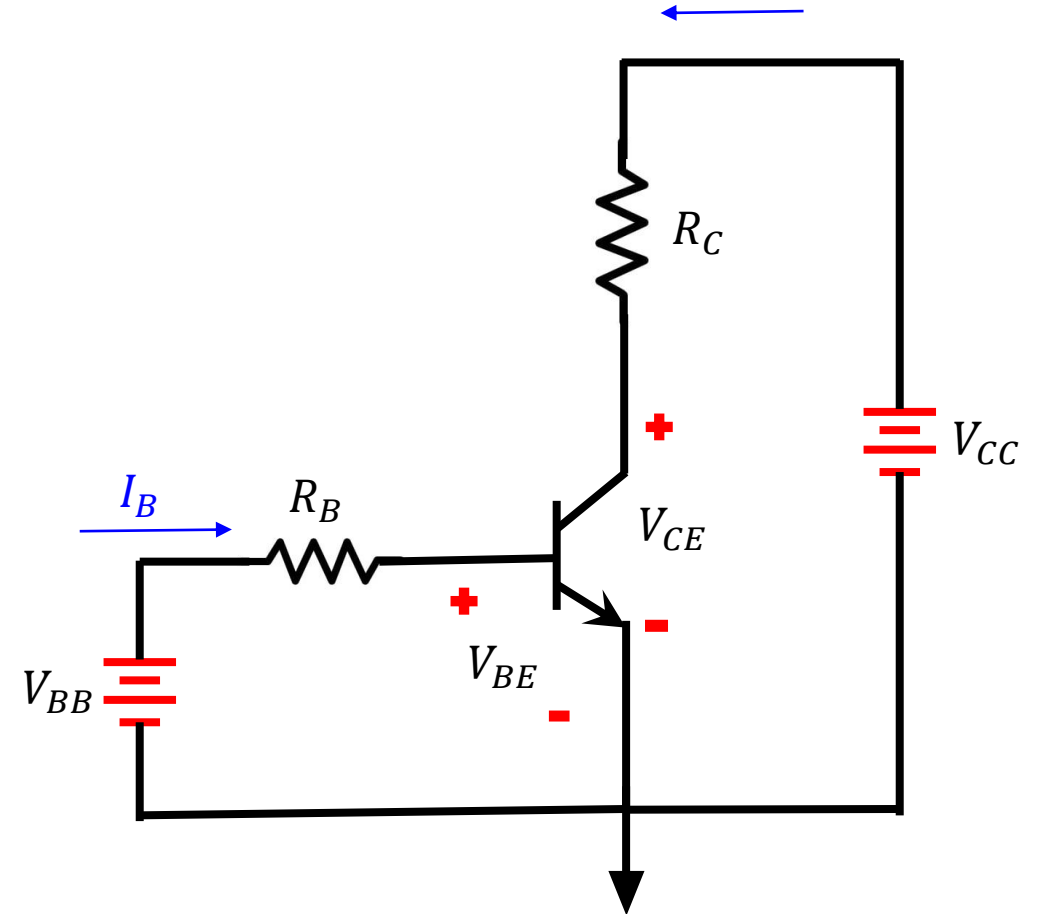
$$ODF = \frac{I_B}{I_{BS}} = 5$$

$$I_B = \frac{V_{BB} - V_{BE,sat}}{R_B} = \frac{10 - 1.5}{R_B}$$

$$I_{BS} = \frac{I_{CS}}{\beta}$$

$$I_{CS} = \frac{V_{CC} - V_{CE,sat}}{R_C} = \frac{200 - 1}{11} = 18.1 \text{ A}$$

$$I_{BS} = \frac{18.1}{8} = 2.2625 \text{ A}$$



BJT Problem

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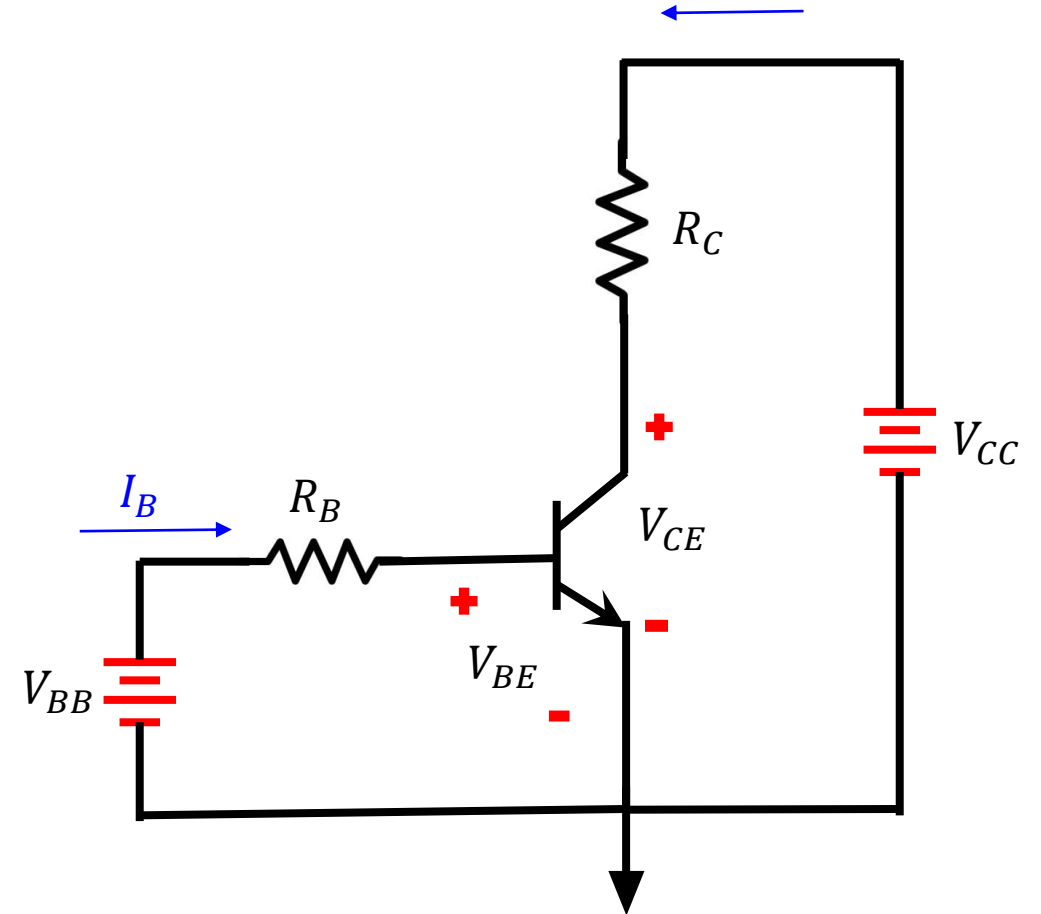
a) The value of R_B that results in saturation with a ODF of 5.

$$ODF = \frac{I_B}{I_{BS}} = 5 \quad I_{BS} = 2.2625\text{ A}$$

$$I_B = 5 \times 2.2625 = 11.3125\text{ A}$$

$$\frac{10 - 1.5}{R_B} = 11.3125$$

$$R_B = \frac{8.5}{11.3125} = 0.715\ \Omega$$



BJT Problem

Q. The BJT is specified to have a range of $\beta = 8 \text{ to } 40$. The load resistance $R_e = 11\Omega$. The dc supply voltage $V_{CC} = 200 \text{ V}$ and the input voltage to the base circuit is $V_{BB} = 10 \text{ V}$. If $V_{CE,sat} = 1.0 \text{ V}$ and $V_{BE,sat} = 1.5 \text{ V}$, Find

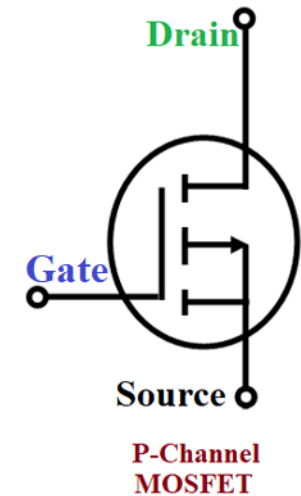
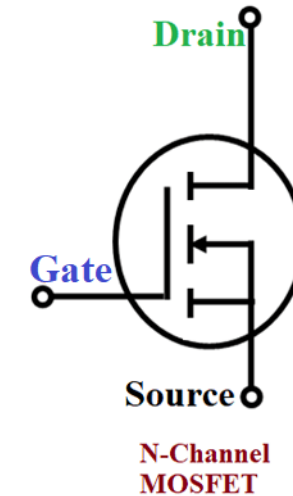
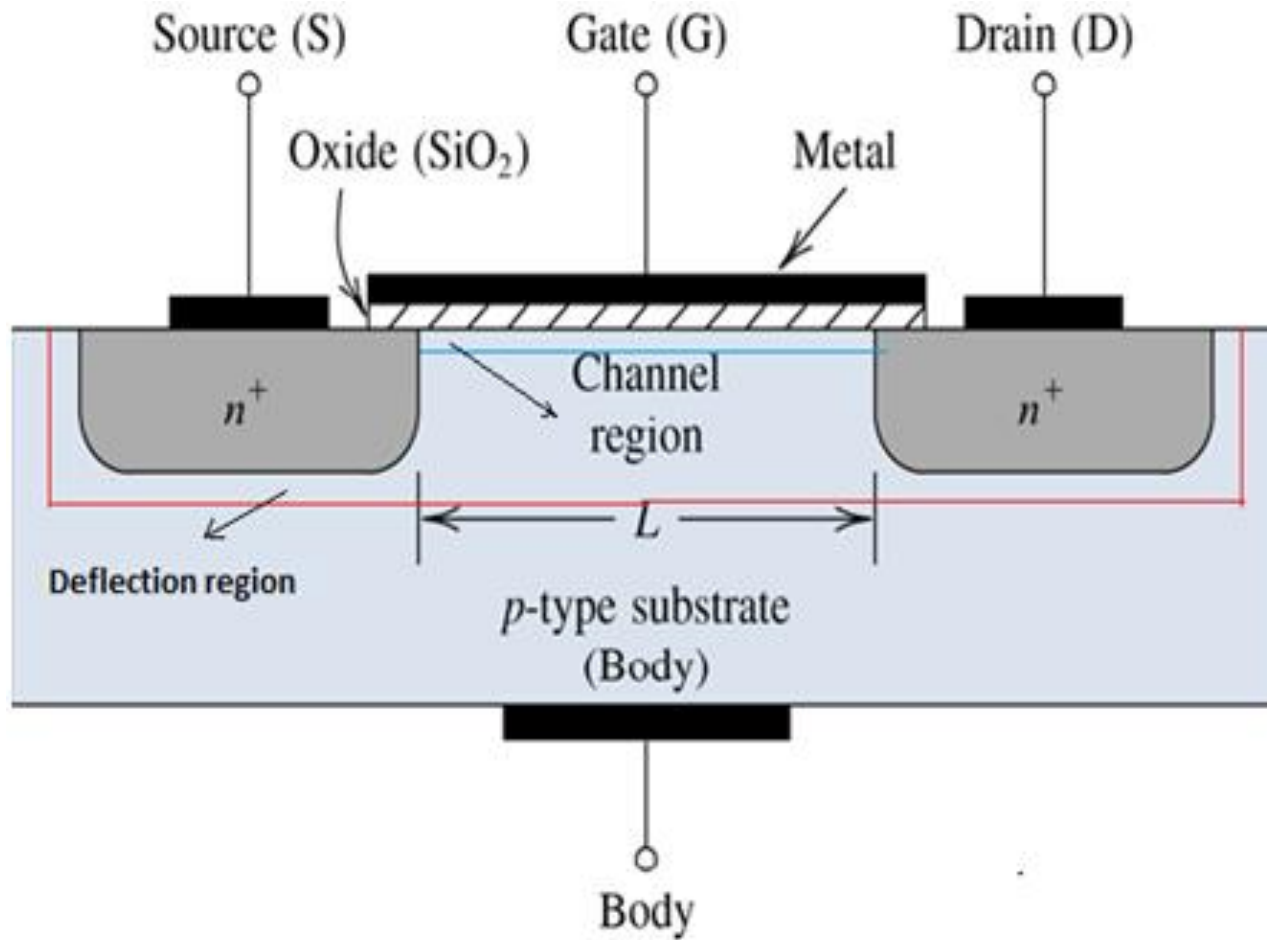
- a) The value of R_B that results in saturation with a ODF of 5.
- b) The forced β_F
- c) The power loss in the transistor

$$\beta_F = \frac{I_{CS}}{I_B} = \frac{18.1}{11.3125} = 1.6$$

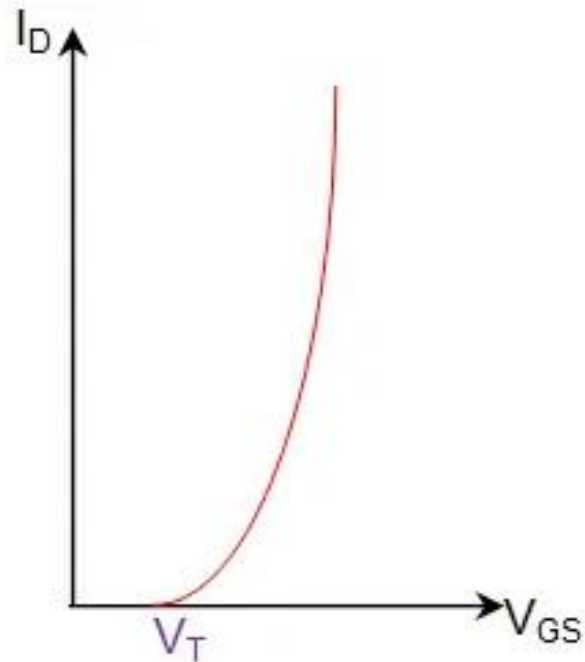
$$P_{loss} = V_{BE} * I_B + V_{CE} * I_C = 1.5 * 11.3125 + 1 * 18.1$$

$$P_{loss} = 35.07 \text{ W}$$

MOSFET SCHEMATIC

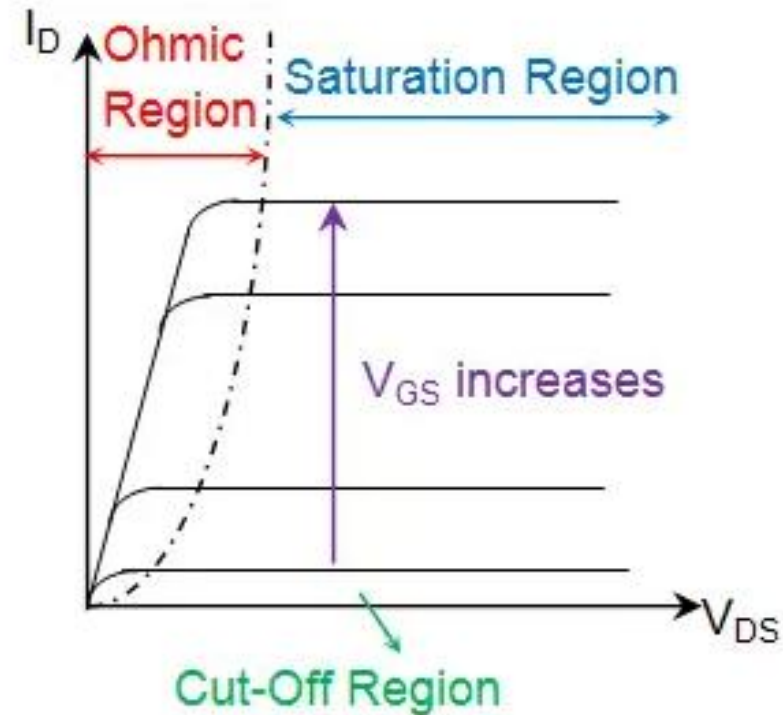


TX AND O/P CHARACTERISTICS (MOSFET)



(a)

Transfer Characteristics



(b)

Output Characteristics

A MOSFET operates as a switch using its **Cutoff** and **Saturation** regions:

- Cutoff Region (OFF state):** When $V_{GS} < V_{th}$ (for NMOS), the MOSFET is OFF, acting as an open switch with no current flow.
- Saturation Region (ON state):** When $V_{GS} > V_{th}$ and $V_{DS} \geq (V_{GS} - V_{th})$, the MOSFET is fully ON, allowing maximum current flow like a closed switch.

POWER MOSFET

- A Power MOSFET is a specialized MOSFET designed for high power applications.
- It offers fast switching speeds and superior performance at low voltages compared to standard MOSFETs.
- Its fundamental operating principle remains the same as that of general MOSFETs.
- The most common types are n-channel Enhancement-mode, p-channel Enhancement-mode, and n-channel Depletion-mode.
- Further, there are a wide variety of power MOSFET structures like Vertical Diffused MOS (VDMOS) or Double-Diffused MOS or DMOS, etc.
- In vertical-structure Power MOSFETs, the doping and thickness of the epitaxial layer determine the voltage rating, while the channel width sets the current rating. These characteristics allow them to support high blocking voltages and currents, ideal for low power switching applications.
- Power MOSFETs are extensively used as a part of power supplies, DC-DC converters and low-voltage motor controllers.

VDMOS (POWER MOSFET)

Vertically diffused

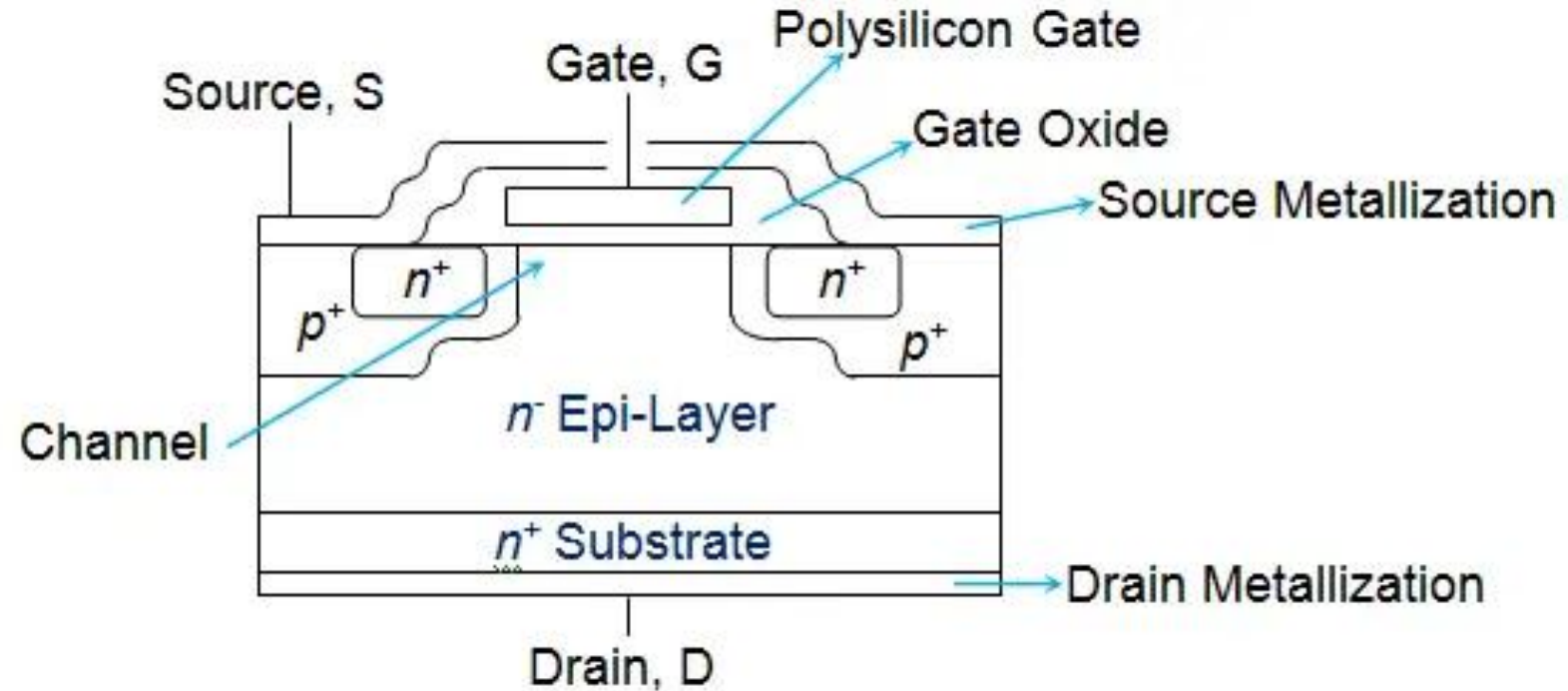
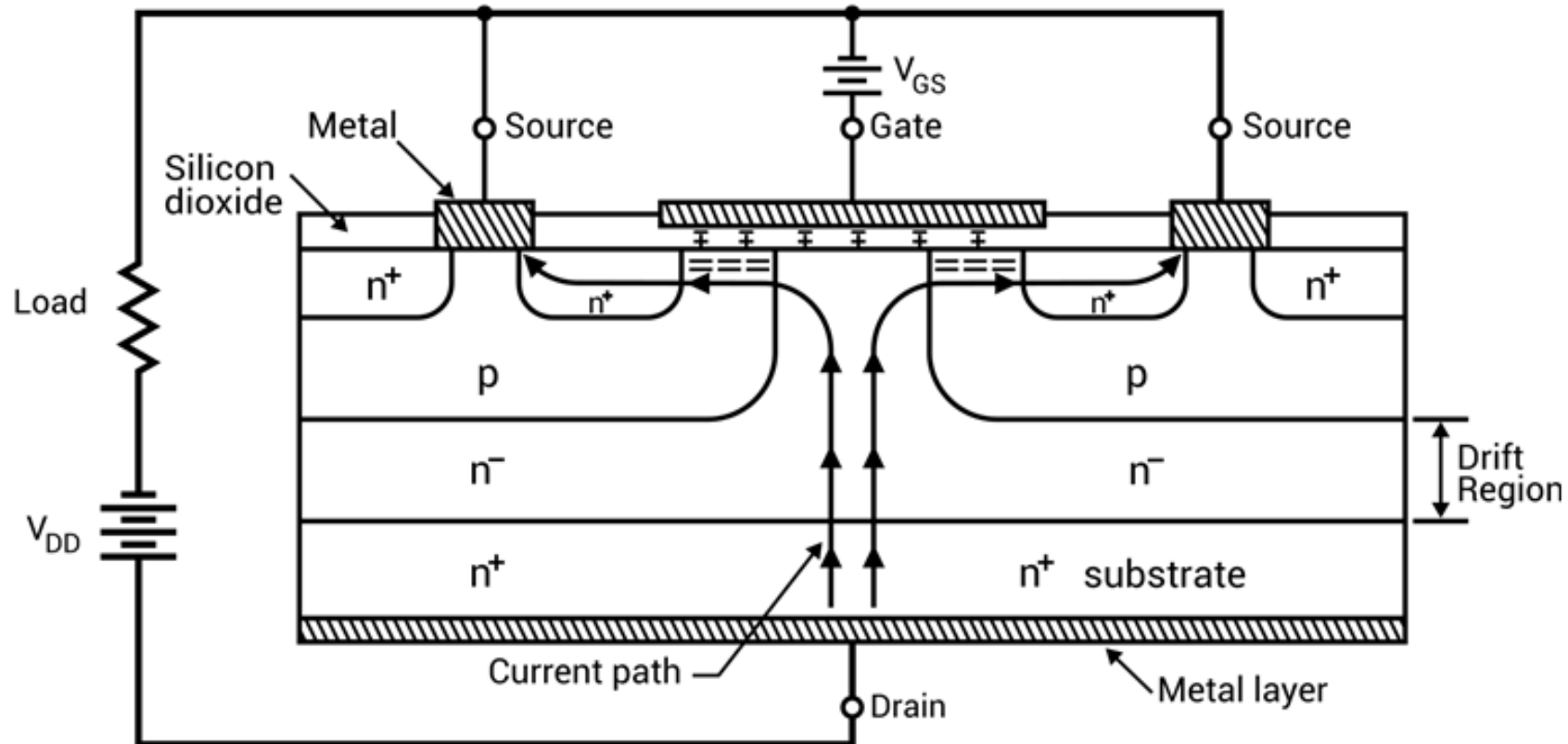


Figure 1 Schematic of n -substrate Power MOSFET

DMOS (POWER MOSFET)

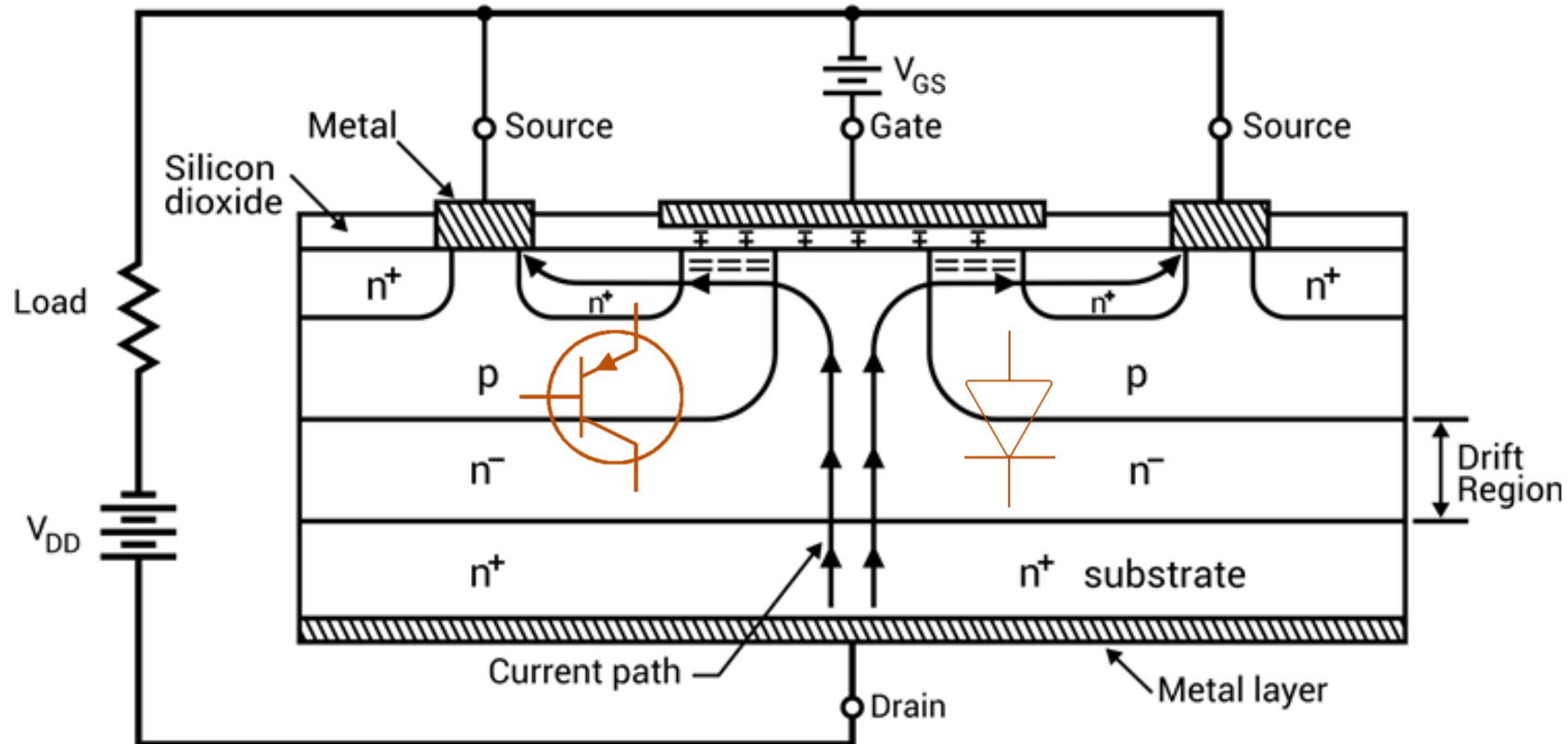
Double diffused



Basic structure of a n-channel DMOS power MOSFET

DMOS (POWER MOSFET)

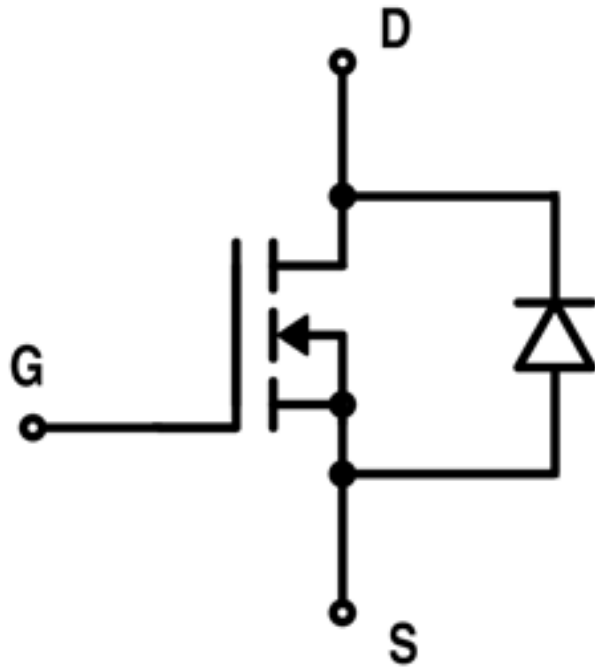
Double diffused



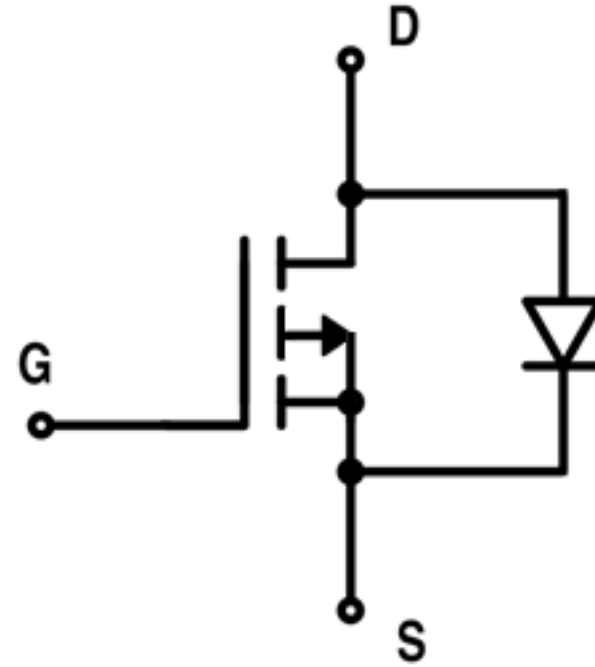
Basic structure of a n-channel DMOS power MOSFET

* Due to the presence of the **parasitic diode**, **POWER MOSFET** doesn't have **reverse voltage blocking** capability.

SYMBOL (POWER MOSFET)

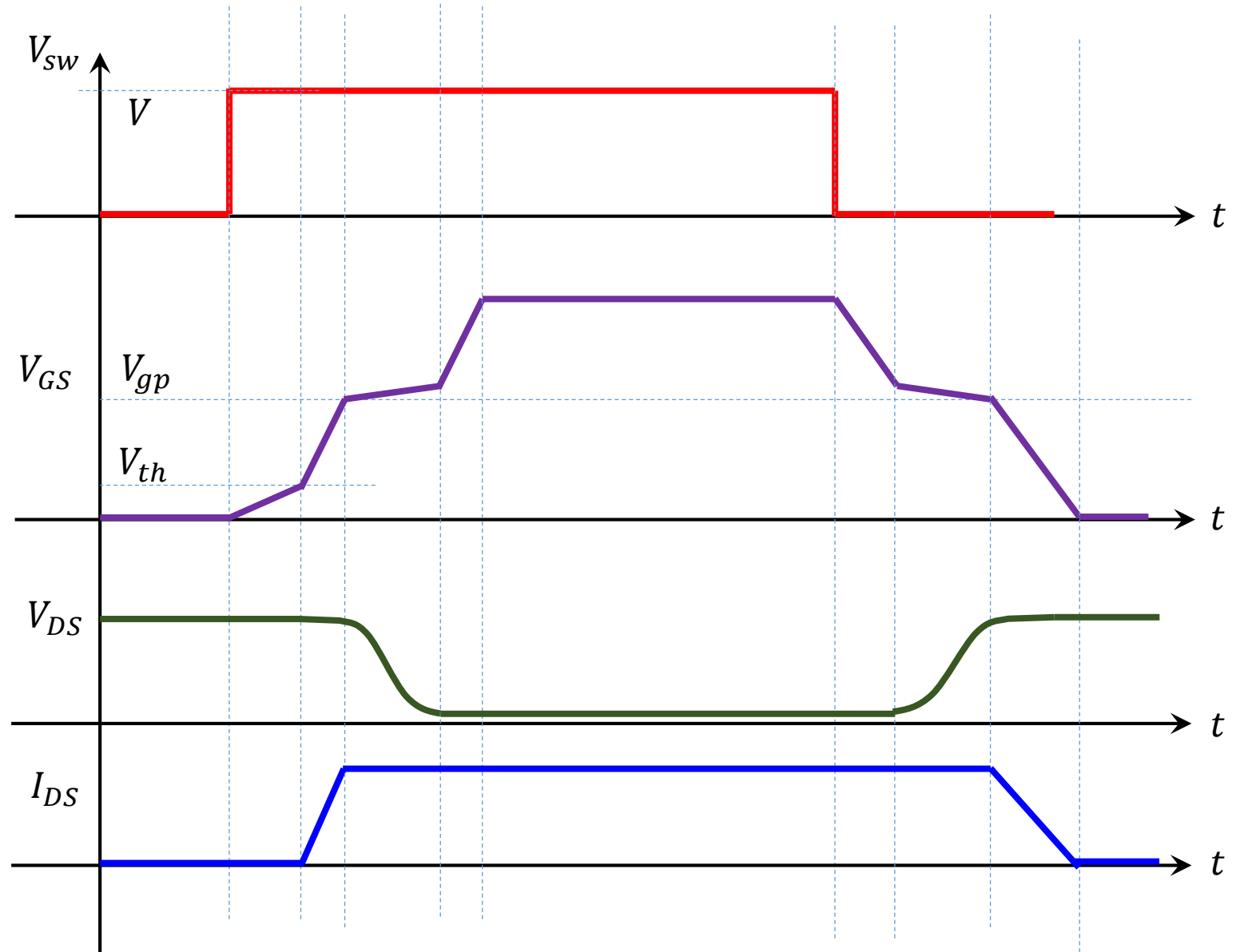
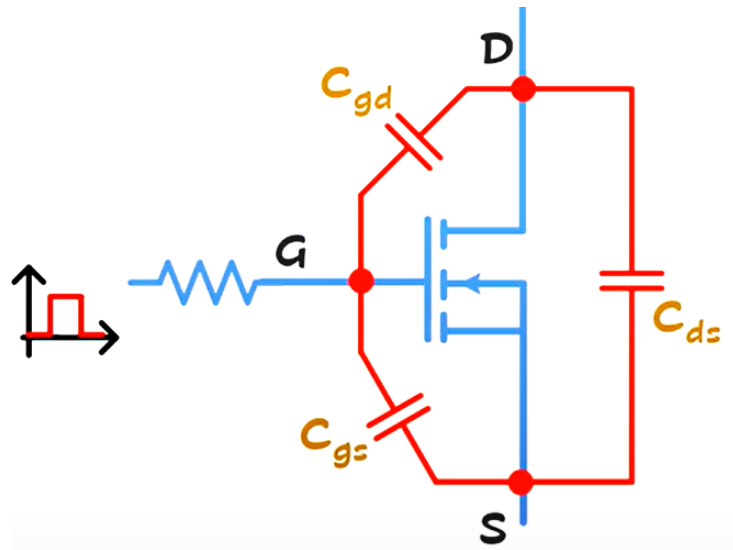


N MOS



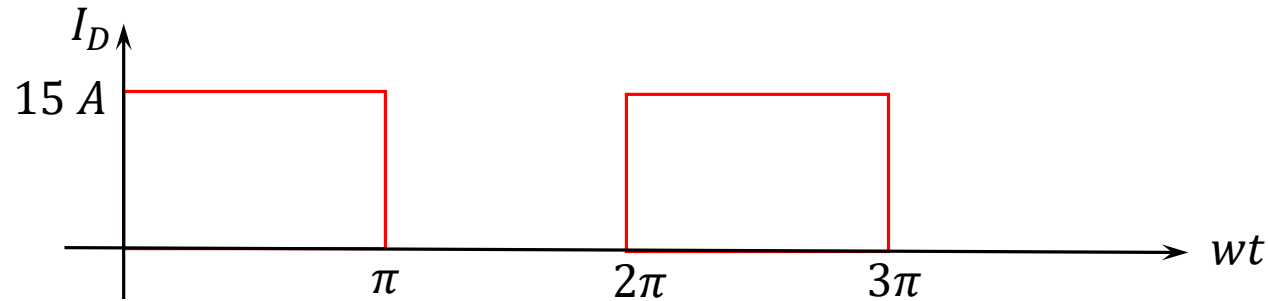
P MOS

SWITCHING CHARACTERISTICS (POWER MOSFET)



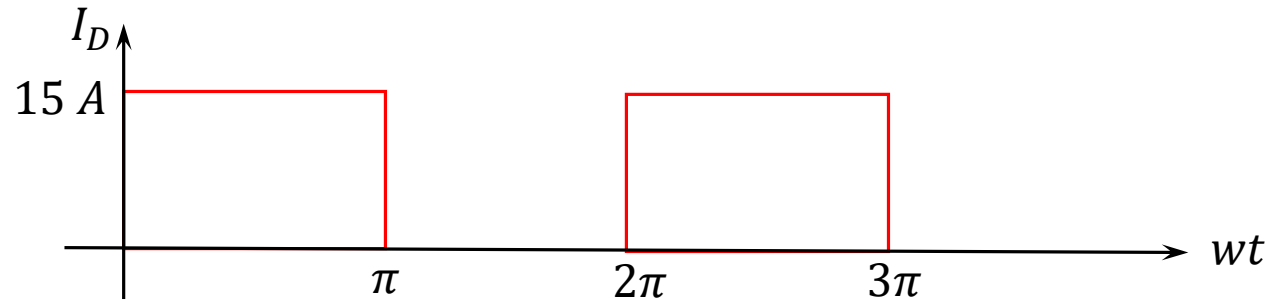
NUMERICALS (POWER MOSFET)

Q.1. A power MOSFET rated for 20A is carries a periodic current as shown in the figure below. The ON state resistance of the MOSFET is 0.12 ohm. Draw the waveform of power loss and calculate the average ON state power loss.

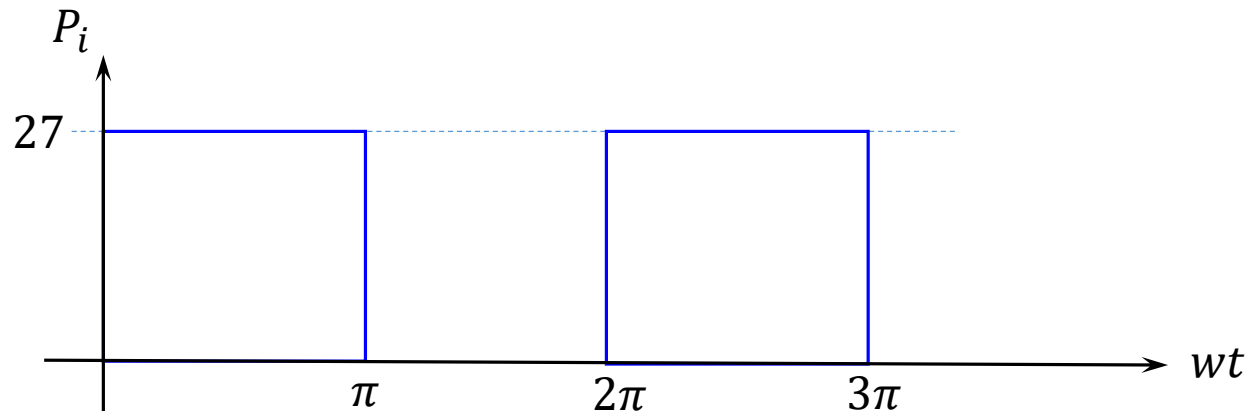


NUMERICALS (POWER MOSFET)

Q.1. A power MOSFET rated for 20A is carries a periodic current as shown in the figure below. The ON state resistance of the MOSFET is 0.12 ohm. Draw the waveform of power loss and calculate the average ON state power loss.



$$P_i = vi = i^2 R \quad 15^2 \times 0.12 = 27 \text{ W}$$

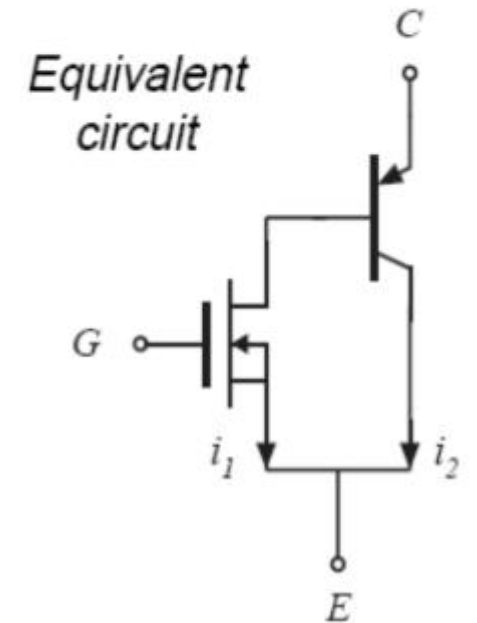
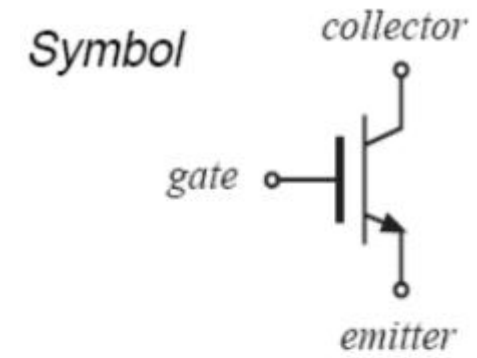
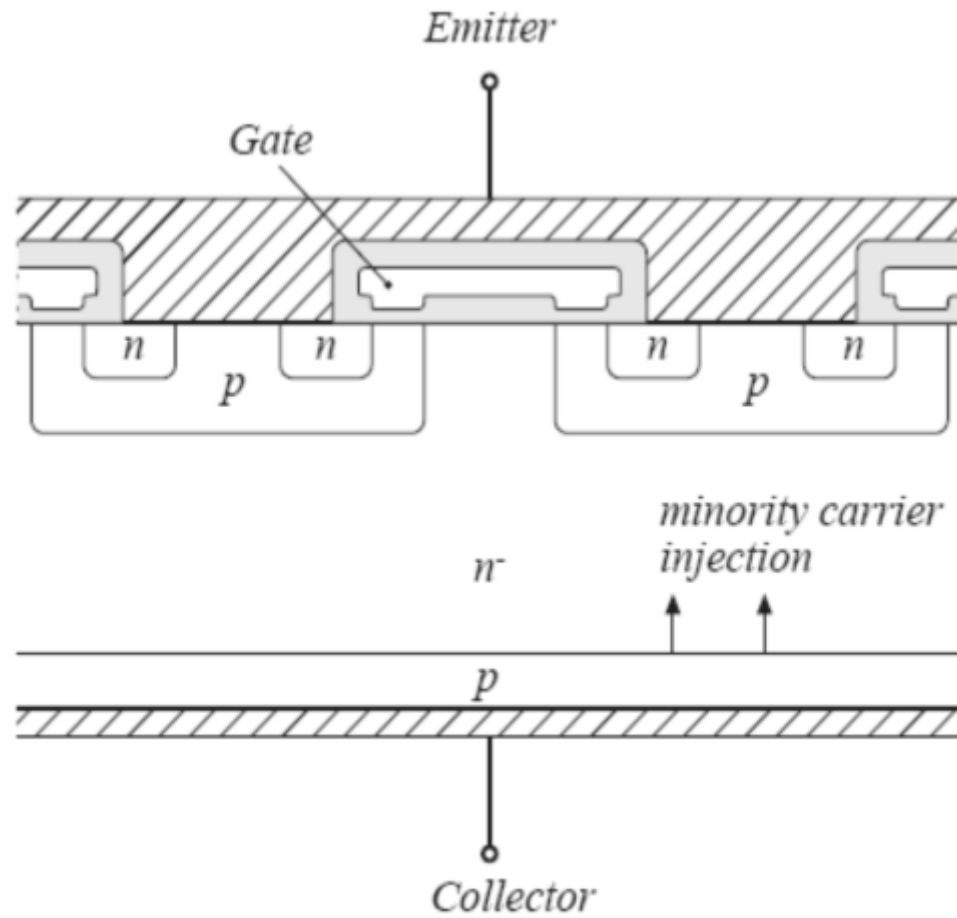


$$P_i = 27 \text{ W}, 0 < wt < \pi$$

$$P_i = 0, \pi < wt < 2\pi$$

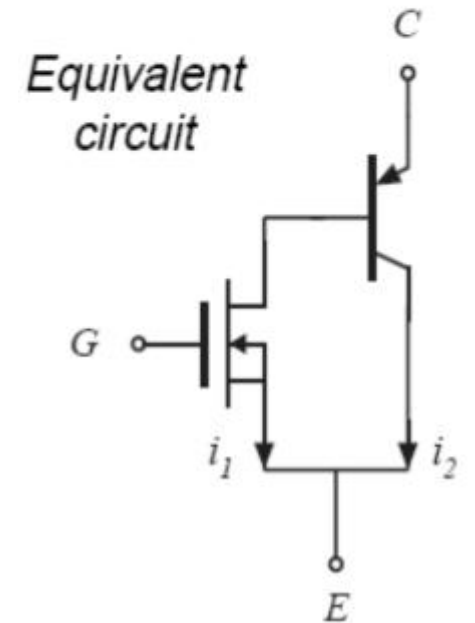
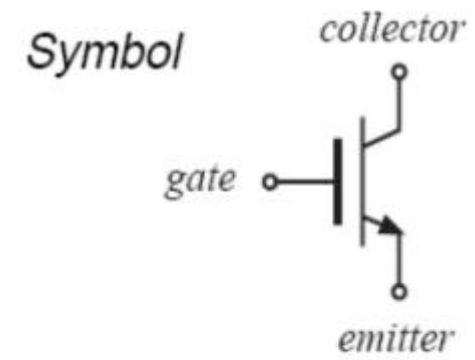
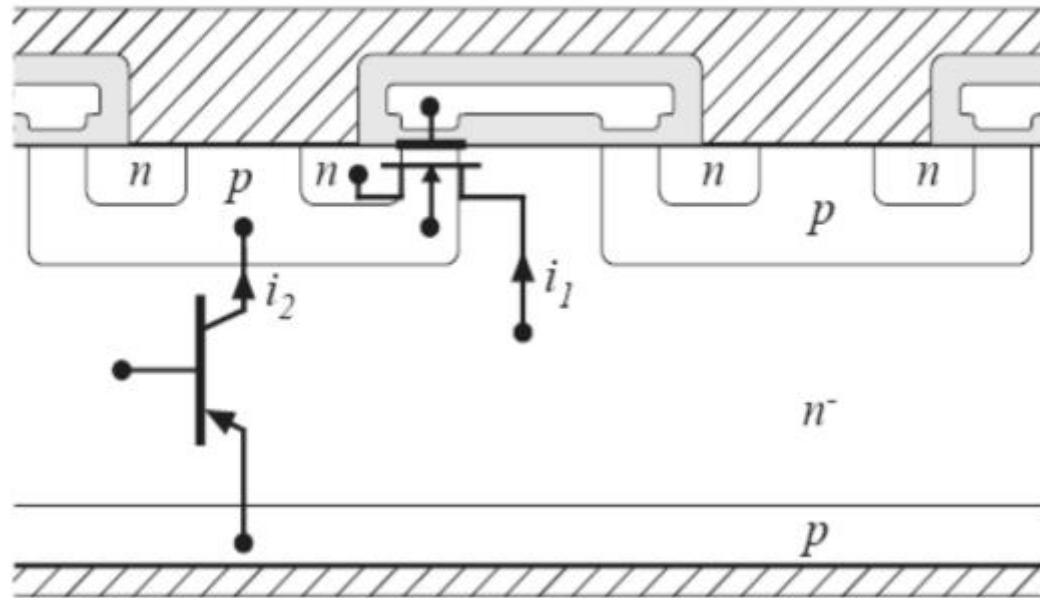
$$P_{avg} = \frac{1}{2\pi} \int_0^{\pi} 27 \, d(wt) = \frac{27}{2} = 13.5 \text{ W}$$

IGBT:



IGBT:

Location of equivalent devices



ADVANTAGES OF IGBT:

Combines the advantages of BJT & MOSFET

- High input impedance like MOSFET
- Voltage controlled device like MOSFET
- Simple gate drive, Lower switching loss
- Low on state conduction power loss like BJT
- Higher current capability & higher switching speed than a BJT. (Switching speed lower than MOSFET)

Applications of IGBT

- ac and dc motor controls.
- General purpose inverters.
- Uninterrupted Power Supply (UPS).
- Welding Equipments.
- Numerical control, Cutting tools.
- Robotics & Induction heating.

POWER TRANSISTOR COMPARISON

Power BJT	Power MOSFET	IGBT
Current Controlled	Voltage Controlled	Voltage Controlled
Minority carrier device	Majority carrier device	Minority carrier device
Low input impedance	High input impedance	High input impedance
Low conduction loss	High conduction loss	Low conduction loss
High Switching loss (High t_{ON} , t_{OFF})	Low Switching loss (Low t_{ON} , t_{OFF})	Moderate (moderate t_{ON} , t_{OFF})
Negative Temperature coefficient, Secondary Breakdown	Positive Temperature coefficient, No Secondary Breakdown	Positive Temperature coefficient, No Secondary Breakdown